

CMakeLists.txt

```
1 cmake_minimum_required(VERSION 3.16)
2 project(slipier VERSION 1.0)
3
4 set(CMAKE_CXX_STANDARD 17)
5 set(CMAKE_PREFIX_PATH "C:/Qt/6.7.3/mingw_64")
6
7 find_package(Qt6 REQUIRED COMPONENTS Core Quick Charts Widgets Sql PrintSupport Network Multimedia)
8
9 qt_standard_project_setup()
10 qt_policy(SET QTP0001 NEW)
11
12 qt_add_executable(slipier
13     main.cpp
14     src/Backend.h
15     src/Backend.cpp
16     src/SensorManager.h
17     src/SensorManager.cpp
18     src/Calculator.h
19     src/Calculator.cpp
20     src/Database.h
21     src/Database.cpp
22     src/ReportManager.h
23     src/ReportManager.cpp
24     src/WifiManager.h
25     src/WifiManager.cpp
26     src/VoiceCommandManager.h
27     src/VoiceCommandManager.cpp
28     src/VoiceRecognitionWorker.h
29     src/VoiceRecognitionWorker.cpp
30 )
31
32 set_source_files_properties(qml/Translations.qml PROPERTIES
33     QT_QML_SINGLETON_TYPE TRUE
34 )
35
36 qt_add_qml_module(slipier
37     URI slipier
38     VERSION 1.0
39
40     QML_FILES
41         qml/Main.qml
42         qml/Translations.qml
43         qml/pages/LoginPage.qml
44         qml/pages/DashboardPage.qml
45         qml/pages/OlcumPage.qml
46         qml/pages/SonuclarPage.qml
47         qml/pages/GecmisPage.qml
48         qml/pages/KalibrasyonPage.qml
49         qml/components/SuTerazisi.qml
50         qml/components/TarihSecici.qml
51
52     RESOURCES
53         qml/assets/logo.png
54         qml/assets/egim_sensor.png
55         qml/assets/loadcell.png
56         qml/assets/mesafe_sensor.png
57         qml/assets/slipergif.mp4
58 )
59
60 target_link_libraries(slipier PRIVATE
61     Qt6::Core
62     Qt6::Quick
63     Qt6::Charts
64     Qt6::Widgets
65     Qt6::Sql
66     Qt6::PrintSupport
67     Qt6::Network
68     Qt6::Multimedia
69 )
70
71 # --- Sesli komut (Vosk, offline Turkce konusma tanima) ---
72 set(VOSK_DIR "${CMAKE_SOURCE_DIR}/third_party/vosk")
73 set(VOSK_MODEL_DIR "${CMAKE_SOURCE_DIR}/resources/vosk-model-small-tr-0.3")
74
75 target_include_directories(slipier PRIVATE ${VOSK_DIR})
76 target_link_libraries(slipier PRIVATE "${VOSK_DIR}/libvosk.lib")
77
```

```
78 # libvosk.dll kendi MinGW calisma zamanini (libstdc++/libwinpthread/libgcc_s)
79 # gerektirir. Vosk paketiyle gelen bu DLL'leri DEGIL, projenin kullandigi
80 # Qt MinGW derleyicisiyle ayni surumden gelenleri kopyaliyoruz; aksi halde
81 # surum uyumsuzlugu uygulamanin ENTRYPOINT_NOT_FOUND ile cokmesine yol acar.
82 get_filename_component(MINGW_BIN_DIR "${CMAKE_CXX_COMPILER}" DIRECTORY)
83
84 add_custom_command(TARGET sliper POST_BUILD
85     COMMAND ${CMAKE_COMMAND} -E copy_if_different
86         "${VOSK_DIR}/libvosk.dll"
87         "${MINGW_BIN_DIR}/libstdc++-6.dll"
88         "${MINGW_BIN_DIR}/libwinpthread-1.dll"
89         "${MINGW_BIN_DIR}/libgcc_s_seh-1.dll"
90         "${TARGET_FILE_DIR:sliper}"
91     COMMENT "Vosk kutuphaneleri kopyalaniyor"
92 )
93
94 add_custom_command(TARGET sliper POST_BUILD
95     COMMAND ${CMAKE_COMMAND} -E copy_directory
96         "${VOSK_MODEL_DIR}"
97         "${TARGET_FILE_DIR:sliper}/vosk-model-small-tr-0.3"
98     COMMENT "Vosk Turkce ses tanima modeli kopyalaniyor"
99 )
```

main.cpp

```
1 #include <QmlContext>
2 #include <QApplication>
3 #include <QmlApplicationEngine>
4 #include <QDebug>
5 #include <QtGlobal>
6 #include "src/Backend.h"
7 #include "src/SensorManager.h"
8 #include "src/Calculator.h"
9 #include "src/Database.h"
10 #include "src/ReportManager.h"
11 #include "src/WifiManager.h"
12 #include "src/VoiceCommandManager.h"
13
14 int main(int argc, char *argv[])
15 {
16     qputenv("QT_QUICK_CONTROLS_STYLE", "Basic");
17
18     QApplication app(argc, argv);
19     QApplication::setApplicationName("SLIPER");
20
21     Backend backend;
22     SensorManager sensorManager;
23     Calculator calculator;
24     Database database;
25     ReportManager reportManager;
26     WifiManager wifiManager(&sensorManager, &database);
27     VoiceCommandManager voiceCommandManager;
28
29     QmlApplicationEngine engine;
30
31     engine.rootContext()->setContextProperty("backend", &backend);
32     engine.rootContext()->setContextProperty("sensorManager", &sensorManager);
33     engine.rootContext()->setContextProperty("calculator", &calculator);
34     engine.rootContext()->setContextProperty("database", &database);
35     engine.rootContext()->setContextProperty("reportManager", &reportManager);
36     engine.rootContext()->setContextProperty("wifiManager", &wifiManager);
37     engine.rootContext()->setContextProperty("voiceCommandManager", &voiceCommandManager);
38
39     QObject::connect(
40         &engine,
41         &QmlApplicationEngine::objectCreationFailed,
42         &app,
43         []() {
44             qWarning() << "QML nesnesi olusturulamadi!";
45             QCoreApplication::exit(-1);
46         },
47         Qt::QueuedConnection);
48
49     engine.loadFromModule("sliper", "Main");
50
51     return app.exec();
52 }
```

```
1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Window 6.7
4 import QtQuick.Layouts 6.7
5 import "pages"
6
7 ApplicationWindow {
8     id: pencere
9     visibility: Window.Maximized
10    title: "SLIPER Analiz"
11
12    background: Rectangle {
13        color: "#060d17"
14    }
15
16    StackLayout {
17        id: anaStack
18        anchors.fill: parent
19        currentIndex: 0
20
21        LoginPage {
22            onGirisBasarili: anaStack.currentIndex = 1
23        }
24
25        DashboardPage {
26            onCikisYapildi: anaStack.currentIndex = 0
27        }
28    }
29 }
```

```
1 pragma Singleton
2 import QtQuick
3
4 QObject {
5     // true = Türkçe, false = English
6     property bool turkish: true
7
8     function toggle() {
9         turkish = !turkish
10    }
11 }
12
```

```

1  import QtQuick 6.7
2  import sliper
3
4  Item {
5      id: root
6      property real egimX: 0
7      property real egimY: 0
8      property real maxAci: 15
9
10     readonly property var metinler: ({
11         duzMetni: { tr: "Düz", en: "Level" },
12         ayarlaniyorMetni: { tr: "Ayarlanıyor...", en: "Adjusting..." }
13     })
14
15     function txt(anahtar) {
16         return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
17     }
18
19     width: 200
20     height: 200
21
22     readonly property real yaricap: (width / 2) - 20
23     readonly property bool duz: Math.abs(egimX) < 0.5 && Math.abs(egimY) < 0.5
24
25     readonly property real kabarcikOffsetX: Math.max(-1, Math.min(1, egimY / maxAci)) * yaricap
26     readonly property real kabarcikOffsetY: Math.max(-1, Math.min(1, egimX / maxAci)) * yaricap
27
28     Rectangle {
29         anchors.fill: parent
30         radius: width / 2
31         color: "#0f1420"
32         border.color: "#1e2a3f"
33         border.width: 2
34     }
35
36     Rectangle {
37         anchors.centerIn: parent
38         width: root.yaricap * 1.2
39         height: width
40         radius: width / 2
41         color: "transparent"
42         border.color: "#1e2a3f"
43         border.width: 1
44     }
45
46     Rectangle {
47         anchors.centerIn: parent
48         width: parent.width - 20
49         height: 1
50         color: "#1e2a3f"
51     }
52
53     Rectangle {
54         anchors.centerIn: parent
55         width: 1
56         height: parent.height - 20
57         color: "#1e2a3f"
58     }
59
60     Rectangle {
61         id: kabarcik
62         width: 28
63         height: 28
64         radius: 14
65         color: root.duz ? "#16a34a" : "#dc2626"
66         border.color: root.duz ? "#4ade80" : "#f87171"
67         border.width: 2
68
69         x: (root.width / 2) - (width / 2) + root.kabarcikOffsetX
70         y: (root.height / 2) - (height / 2) + root.kabarcikOffsetY
71
72         Behavior on x { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }
73         Behavior on y { NumberAnimation { duration: 150; easing.type: Easing.OutQuad } }
74         Behavior on color { ColorAnimation { duration: 200 } }
75         Behavior on border.color { ColorAnimation { duration: 200 } }
76     }
77

```

```
78     Text {
79         anchors.top: parent.bottom
80         anchors.horizontalCenter: parent.horizontalCenter
81         anchors.topMargin: 8
82         text: root.duz ? root.txt("duzMetni") : root.txt("ayarlaniyorMetni")
83         color: root.duz ? "#4ade80" : "#6b7280"
84         font.family: "Segoe UI"
85         font.pixelSize: 12
86         font.bold: root.duz
87     }
88 }
```



```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import sliper
4
5 Item {
6     id: root
7     width: 130
8     height: 36
9
10    readonly property var metinler: ({
11        placeholderMetni: { tr: "gg.aa.yyyy", en: "dd.mm.yyyy" },
12        temizleButon: { tr: "Temizle", en: "Clear" }
13    })
14
15    function txt(anahtar) {
16        return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
17    }
18
19    // Kontrollü bileşen: seçili tarih dışarıdan (ör. gecmisSayfasi.baslangicTarihi)
20    // bağlanır, kullanıcı bir gün seçtiğinde sadece tarihSecildi sinyali yayılır –
21    // root.secilenTarih burada asla doğrudan atanmaz ki dışarıdaki binding kopmasın.
22    property string secilenTarih: ""
23    property string placeholderText: txt("placeholderMetni")
24    property date gorunenAy: new Date()
25
26    readonly property int hucreBoyutu: 28
27    readonly property int hucreAraligi: 4
28    readonly property int gridGenisligi: 7 * hucreBoyutu + 6 * hucreAraligi
29
30    signal tarihSecildi(string tarihStr)
31
32    function ikiHane(n) {
33        return n < 10 ? "0" + n : "" + n
34    }
35
36    function tarihStr(d) {
37        return ikiHane(d.getDate()) + "." + ikiHane(d.getMonth() + 1) + "." + d.getFullYear()
38    }
39
40    function gunSayisi(yil, ay) {
41        return new Date(yil, ay + 1, 0).getDate()
42    }
43
44    // Takvim ızgarasını (önceki/sonraki aydan taşan günler dahil, Pazartesi başlangıçlı,
45    // her zaman 6 hafta = 42 hücre) saf QtQuick ile hesaplar; Qt.labs.calendar modülüne
46    // bağımlı değildir çünkü bu Qt kurulumunda mevcut değil.
47    function ayGunleriniHesapla(ay) {
48        var yil = ay.getFullYear()
49        var ayIndeks = ay.getMonth()
50        var ilkGun = new Date(yil, ayIndeks, 1)
51        var haftaBaslangicKaymasi = (ilkGun.getDay() + 6) % 7 // Pazartesi = 0
52        var baslangic = new Date(yil, ayIndeks, 1 - haftaBaslangicKaymasi)
53
54        var gunler = []
55        for (var i = 0; i < 42; i++) {
56            gunler.push(new Date(baslangic.getFullYear(), baslangic.getMonth(), baslangic.getDate() + i))
57        }
58        return gunler
59    }
60
61    property var gunHucreleri: ayGunleriniHesapla(gorunenAy)
62
63    readonly property var ayAdlari: [
64        { tr: "Ocak", en: "January" }, { tr: "Şubat", en: "February" }, { tr: "Mart", en: "March" },
65        { tr: "Nisan", en: "April" }, { tr: "Mayıs", en: "May" }, { tr: "Haziran", en: "June" },
66        { tr: "Temmuz", en: "July" }, { tr: "Ağustos", en: "August" }, { tr: "Eylül", en: "September" },
67        { tr: "Ekim", en: "October" }, { tr: "Kasım", en: "November" }, { tr: "Aralık", en: "December" }
68    ]
69    readonly property var haftaGunleri: [
70        { tr: "Pt", en: "Mo" }, { tr: "Sa", en: "Tu" }, { tr: "Ça", en: "We" },
71        { tr: "Pe", en: "Th" }, { tr: "Cu", en: "Fr" }, { tr: "Ct", en: "Sa" }, { tr: "Pz", en: "Su" }
72    ]
73
74    Rectangle {
75        id: kutu
76        anchors.fill: parent
77        radius: 8

```

```

78     color: "#0f1420"
79     border.color: (alan.containsMouse || takvimPopup.visible) ? "#3b82f6" : "#1e2a3f"
80     border.width: 1
81     Behavior on border.color { ColorAnimation { duration: 120 } }
82
83     Text {
84         anchors.left: parent.left
85         anchors.right: simgeMetni.left
86         anchors.leftMargin: 10
87         anchors.verticalCenter: parent.verticalCenter
88         elide: Text.ElideRight
89         text: root.secilenTarih.length > 0 ? root.secilenTarih : root.placeholderText
90         color: root.secilenTarih.length > 0 ? "#dce8f5" : "#4b5563"
91         font.family: "Segoe UI"
92         font.pixelSize: 13
93     }
94
95     Text {
96         id: simgeMetni
97         anchors.right: parent.right
98         anchors.rightMargin: 10
99         anchors.verticalCenter: parent.verticalCenter
100        text: "📅"
101        font.pixelSize: 12
102        color: "#6b7280"
103    }
104
105    MouseArea {
106        id: alan
107        anchors.fill: parent
108        hoverEnabled: true
109        cursorShape: Qt.PointingHandCursor
110        onClicked: {
111            if (root.secilenTarih.length === 10) {
112                var parcalar = root.secilenTarih.split(".")
113                root.gorunenAy = new Date(parseInt(parcalar[2], 10), parseInt(parcalar[1], 10) - 1, 1)
114            } else {
115                root.gorunenAy = new Date()
116            }
117            takvimPopup.open()
118        }
119    }
120 }
121
122 Popup {
123     id: takvimPopup
124     y: kutu.height + 6
125     width: root.gridGenisligi + 28
126     padding: 14
127     modal: true
128     focus: true
129     closePolicy: Popup.CloseOnEscape | Popup.CloseOnPressOutside
130
131     background: Rectangle {
132         color: "#0f1420"
133         radius: 12
134         border.color: "#1e2a3f"
135         border.width: 1
136     }
137
138     Overlay.modal: Rectangle {
139         color: "#a6000000"
140     }
141
142     contentItem: Column {
143         spacing: 10
144
145         Row {
146             width: root.gridGenisligi
147             height: 28
148
149             Rectangle {
150                 width: 28
151                 height: 28
152                 radius: 6
153                 anchors.verticalCenter: parent.verticalCenter
154                 color: oncekiAlan.containsMouse ? "#1e2a3f" : "transparent"
155                 Behavior on color { ColorAnimation { duration: 120 } }
156
157                 Text {

```

```

158         anchors.centerIn: parent
159         text: "<"
160         color: "#9ca3af"
161         font.pixelSize: 16
162     }
163
164     MouseArea {
165         id: oncekiAlan
166         anchors.fill: parent
167         hoverEnabled: true
168         cursorShape: Qt.PointingHandCursor
169         onClicked: root.gorunenAy = new Date(root.gorunenAy.getFullYear(), root.gorunenAy.getMonth() - 1, 1)
170     }
171 }
172
173 Text {
174     width: parent.width - 56
175     anchors.verticalCenter: parent.verticalCenter
176     horizontalAlignment: Text.AlignHCenter
177     text: (Translations.turkish ? root.ayAdlari[root.gorunenAy.getMonth()].tr :
root.ayAdlari[root.gorunenAy.getMonth()].en) + " " + root.gorunenAy.getFullYear()
178     color: "#dce8f5"
179     font.family: "Segoe UI"
180     font.pixelSize: 13
181     font.bold: true
182 }
183
184 Rectangle {
185     width: 28
186     height: 28
187     radius: 6
188     anchors.verticalCenter: parent.verticalCenter
189     color: sonrakiAlan.containsMouse ? "#1e2a3f" : "transparent"
190     Behavior on color { ColorAnimation { duration: 120 } }
191
192     Text {
193         anchors.centerIn: parent
194         text: ">"
195         color: "#9ca3af"
196         font.pixelSize: 16
197     }
198
199     MouseArea {
200         id: sonrakiAlan
201         anchors.fill: parent
202         hoverEnabled: true
203         cursorShape: Qt.PointingHandCursor
204         onClicked: root.gorunenAy = new Date(root.gorunenAy.getFullYear(), root.gorunenAy.getMonth() + 1, 1)
205     }
206 }
207
208
209 Row {
210     width: root.gridGenisligi
211     spacing: root.hucreAraligi
212
213     Repeater {
214         model: root.haftaGunleri
215
216         delegate: Text {
217             width: root.hucreBoyutu
218             horizontalAlignment: Text.AlignHCenter
219             text: Translations.turkish ? modelData.tr : modelData.en
220             color: "#6b7280"
221             font.family: "Segoe UI"
222             font.pixelSize: 10
223             font.bold: true
224         }
225     }
226 }
227
228 Grid {
229     width: root.gridGenisligi
230     columns: 7
231     rowSpacing: root.hucreAraligi
232     columnSpacing: root.hucreAraligi
233
234     Repeater {
235         model: root.gunHucreleri

```

```

237     delegate: Rectangle {
238         id: gunHucreresi
239         width: root.hucreBoyutu
240         height: root.hucreBoyutu
241         radius: root.hucreBoyutu / 2
242         property bool buAyMi: modelData.getMonth() === root.gorunenAy.getMonth()
243         property bool seciliMi: root.secilenTarih === root.tarihStr(modelData)
244         property bool bugunMu: root.tarihStr(modelData) === root.tarihStr(new Date())
245         color: seciliMi ? "#3b82f6" : (gunAlani.containsMouse ? "#1e2a3f" : "transparent")
246         border.color: (bugunMu && !seciliMi) ? "#3b82f6" : "transparent"
247         border.width: 1
248
249         Text {
250             anchors.centerIn: parent
251             text: modelData.getDate()
252             color: gunHucreresi.buAyMi ? (gunHucreresi.seciliMi ? "ffffff" : "#dce8f5") : "#374151"
253             font.family: "Segoe UI"
254             font.pixelSize: 12
255             font.bold: gunHucreresi.seciliMi
256         }
257
258         MouseArea {
259             id: gunAlani
260             anchors.fill: parent
261             hoverEnabled: true
262             cursorShape: Qt.PointingHandCursor
263             onClicked: {
264                 root.tarihSecildi(root.tarihStr(modelData))
265                 takvimPopup.close()
266             }
267         }
268     }
269 }
270
271 Rectangle {
272     width: root.gridGenisligi
273     height: 1
274     color: "#1e2a3f"
275 }
276
277 Rectangle {
278     width: root.gridGenisligi
279     height: 30
280     radius: 6
281     color: temizleAlani.containsMouse ? "#1e2a3f" : "transparent"
282     Behavior on color { ColorAnimation { duration: 120 } }
283
284     Text {
285         anchors.centerIn: parent
286         text: root.txt("temizleButon")
287         color: "#9ca3af"
288         font.family: "Segoe UI"
289         font.pixelSize: 12
290     }
291
292     MouseArea {
293         id: temizleAlani
294         anchors.fill: parent
295         hoverEnabled: true
296         cursorShape: Qt.PointingHandCursor
297         onClicked: {
298             root.tarihSecildi("")
299             takvimPopup.close()
300         }
301     }
302 }
303
304 }
305
306 }
307

```

```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Layouts 6.7
4 import QtMultimedia
5 import sliper
6
7 Rectangle {
8     color: "#0a0e17"
9
10    signal cikisYapildi()
11
12    readonly property var navOgeleri: [
13        { etiketTr: "Ölçüm", etiketEn: "Measurement", ikon: "📏" },
14        { etiketTr: "Sonuçlar", etiketEn: "Results", ikon: "📊" },
15        { etiketTr: "Geçmiş", etiketEn: "History", ikon: "🕒" },
16        { etiketTr: "Kalibrasyon", etiketEn: "Calibration", ikon: "⚙️" }
17    ]
18
19    readonly property var metinler: ({
20        baglandi: { tr: "SLIPER-ESP32 Bağlı", en: "SLIPER-ESP32 Connected" },
21        baglaniyor: { tr: "Bağlanıyor...", en: "Connecting..." },
22        bagliDegil: { tr: "Bağlı Değil", en: "Not Connected" },
23        batarya: { tr: "🔋 Batarya: ", en: "🔋 Battery: " },
24        sesliKomut: { tr: "🗣 Sesli Komut: ", en: "🗣 Voice Command: " },
25        yukleniyor: { tr: "Yükleniyor...", en: "Loading..." },
26        acik: { tr: "Açık", en: "On" },
27        kapali: { tr: "Kapalı", en: "Off" },
28        cikisYap: { tr: "Çıkış Yap", en: "Log Out" }
29    })
30
31    function txt(anahtar) {
32        return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
33    }
34
35    Row {
36        anchors.fill: parent
37        spacing: 0
38
39        Rectangle {
40            id: sidebar
41            width: 230
42            height: parent.height
43
44            gradient: Gradient {
45                GradientStop { position: 0.0; color: "#121b30" }
46                GradientStop { position: 1.0; color: "#080b12" }
47            }
48
49            Image {
50                anchors.top: parent.top
51                anchors.horizontalCenter: parent.horizontalCenter
52                anchors.topMargin: 24
53                source: "../assets/logo.png"
54                width: 144
55                height: 144
56                fillMode: Image.PreserveAspectFit
57            }
58
59            Rectangle {
60                anchors.top: parent.top
61                anchors.left: parent.left
62                anchors.right: parent.right
63                anchors.leftMargin: 24
64                anchors.rightMargin: 24
65                anchors.topMargin: 172
66                height: 1
67                color: "#1f2c46"
68            }
69
70            Row {
71                anchors.top: parent.top
72                anchors.topMargin: 184
73                anchors.horizontalCenter: parent.horizontalCenter
74                spacing: 6
75
76                Rectangle {
77                    width: 32

```

```

78         height: 22
79         radius: 5
80         color: "#0f1420"
81         border.color: Translations.turkish ? "#3b82f6" : "#1e2a3f"
82         border.width: 1
83
84         Text {
85             anchors.centerIn: parent
86             text: "TR"
87             color: Translations.turkish ? "ffffff" : "#6b7280"
88             font.pixelSize: 10
89             font.bold: Translations.turkish
90         }
91
92         MouseArea {
93             anchors.fill: parent
94             cursorShape: Qt.PointingHandCursor
95             onClicked: Translations.turkish = true
96         }
97     }
98
99     Rectangle {
100         width: 32
101         height: 22
102         radius: 5
103         color: "#0f1420"
104         border.color: !Translations.turkish ? "#3b82f6" : "#1e2a3f"
105         border.width: 1
106
107         Text {
108             anchors.centerIn: parent
109             text: "EN"
110             color: !Translations.turkish ? "ffffff" : "#6b7280"
111             font.pixelSize: 10
112             font.bold: !Translations.turkish
113         }
114
115         MouseArea {
116             anchors.fill: parent
117             cursorShape: Qt.PointingHandCursor
118             onClicked: Translations.turkish = false
119         }
120     }
121
122     Rectangle {
123         id: cikisButonu
124         width: cikisMetni.implicitWidth + 16
125         height: 22
126         radius: 5
127         color: cikisAlani.pressed ? "#4c1414" : (cikisAlani.containsMouse ? "#3a1414" : "#1f1414")
128         border.color: "#7f1d1d"
129         border.width: 1
130         Behavior on color { ColorAnimation { duration: 120 } }
131
132         Text {
133             id: cikisMetni
134             anchors.centerIn: parent
135             text: txt("cikisYap")
136             color: "#f87171"
137             font.family: "Segoe UI"
138             font.pixelSize: 10
139             font.bold: true
140         }
141
142         MouseArea {
143             id: cikisAlani
144             anchors.fill: parent
145             hoverEnabled: true
146             cursorShape: Qt.PointingHandCursor
147             onClicked: cikisYapildi()
148         }
149     }
150 }
151
152 Rectangle {
153     id: baglantiDurumu
154     anchors.top: parent.top
155     anchors.left: parent.left
156     anchors.right: parent.right
157     anchors.leftMargin: 24

```

```

158     anchors.rightMargin: 24
159     anchors.topMargin: 224
160     height: 36
161     radius: 6
162     color: baglantiAlani.hoverli ? "#101a2c" : "transparent"
163     Behavior on color { ColorAnimation { duration: 120 } }
164
165     Row {
166         anchors.left: parent.left
167         anchors.leftMargin: 8
168         anchors.verticalCenter: parent.verticalCenter
169         spacing: 8
170
171         Rectangle {
172             width: 8
173             height: 8
174             radius: 4
175             anchors.verticalCenter: parent.verticalCenter
176             color: wifiManager.baglandi ? "#16a34a" : (wifiManager.baglaniyor ? "#f59e0b" : "#dc2626")
177         }
178
179         Text {
180             text: wifiManager.baglandi ? txt("baglandi") : (wifiManager.baglaniyor ? txt("baglaniyor") : txt("bagliDegil"))
181             color: "#9ca3af"
182             font.family: "Segoe UI"
183             font.pixelSize: 12
184             anchors.verticalCenter: parent.verticalCenter
185         }
186     }
187
188     MouseArea {
189         id: baglantiAlani
190         anchors.fill: parent
191         hoverEnabled: true
192         cursorShape: Qt.PointingHandCursor
193         enabled: !wifiManager.baglandi && !wifiManager.baglaniyor
194         onClicked: wifiManager.baglan()
195     }
196 }
197
198 Rectangle {
199     id: bataryaGostergesi
200     anchors.top: parent.top
201     anchors.left: parent.left
202     anchors.right: parent.right
203     anchors.leftMargin: 24
204     anchors.rightMargin: 24
205     anchors.topMargin: 236
206     height: 22
207     radius: 4
208     visible: wifiManager.baglandi && sensorManager.bataryaVoltaj > 0.1
209     color: {
210         if (sensorManager.bataryaVoltaj >= 14.5) return "#0f2417"
211         if (sensorManager.bataryaVoltaj >= 14.0) return "#2a2414"
212         return "#2a1414"
213     }
214
215     Text {
216         anchors.left: parent.left
217         anchors.leftMargin: 8
218         anchors.verticalCenter: parent.verticalCenter
219         text: txt("batarya") + sensorManager.bataryaVoltaj.toFixed(2) + " V"
220         color: {
221             if (sensorManager.bataryaVoltaj >= 14.5) return "#4ade80"
222             if (sensorManager.bataryaVoltaj >= 14.0) return "#e8a020"
223             return "#f87171"
224         }
225         font.family: "Segoe UI"
226         font.pixelSize: 11
227         font.bold: true
228     }
229 }
230
231 Rectangle {
232     id: sesKontrolu
233     anchors.top: parent.top
234     anchors.left: parent.left
235     anchors.right: parent.right
236     anchors.leftMargin: 24
237     anchors.rightMargin: 24

```

```

238         anchors.topMargin: 264
239         height: 30
240         radius: 6
241         color: sesKontroluAlani.hoverli ? "#101a2c" : "transparent"
242         Behavior on color { ColorAnimation { duration: 120 } }
243
244         Row {
245             anchors.left: parent.left
246             anchors.leftMargin: 8
247             anchors.verticalCenter: parent.verticalCenter
248             spacing: 8
249
250             Rectangle {
251                 width: 8
252                 height: 8
253                 radius: 4
254                 anchors.verticalCenter: parent.verticalCenter
255                 color: !voiceCommandManager.modelHazir ? "#6b7280" : (voiceCommandManager.etkin ? "#16a34a" : "#dc2626")
256             }
257
258             Text {
259                 text: txt("sesliKomut") + (!voiceCommandManager.modelHazir ? txt("yukleniyor") : (voiceCommandManager.etkin ?
txt("acik") : txt("kapali")))
260                 color: "#9ca3af"
261                 font.family: "Segoe UI"
262                 font.pixelSize: 12
263                 anchors.verticalCenter: parent.verticalCenter
264             }
265         }
266
267         MouseArea {
268             id: sesKontroluAlani
269             property bool hoverli: false
270             anchors.fill: parent
271             hoverEnabled: true
272             cursorShape: Qt.PointingHandCursor
273             enabled: voiceCommandManager.modelHazir
274             onEntered: hoverli = true
275             onExited: hoverli = false
276             onClicked: voiceCommandManager.etkin = !voiceCommandManager.etkin
277         }
278     }
279
280     Rectangle {
281         anchors.top: parent.top
282         anchors.left: parent.left
283         anchors.right: parent.right
284         anchors.leftMargin: 24
285         anchors.rightMargin: 24
286         anchors.topMargin: 302
287         height: 1
288         color: "#1f2c46"
289     }
290
291     Column {
292         id: navColumn
293         anchors.top: parent.top
294         anchors.topMargin: 320
295         width: parent.width
296         spacing: 2
297
298         Repeater {
299             model: navOgeleri
300
301             Rectangle {
302                 id: navOgesi
303                 width: parent.width
304                 height: 46
305                 property bool aktif: icerikStack.currentIndex === index
306                 property bool hoverli: false
307                 color: aktif ? "#182644" : (hoverli ? "#101a2c" : "transparent")
308                 Behavior on color { ColorAnimation { duration: 120 } }
309
310                 Rectangle {
311                     width: 3
312                     height: parent.height
313                     color: "#3b82f6"
314                     visible: navOgesi.aktif
315                 }
316

```



```

317         Row {
318             anchors.left: parent.left
319             anchors.leftMargin: 24
320             anchors.verticalCenter: parent.verticalCenter
321             spacing: 12
322
323             Text {
324                 text: modelData.ikon
325                 font.pixelSize: 15
326                 anchors.verticalCenter: parent.verticalCenter
327             }
328
329             Text {
330                 text: Translations.turkish ? modelData.etiketTr : modelData.etiketEn
331                 color: navOgesi.aktif ? "#6ea8ff" : (navOgesi.hoverli ? "#dce8f5" : "#9ca3af")
332                 font.family: "Segoe UI"
333                 font.pixelSize: 14
334                 font.bold: navOgesi.aktif
335                 anchors.verticalCenter: parent.verticalCenter
336                 Behavior on color { ColorAnimation { duration: 120 } }
337             }
338         }
339
340         MouseArea {
341             anchors.fill: parent
342             hoverEnabled: true
343             cursorShape: Qt.PointingHandCursor
344             onEntered: navOgesi.hoverli = true
345             onExited: navOgesi.hoverli = false
346             onClicked: icerikStack.currentIndex = index
347         }
348     }
349 }
350
351 MediaPlayer {
352     id: simulasyonOynatici
353     source: "../assets/slipergif.mp4"
354     videoOutput: simulasyonVideo
355     loops: 1
356 }
357
358 Item {
359     id: simulasyonAlani
360     anchors.top: navColumn.bottom
361     anchors.left: parent.left
362     anchors.right: parent.right
363     anchors.bottom: parent.bottom
364     anchors.margins: 16
365     anchors.topMargin: 20
366     clip: true
367
368     VideoOutput {
369         id: simulasyonVideo
370         anchors.fill: parent
371     }
372 }
373
374 }
375
376 Rectangle {
377     id: sidebarAyraci
378     width: 3
379     height: parent.height
380     color: "#05070b"
381
382     Rectangle {
383         anchors.right: parent.right
384         anchors.top: parent.top
385         anchors.bottom: parent.bottom
386         width: 1
387         color: "#403b82f6"
388     }
389 }
390
391 StackLayout {
392     id: icerikStack
393     width: parent.width - sidebar.width - sidebarAyraci.width
394     height: parent.height
395     currentIndex: 0
396

```

```
397         OlcumPage {
398             onOlcumTamamlandi: function(id) {
399                 sonuclarSayfasi.olcumId = id
400                 icerikStack.currentIndex = 1
401             }
402             onAgirlikEklendi: {
403                 simulasyonOynatici.stop()
404                 simulasyonOynatici.play()
405             }
406         }
407
408         SonuclarPage {
409             id: sonuclarSayfasi
410         }
411
412         GecmisPage {
413             onTestSecildi: function(id) {
414                 sonuclarSayfasi.olcumId = id
415                 icerikStack.currentIndex = 1
416             }
417         }
418
419         KalibrasyonPage {}
420     }
421 }
422 }
```

```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import "../components"
4 import sliper
5
6 Rectangle {
7     id: gecmisSayfasi
8     color: "#0a0e17"
9
10    signal testSecildi(int id)
11
12    property var olcumListesi: []
13    property string baslangicTarihi: ""
14    property string bitisTarihi: ""
15
16    readonly property var filtreAnahtarları: ["tumTestler", "son7Gun", "son30Gun"]
17
18    readonly property var metinler: ({
19        aramaPlaceholder: { tr: "Müşteri veya reçete ara...", en: "Search by customer or recipe..." },
20        tumTestler: { tr: "Tüm Testler", en: "All Tests" },
21        son7Gun: { tr: "Son 7 Gün", en: "Last 7 Days" },
22        son30Gun: { tr: "Son 30 Gün", en: "Last 30 Days" },
23        ozelTarih: { tr: "Özel Tarih:", en: "Custom Date:" },
24        temizle: { tr: "Temizle", en: "Clear" },
25        musteriEtiket: { tr: "Müşteri: ", en: "Customer: " },
26        receteEtiket: { tr: " • Reçete: ", en: " • Recipe: " },
27        goruntule: { tr: "Görüntüle", en: "View" },
28        kayitSilBaslik: { tr: "Kaydı Sil", en: "Delete Record" },
29        kayitSilOnSoru: { tr: "", en: "Are you sure you want to delete the measurement for '" },
30        kayitSilSonSoru: { tr: "' ölçümünü silmek istediğinize emin misiniz? Bu işlem geri alınamaz.", en: "'? This action cannot be
undone." },
31        sifreEtiket: { tr: "ŞİFRE", en: "PASSWORD" },
32        sifrenizGirin: { tr: "Şifrenizi girin", en: "Enter your password" },
33        sifreHatali: { tr: "Şifre hatalı, tekrar deneyin.", en: "Incorrect password, please try again." },
34        sil: { tr: "Sil", en: "Delete" },
35        iptal: { tr: "İptal", en: "Cancel" },
36        gelismisAyarlarBaslik: { tr: "Gelişmiş Ayarlar", en: "Advanced Settings" },
37        onayla: { tr: "Onayla", en: "Confirm" },
38        veriTabaniDisaAktar: { tr: "Veritabanını Dışa Aktar", en: "Export Database" },
39        disaAktarildi: { tr: "✓ Dışa aktarıldı: ", en: "✓ Exported: " },
40        disaAktarilamadi: { tr: "× Dışa aktarılamadı", en: "× Export failed" },
41        icaAktarYoluEtiket: { tr: "İÇE AKTARILACAK .DB DOSYA YOLU", en: ".DB FILE PATH TO IMPORT" },
42        icaAktarPlaceholder: { tr: "ör. C:/Users/.../SliperRaporlari/sliper_yedek_....db", en: "e.g.
C:/Users/.../SliperRaporlari/sliper_yedek_....db" },
43        veriTabaniIcaAktar: { tr: "Veritabanını İçe Aktar", en: "Import Database" },
44        icaAktarildi: { tr: "✓ İçe aktarıldı, uygulamayı yeniden başlatmanız önerilir.", en: "✓ Imported, it is recommended to restart
the application." },
45        icaAktarilamadi: { tr: "× İçe aktarılamadı, dosya yolunu kontrol edin.", en: "× Import failed, please check the file path." },
46        tumVeriyiSil: { tr: "Tüm Verileri Sil", en: "Delete All Data" },
47        kapat: { tr: "Kapat", en: "Close" },
48        tumVeriyiSilMesaj: { tr: "Tüm ölçüm ve stroke verileri kalıcı olarak silinecek. Sensör kalibrasyonları etkilenmez. Bu işlem GERİ
ALINAMAZ, emin misiniz?", en: "All measurement and stroke data will be permanently deleted. Sensor calibrations will not be affected. This action
CANNOT BE UNDONE, are you sure?" },
49        evetHepsiniSil: { tr: "Evet, Hepsini Sil", en: "Yes, Delete All" },
50        tumVerilerSilindi: { tr: "✓ Tüm veriler silindi.", en: "✓ All data deleted." }
51    })
52
53    function txt(anahtar) {
54        return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
55    }
56
57    function listeyiYenile() {
58        olcumListesi = database.tumOlcumleriGetir()
59    }
60
61    onVisibleChanged: {
62        if (visible) {
63            listeyiYenile()
64        }
65    }
66
67    Component.onCompleted: listeyiYenile()
68
69    function tarihiAyristir(tarihStr) {
70        if (!tarihStr) return null
71        var tarihKismi = tarihStr.split(" ")[0].split(".")
72        if (tarihKismi.length != 3) return null

```

```

73     var gun = parseInt(tarihKismi[0], 10)
74     var ay = parseInt(tarihKismi[1], 10)
75     var yil = parseInt(tarihKismi[2], 10)
76     if (isNaN(gun) || isNaN(ay) || isNaN(yil)) return null
77     return new Date(yil, ay - 1, gun)
78 }
79
80 function gecerliOzelTarih(metin) {
81     if (!metin || metin.length !== 10) return null
82     return tarihiAyristir(metin)
83 }
84
85 function filtrelenmisListeyiHesapla() {
86     var sonuc = []
87     var aramaMetni = aramaKutusu.text.toLowerCase()
88     var ozelBaslangic = gecerliOzelTarih(gecmisSayfasi.baslangicTarihi)
89     var ozelBitis = gecerliOzelTarih(gecmisSayfasi.bitisTarihi)
90     var simdi = new Date()
91
92     for (var i = 0; i < olcumListesi.length; i++) {
93         var kayit = olcumListesi[i]
94         var tarihObj = tarihiAyristir(kayit.tarih)
95
96         if (aramaMetni.length > 0) {
97             var musterislesir = kayit.musteri.toLowerCase().indexOf(aramaMetni) !== -1
98             var receteslesir = kayit.recete.toLowerCase().indexOf(aramaMetni) !== -1
99             if (!musterislesir && !receteslesir) continue
100         }
101
102         if (ozelBaslangic || ozelBitis) {
103             if (ozelBaslangic && tarihObj && tarihObj < ozelBaslangic) continue
104             if (ozelBitis && tarihObj) {
105                 var bitisSonu = new Date(ozelBitis.getFullYear(), ozelBitis.getMonth(), ozelBitis.getDate(), 23, 59, 59)
106                 if (tarihObj > bitisSonu) continue
107             }
108         } else if (tarihObj) {
109             if (filtreKutusu.currentIndex === 1) {
110                 var esik7 = new Date(simdi.getTime() - 7 * 24 * 60 * 60 * 1000)
111                 if (tarihObj < esik7) continue
112             } else if (filtreKutusu.currentIndex === 2) {
113                 var esik30 = new Date(simdi.getTime() - 30 * 24 * 60 * 60 * 1000)
114                 if (tarihObj < esik30) continue
115             }
116         }
117
118         sonuc.push(kayit)
119     }
120
121     return sonuc
122 }
123
124 property var filtrelenmisListe: filtrelenmisListeyiHesapla()
125
126 Column {
127     anchors.fill: parent
128     anchors.margins: 20
129     spacing: 16
130
131     Row {
132         width: parent.width
133         spacing: 12
134
135         TextField {
136             id: aramaKutusu
137             width: parent.width - filtreKutusu.width - gelismisAyarlarButonu.width - parent.spacing * 2
138             height: 40
139             placeholderText: txt("aramaPlaceholder")
140             placeholderTextColor: "#4b5563"
141             color: "#ffffff"
142             font.pixelSize: 13
143             leftPadding: 34
144             verticalAlignment: TextInput.AlignVCenter
145             background: Rectangle {
146                 color: "#0f1420"
147                 radius: 8
148                 border.color: aramaKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
149                 border.width: 1
150             }
151
152             Text {

```

```

153         anchors.left: parent.left
154         anchors.leftMargin: 12
155         anchors.verticalCenter: parent.verticalCenter
156         text: "🔍"
157         color: "#4b5563"
158         font.pixelSize: 13
159     }
160 }
161
162 ComboBox {
163     id: filtreKutusu
164     width: 160
165     height: 40
166     model: filtreAnahtarlar.map(function(anahtar) { return txt(anahtar) })
167
168     background: Rectangle {
169         color: "#0f1420"
170         radius: 8
171         border.color: filtreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
172         border.width: 1
173     }
174
175     contentItem: Text {
176         text: filtreKutusu.displayText
177         color: "#9ca3af"
178         font.pixelSize: 13
179         leftPadding: 10
180         verticalAlignment: Text.AlignVCenter
181     }
182 }
183
184 Button {
185     id: gelismisAyarlarButonu
186     width: 44
187     height: 40
188     text: "⚙️"
189     font.pixelSize: 15
190
191     onClicked: {
192         gelismisAyarlarPopup.pinDogrulandi = false
193         pinKutusu.text = ""
194         gelismisAyarlarBildirim.text = ""
195         gelismisAyarlarPopup.open()
196     }
197
198     background: Rectangle {
199         radius: 8
200         color: parent.hovered ? "#162033" : "#0f1420"
201         border.color: "#1e2a3f"
202         border.width: 1
203         Behavior on color { ColorAnimation { duration: 120 } }
204     }
205
206     contentItem: Text {
207         text: parent.text
208         color: "#9ca3af"
209         font: parent.font
210         horizontalAlignment: Text.AlignHCenter
211         verticalAlignment: Text.AlignVCenter
212     }
213 }
214 }
215
216 Row {
217     width: parent.width
218     spacing: 10
219
220     Text {
221         text: txt("ozelTarih")
222         color: "#9ca3af"
223         font.family: "Segoe UI"
224         font.pixelSize: 12
225         anchors.verticalCenter: parent.verticalCenter
226     }
227
228     TarihSecici {
229         id: baslangicSecici
230         anchors.verticalCenter: parent.verticalCenter
231         secilenTarih: gecmisSayfasi.baslangicTarihi
232         onTarihSecildi: function(t) { gecmisSayfasi.baslangicTarihi = t }

```

```

233     }
234
235     Text {
236         text: "-"
237         color: "#4b5563"
238         font.pixelSize: 13
239         anchors.verticalCenter: parent.verticalCenter
240     }
241
242     TarihSecici {
243         id: bitisSecici
244         anchors.verticalCenter: parent.verticalCenter
245         secilenTarih: gecmisSayfasi.bitisTarihi
246         onTarihSecildi: function(t) { gecmisSayfasi.bitisTarihi = t }
247     }
248
249     Rectangle {
250         width: temizleMetni.implicitWidth + 20
251         height: 36
252         radius: 8
253         color: temizleMouse.containsMouse ? "#162033" : "#0f1420"
254         border.color: "#1e2a3f"
255         border.width: 1
256         visible: gecmisSayfasi.baslangicTarihi.length > 0 || gecmisSayfasi.bitisTarihi.length > 0
257         anchors.verticalCenter: parent.verticalCenter
258
259         Text {
260             id: temizleMetni
261             anchors.centerIn: parent
262             text: txt("temizle")
263             color: "#9ca3af"
264             font.family: "Segoe UI"
265             font.pixelSize: 12
266         }
267
268         MouseArea {
269             id: temizleMouse
270             anchors.fill: parent
271             hoverEnabled: true
272             cursorShape: Qt.PointingHandCursor
273             onClicked: {
274                 gecmisSayfasi.baslangicTarihi = ""
275                 gecmisSayfasi.bitisTarihi = ""
276             }
277         }
278     }
279 }
280
281 ListView {
282     width: parent.width
283     height: parent.height - 96
284     clip: true
285     spacing: 10
286
287     model: gecmisSayfasi.filtrelenmisListe
288
289     delegate: Rectangle {
290         id: gecmisKarti
291         width: parent.width
292         height: 76
293         radius: 10
294         color: "#0f1420"
295         border.color: kartAlani.containsMouse ? "#3b82f6" : "#1e2a3f"
296         border.width: 1
297         Behavior on border.color { ColorAnimation { duration: 120 } }
298
299         MouseArea {
300             id: kartAlani
301             anchors.fill: parent
302             hoverEnabled: true
303             acceptedButtons: Qt.NoButton
304         }
305
306         Rectangle {
307             id: rozet
308             width: 44
309             height: 44
310             radius: 10
311             anchors.left: parent.left
312             anchors.verticalCenter: parent.verticalCenter

```

```

313         anchors.leftMargin: 16
314         color: "#132335"
315
316         Text {
317             anchors.centerIn: parent
318             text: modelData.musteri.charAt(0).toUpperCase()
319             color: "#3b82f6"
320             font.family: "Segoe UI"
321             font.pixelSize: 18
322             font.bold: true
323         }
324     }
325
326     Column {
327         anchors.left: rozet.right
328         anchors.verticalCenter: parent.verticalCenter
329         anchors.leftMargin: 14
330         spacing: 4
331
332         Row {
333             spacing: 8
334
335             Text {
336                 text: modelData.tarih
337                 color: "#6b7280"
338                 font.family: "Segoe UI"
339                 font.pixelSize: 11
340             }
341
342             Rectangle {
343                 width: strokeMetni.implicitWidth + 12
344                 height: 16
345                 radius: 8
346                 color: "#2a2010"
347                 anchors.verticalCenter: parent.verticalCenter
348
349                 Text {
350                     id: strokeMetni
351                     anchors.centerIn: parent
352                     text: modelData.strokeSayisi + " stroke"
353                     color: "#e8a020"
354                     font.family: "Segoe UI"
355                     font.pixelSize: 10
356                     font.bold: true
357                 }
358             }
359         }
360
361         Text {
362             text: txt("musteriEtiket") + modelData.musteri + txt("receteEtiket") + modelData.recete
363             color: "#dce8f5"
364             font.family: "Segoe UI"
365             font.pixelSize: 13
366             font.bold: true
367         }
368     }
369
370     Row {
371         anchors.right: parent.right
372         anchors.verticalCenter: parent.verticalCenter
373         anchors.rightMargin: 16
374         spacing: 8
375
376         Rectangle {
377             id: silButonu
378             width: 34
379             height: 34
380             radius: 6
381             color: silMouse.containsMouse ? "#3f1d24" : "#1a1420"
382             border.color: silMouse.containsMouse ? "#ef4444" : "#2a2030"
383             border.width: 1
384             Behavior on color { ColorAnimation { duration: 120 } }
385
386             Text {
387                 anchors.centerIn: parent
388                 text: "■"
389                 font.pixelSize: 14
390             }
391
392             MouseArea {

```

```

393         id: silMouse
394         anchors.fill: parent
395         hovered: true
396         cursorShape: Qt.PointingHandCursor
397         onClicked: {
398             silOnayPopup.hedefId = modelData.id
399             silOnayPopup.hedefMusteri = modelData.musteri
400             silOnayPopup.open()
401         }
402     }
403 }
404
405 Button {
406     anchors.verticalCenter: parent.verticalCenter
407     height: 34
408     text: txt("goruntule")
409     font.pixelSize: 12
410
411     onClicked: testSecildi(modelData.id)
412
413     background: Rectangle {
414         radius: 6
415         color: parent.hovered ? "#4f8cf7" : "#3b82f6"
416         Behavior on color { ColorAnimation { duration: 120 } }
417     }
418
419     contentItem: Text {
420         text: parent.text
421         color: "ffffff"
422         font: parent.font
423         horizontalAlignment: Text.AlignHCenter
424         verticalAlignment: Text.AlignVCenter
425         leftPadding: 8
426         rightPadding: 8
427     }
428 }
429 }
430 }
431 }
432 }
433
434 Popup {
435     id: silOnayPopup
436     anchors.centerIn: parent
437     width: 380
438     padding: 24
439     modal: true
440     focus: true
441     closePolicy: Popup.CloseOnEscape
442
443     property int hedefId: -1
444     property string hedefMusteri: ""
445
446     onOpened: {
447         sifreGirisAlani.text = ""
448         hataMetni.visible = false
449         sifreGirisAlani.forceActiveFocus()
450     }
451
452     background: Rectangle {
453         color: "#0f1420"
454         radius: 14
455         border.color: "#1e2a3f"
456         border.width: 1
457     }
458
459     contentItem: Column {
460         width: silOnayPopup.availableWidth
461         spacing: 16
462
463         Text {
464             text: txt("kayitSilBaslik")
465             color: "ffffff"
466             font.family: "Segoe UI"
467             font.pixelSize: 17
468             font.bold: true
469         }
470
471         Text {
472             width: parent.width

```



```

473         wrapMode: Text.WordWrap
474         text: txt("kayitSilOnSoru") + silOnayPopup.hedefMusteri + txt("kayitSilSonSoru")
475         color: "#9ca3af"
476         font.family: "Segoe UI"
477         font.pixelSize: 13
478     }
479
480     Column {
481         width: parent.width
482         spacing: 6
483
484         Text {
485             text: txt("sifreEtiket")
486             color: "#6b7280"
487             font.family: "Segoe UI"
488             font.pixelSize: 10
489             font.letterSpacing: 1
490         }
491
492         TextField {
493             id: sifreGirisAlani
494             width: parent.width
495             height: 40
496             echoMode: TextInput.Password
497             placeholderText: txt("sifrenizGirin")
498             placeholderTextColor: "#4b5563"
499             color: "ffffff"
500             font.pixelSize: 13
501             leftPadding: 12
502             verticalAlignment: TextInput.AlignVCenter
503             background: Rectangle {
504                 color: "#0a0e17"
505                 radius: 6
506                 border.color: sifreGirisAlani.activeFocus ? "#3b82f6" : "#1e2a3f"
507                 border.width: 1
508             }
509             Keys.onReturnPressed: silOnaylaButonu.clicked()
510         }
511
512         Text {
513             id: hataMetni
514             text: txt("sifreHatali")
515             color: "#ef4444"
516             font.family: "Segoe UI"
517             font.pixelSize: 11
518             visible: false
519         }
520     }
521
522     Row {
523         width: parent.width
524         spacing: 10
525         layoutDirection: Qt.RightToLeft
526
527         Button {
528             id: silOnaylaButonu
529             width: (parent.width - 10) / 2
530             height: 40
531             text: txt("sil")
532             font.pixelSize: 13
533             font.bold: true
534
535             onClicked: {
536                 if (backend.sifreDogrula(sifreGirisAlani.text)) {
537                     database.olcumSil(silOnayPopup.hedefId)
538                     gecmisSayfasi.listeyiYenile()
539                     silOnayPopup.close()
540                 } else {
541                     hataMetni.visible = true
542                     sifreGirisAlani.text = ""
543                     sifreGirisAlani.forceActiveFocus()
544                 }
545             }
546
547             background: Rectangle {
548                 radius: 8
549                 color: parent.pressed ? "#b91c1c" : (parent.hovered ? "#ef4444" : "#dc2626")
550                 Behavior on color { ColorAnimation { duration: 120 } }
551             }
552

```

```

553         contentItem: Text {
554             text: parent.text
555             color: "#ffffff"
556             font: parent.font
557             horizontalAlignment: Text.AlignHCenter
558             verticalAlignment: Text.AlignVCenter
559         }
560     }
561
562     Button {
563         width: (parent.width - 10) / 2
564         height: 40
565         text: txt("iptal")
566         font.pixelSize: 13
567
568         onClicked: silOnayPopup.close()
569
570         background: Rectangle {
571             radius: 8
572             color: parent.hovered ? "#162033" : "#0f1420"
573             border.color: "#1e2a3f"
574             border.width: 1
575             Behavior on color { ColorAnimation { duration: 120 } }
576         }
577
578         contentItem: Text {
579             text: parent.text
580             color: "#9ca3af"
581             font: parent.font
582             horizontalAlignment: Text.AlignHCenter
583             verticalAlignment: Text.AlignVCenter
584         }
585     }
586 }
587 }
588 }
589
590 // Orijinal SLIPER'daki "Expert Settings > Database Export / Import / Delete"
591 // özelliklerine karşılık gelir (bkz. slipermanV13.pdf bölüm 6). Şifre
592 // doğrulaması, uygulamada zaten var olan giriş şifresini (backend.sifreDogrula)
593 // kullanır; orijinal cihazdaki ayrı "2603" servis şifresi yerine bu tercih
594 // edildi çünkü uygulamada zaten tek bir şifre akışı var.
595 Popup {
596     id: gelismisAyarlarPopup
597     anchors.centerIn: parent
598     width: 420
599     padding: 24
600     modal: true
601     focus: true
602     closePolicy: Popup.CloseOnEscape
603
604     property bool pinDogrulandi: false
605
606     background: Rectangle {
607         color: "#0f1420"
608         radius: 14
609         border.color: "#1e2a3f"
610         border.width: 1
611     }
612
613     contentItem: Column {
614         width: gelismisAyarlarPopup.availableWidth
615         spacing: 16
616
617         Text {
618             text: txt("gelismisAyarlarBaslik")
619             color: "#ffffff"
620             font.family: "Segoe UI"
621             font.pixelSize: 17
622             font.bold: true
623         }
624
625         Column {
626             width: parent.width
627             spacing: 6
628             visible: !gelismisAyarlarPopup.pinDogrulandi
629
630             Text {
631                 text: txt("sifreEtiket")
632                 color: "#6b7280"

```

```

633         font.family: "Segoe UI"
634         font.pixelSize: 10
635         font.letterSpacing: 1
636     }
637
638     TextField {
639         id: pinKutusu
640         width: parent.width
641         height: 40
642         echoMode: TextInput.Password
643         placeholderText: txt("sifrenizGirin")
644         placeholderTextColor: "#4b5563"
645         color: "ffffff"
646         font.pixelSize: 13
647         leftPadding: 12
648         verticalAlignment: TextInput.AlignVCenter
649         background: Rectangle {
650             color: "#0a0e17"
651             radius: 6
652             border.color: pinKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
653             border.width: 1
654         }
655         Keys.onReturnPressed: pinDogrulaButonu.clicked()
656     }
657
658     Text {
659         id: pinHataMetni
660         text: txt("sifreHatali")
661         color: "#ef4444"
662         font.family: "Segoe UI"
663         font.pixelSize: 11
664         visible: false
665     }
666
667     Button {
668         id: pinDogrulaButonu
669         width: parent.width
670         height: 40
671         text: txt("onayla")
672         font.pixelSize: 13
673         font.bold: true
674
675         onClicked: {
676             if (backend.sifreDogrula(pinKutusu.text)) {
677                 gelismisAyarlarPopup.pinDogrulandi = true
678                 pinHataMetni.visible = false
679             } else {
680                 pinHataMetni.visible = true
681                 pinKutusu.text = ""
682                 pinKutusu.forceActiveFocus()
683             }
684         }
685
686         background: Rectangle {
687             radius: 8
688             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
689             Behavior on color { ColorAnimation { duration: 120 } }
690         }
691
692         contentItem: Text {
693             text: parent.text
694             color: "ffffff"
695             font: parent.font
696             horizontalAlignment: Text.AlignHCenter
697             verticalAlignment: Text.AlignVCenter
698         }
699     }
700 }
701
702 Column {
703     width: parent.width
704     spacing: 10
705     visible: gelismisAyarlarPopup.pinDogrulandi
706
707     Button {
708         width: parent.width
709         height: 40
710         text: txt("veriTabaniDisaAktar")
711         font.pixelSize: 13

```

```

713         onClicked: {
714             var yol = database.veriTabaniDisaAktar()
715             gelismisAyarlarBildirim.text = yol.length > 0
716                 ? txt("disaAktarildi") + yol
717                 : txt("disaAktarilamadi")
718         }
719
720         background: Rectangle {
721             radius: 8
722             color: parent.hovered ? "#162033" : "#0f1420"
723             border.color: "#1e2a3f"
724             border.width: 1
725             Behavior on color { ColorAnimation { duration: 120 } }
726         }
727
728         contentItem: Text {
729             text: parent.text
730             color: "#dce8f5"
731             font: parent.font
732             horizontalAlignment: Text.AlignHCenter
733             verticalAlignment: Text.AlignVCenter
734         }
735     }
736
737     Column {
738         width: parent.width
739         spacing: 6
740
741         Text {
742             text: txt("icaAktarYoluEtiket")
743             color: "#6b7280"
744             font.family: "Segoe UI"
745             font.pixelSize: 10
746             font.letterSpacing: 1
747         }
748
749         TextField {
750             id: icaAktarYoluKutusu
751             width: parent.width
752             height: 40
753             placeholderText: txt("icaAktarPlaceholder")
754             placeholderTextColor: "#4b5563"
755             color: "ffffff"
756             font.pixelSize: 12
757             leftPadding: 12
758             verticalAlignment: TextInput.AlignVCenter
759             background: Rectangle {
760                 color: "#0a0e17"
761                 radius: 6
762                 border.color: icaAktarYoluKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
763                 border.width: 1
764             }
765         }
766     }
767
768     Button {
769         width: parent.width
770         height: 40
771         text: txt("veriTabaniIcaAktar")
772         font.pixelSize: 13
773
774         onClicked: {
775             var basarili = database.veriTabaniIcaAktar(icaAktarYoluKutusu.text)
776             gelismisAyarlarBildirim.text = basarili
777                 ? txt("icaAktarildi")
778                 : txt("icaAktarilamadi")
779             if (basarili) {
780                 gecmisSayfasi.listeyiYenile()
781             }
782         }
783
784         background: Rectangle {
785             radius: 8
786             color: parent.hovered ? "#162033" : "#0f1420"
787             border.color: "#1e2a3f"
788             border.width: 1
789             Behavior on color { ColorAnimation { duration: 120 } }
790         }
791
792         contentItem: Text {

```

```

793         text: parent.text
794         color: "#dce8f5"
795         font: parent.font
796         horizontalAlignment: Text.AlignHCenter
797         verticalAlignment: Text.AlignVCenter
798     }
799 }
800
801 Button {
802     width: parent.width
803     height: 40
804     text: txt("tumVeriyiSil")
805     font.pixelSize: 13
806     font.bold: true
807
808     onClicked: tumVeriyiSilOnayPopup.open()
809
810     background: Rectangle {
811         radius: 8
812         color: parent.pressed ? "#b91c1c" : (parent.hovered ? "#ef4444" : "#dc2626")
813         Behavior on color { ColorAnimation { duration: 120 } }
814     }
815
816     contentItem: Text {
817         text: parent.text
818         color: "#ffffff"
819         font: parent.font
820         horizontalAlignment: Text.AlignHCenter
821         verticalAlignment: Text.AlignVCenter
822     }
823 }
824
825 Text {
826     id: gelismisAyarlarBildirim
827     width: parent.width
828     text: ""
829     color: "#9ca3af"
830     font.family: "Segoe UI"
831     font.pixelSize: 10
832     wrapMode: Text.WrapAnywhere
833 }
834
835 Button {
836     width: parent.width
837     height: 36
838     text: txt("kapat")
839     font.pixelSize: 12
840
841     onClicked: gelismisAyarlarPopup.close()
842
843     background: Rectangle {
844         radius: 8
845         color: parent.hovered ? "#162033" : "transparent"
846         border.color: "#1e2a3f"
847         border.width: 1
848         Behavior on color { ColorAnimation { duration: 120 } }
849     }
850
851     contentItem: Text {
852         text: parent.text
853         color: "#9ca3af"
854         font: parent.font
855         horizontalAlignment: Text.AlignHCenter
856         verticalAlignment: Text.AlignVCenter
857     }
858 }
859 }
860 }
861 }
862
863 // "Tüm Verileri Sil" için çift onay (orijinal cihazdaki "Confirm if you
864 // are really sure" uyarısına karşılık gelir).
865 Popup {
866     id: tumVeriyiSilOnayPopup
867     anchors.centerIn: parent
868     width: 380
869     padding: 24
870     modal: true
871     focus: true
872     closePolicy: Popup.CloseOnEscape

```

```

873
874 background: Rectangle {
875     color: "#0f1420"
876     radius: 14
877     border.color: "#dc2626"
878     border.width: 1
879 }
880
881 contentItem: Column {
882     width: tumVeriyiSilOnayPopup.availableWidth
883     spacing: 16
884
885     Text {
886         text: txt("tumVeriyiSil")
887         color: "#ffffff"
888         font.family: "Segoe UI"
889         font.pixelSize: 17
890         font.bold: true
891     }
892
893     Text {
894         width: parent.width
895         wrapMode: Text.WordWrap
896         text: txt("tumVeriyiSilMesaj")
897         color: "#9ca3af"
898         font.family: "Segoe UI"
899         font.pixelSize: 13
900     }
901
902     Row {
903         width: parent.width
904         spacing: 10
905         layoutDirection: Qt.RightToLeft
906
907         Button {
908             width: (parent.width - 10) / 2
909             height: 40
910             text: txt("evetHepsiniSil")
911             font.pixelSize: 13
912             font.bold: true
913
914             onClicked: {
915                 database.tumVeriyiSil()
916                 gecmisSayfasi.listeyiYenile()
917                 tumVeriyiSilOnayPopup.close()
918                 gelismisAyarlarBildirim.text = txt("tumVerilerSilindi")
919             }
920
921             background: Rectangle {
922                 radius: 8
923                 color: parent.pressed ? "#b91c1c" : (parent.hovered ? "#ef4444" : "#dc2626")
924                 Behavior on color { ColorAnimation { duration: 120 } }
925             }
926
927             contentItem: Text {
928                 text: parent.text
929                 color: "#ffffff"
930                 font: parent.font
931                 horizontalAlignment: Text.AlignHCenter
932                 verticalAlignment: Text.AlignVCenter
933             }
934         }
935
936         Button {
937             width: (parent.width - 10) / 2
938             height: 40
939             text: txt("iptal")
940             font.pixelSize: 13
941
942             onClicked: tumVeriyiSilOnayPopup.close()
943
944             background: Rectangle {
945                 radius: 8
946                 color: parent.hovered ? "#162033" : "#0f1420"
947                 border.color: "#1e2a3f"
948                 border.width: 1
949                 Behavior on color { ColorAnimation { duration: 120 } }
950             }
951
952             contentItem: Text {

```

```
953         text: parent.text
954         color: "#9ca3af"
955         font: parent.font
956         horizontalAlignment: Text.AlignHCenter
957         verticalAlignment: Text.AlignVCenter
958     }
959 }
960 }
961 }
962 }
963 }
964
```

```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtQuick.Layouts 6.7
4 import "../components"
5 import sliper
6
7 Rectangle {
8     color: "#0a0e17"
9
10     readonly property var metinler: ({
11         baslikSensorSec: { tr: "KALİBRASYON – Sensör Seçin", en: "CALIBRATION – Select Sensor" },
12         egimSensoru: { tr: "Eğim Sensörü", en: "Incline Sensor" },
13         egimAciklama: { tr: "Yatay hizalama kalibrasyonu", en: "Level alignment calibration" },
14         kalibreEdildiOnEk: { tr: "✓ Kalibre Edildi: ", en: "✓ Calibrated: " },
15         loadCellAdi: { tr: "Load Cell", en: "Load Cell" },
16         loadCellAciklama: { tr: "Basınç sensörü kalibrasyonu", en: "Load pressure sensor calibration" },
17         mesafeSensoru: { tr: "Mesafe Sensörü", en: "Distance Sensor" },
18         mesafeAciklama: { tr: "Konum ölçümü kalibrasyonu", en: "Position measurement calibration" },
19         geriButon: { tr: "← Geri", en: "← Back" },
20         egimBaslik: { tr: "Eğim Sensörü Kalibrasyonu", en: "Incline Sensor Calibration" },
21         loadCellBaslik: { tr: "Load Cell Kalibrasyonu", en: "Load Cell Calibration" },
22         mesafeBaslik: { tr: "Mesafe Sensörü Kalibrasyonu", en: "Distance Sensor Calibration" },
23         sonKalibrasyonOnEk: { tr: "Son kalibrasyon: ", en: "Last calibration: " },
24         cihazVerlestir: { tr: "Cihazı düz ve sabit bir yüzeye yerleştirin", en: "Place the device on a flat, stable surface" },
25         cihazBagliDegilKalibrasyon: { tr: "Cihaz bağlı değil – kalibrasyon kaydedilemez", en: "Device not connected – calibration cannot
be saved" },
26         sifirlaButon: { tr: "Sıfırla (Zero)", en: "Zero (Reset)" },
27         kalibrasyonKaydedildiMesaj: { tr: "✓ Kalibrasyon kaydedildi", en: "✓ Calibration saved" },
28         kalibrasyonKaydedilemediMesaj: { tr: "✗ Kalibrasyon kaydedilemedi, tekrar deneyin", en: "✗ Calibration could not be saved, try
again" },
29         kayitliOnEk: { tr: "Kayıtlı: ", en: "Saved: " },
30         noktaSonEk: { tr: " nokta", en: " points" },
31         kayitliKalibrasyonBulundu: { tr: "Bu sensör için kayıtlı bir kalibrasyon bulundu", en: "A saved calibration was found for this
sensor" },
32         mevcutGoruntule: { tr: "Mevcut Kalibrasyonu Görüntüle", en: "View Current Calibration" },
33         kayitliNoktalarıIncele: { tr: "Kayıtlı noktaları incele", en: "Review saved points" },
34         yenidenKalibreEt: { tr: "Yeniden Kalibre Et", en: "Recalibrate" },
35         sifirdanYeniOlcum: { tr: "Sıfırdan yeni ölçüm al", en: "Take a fresh measurement" },
36         kayitliNoktalarBaslik: { tr: "KAYITLI KALİBRASYON NOKTALARI", en: "SAVED CALIBRATION POINTS" },
37         duzenleButon: { tr: "Düzenle", en: "Edit" },
38         silinecekOnay: { tr: "Mevcut kalibrasyon silinecek, emin misiniz?", en: "The current calibration will be deleted, are you sure?"
},
39         vazgecButon: { tr: "Vazgeç", en: "Cancel" },
40         evetYenidenBaslaButon: { tr: "Evet, Yeniden Başla", en: "Yes, Start Over" },
41         kalibrasyonTamamlandiBaslik: { tr: "Kalibrasyon Tamamlandı", en: "Calibration Complete" },
42         toplamOnEk: { tr: "Toplam ", en: "Total " },
43         noktaGirildiSonEk: { tr: " nokta girildi. Bu kalibrasyonu kaydetmek istiyor musunuz?", en: " points entered. Do you want to save
this calibration?" },
44         noktalarıKontrolEt: { tr: "Noktaları Kontrol Et", en: "Review Points" },
45         kaydetVeBitir: { tr: "Kaydet ve Bitir", en: "Save and Finish" },
46         kacNoktaSoru: { tr: "Kaç noktalı kalibrasyon yapmak istersiniz?", en: "How many calibration points would you like to use?" },
47         dahaFazlaNokta: { tr: "Daha fazla nokta, daha hassas kalibrasyon sağlar", en: "More points provide a more precise calibration" },
48         hamDeğerOnEk: { tr: "Ham değer: ", en: "Raw value: " },
49         hamDeğerSifirUyari: { tr: "Cihaz bağlı değil – ham değer 0 olarak kaydedilecek", en: "Device not connected – raw value will be
saved as 0" },
50         degisiklikleriKaydet: { tr: "Değişiklikleri Kaydet", en: "Save Changes" },
51         kaydedilenNoktalarBaslik: { tr: "KAYDEDİLEN NOKTALAR", en: "SAVED POINTS" },
52         duzeltmekIcinTikla: { tr: "Düzeltilmek için bir noktaya tıkla", en: "Click a point to correct it" },
53         kalibrasyonuDuzenle: { tr: "Kalibrasyonu Düzenle", en: "Edit Calibration" },
54         noktaOnEk: { tr: "Nokta ", en: "Point " },
55         duzenleniyorSonEk: { tr: " düzenleniyor", en: " being edited" },
56         duzenlemekIcinTikla: { tr: "Düzenlemek istediğiniz noktaya sağdaki listeden tıklayın", en: "Click the point you want to edit from
the list on the right" },
57         hedefAgirlikYonerge: { tr: "Hedef ağırlığı gir, sonra o ağırlığı load cell'e koy", en: "Enter the target weight, then place that
weight on the load cell" },
58         hedefMesafeYonerge: { tr: "Hedef mesafeyi gir, sonra boruyu o mesafeye getir", en: "Enter the target distance, then move the tube
to that distance" },
59         noktayıGuncelle: { tr: "Noktayı Güncelle", en: "Update Point" },
60         kalibrasyonuTamamla: { tr: "Kalibrasyonu Tamamla", en: "Complete Calibration" },
61         sonNoktayıKaydet: { tr: "Son Noktayı Kaydet", en: "Save Last Point" },
62         buNoktayıKaydet: { tr: "Bu Noktayı Kaydet", en: "Save This Point" }
63     })
64
65     function txt(anahtar) {
66         return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
67     }
68

```



```

69     property int seciliSensor: -1
70     property int loadCellAdimi: 0
71
72     property bool egimKayitliVar: false
73     property string egimSonTarih: ""
74     property bool egimBildirimGoster: false
75     property bool egimBildirimBasarili: true
76     property string egimBildirimMesaj: ""
77
78     property bool loadCellKayitliVar: false
79     property string loadCellSonTarih: ""
80     property int loadCellNoktaSayisi: 0
81     property var loadCellNoktalar: []
82     onLoadCellNoktalarChanged: loadCellAdimi = loadCellNoktalar.length
83     property int loadCellToplamNokta: 10
84     property bool loadCellSayiSecildi: false
85     property int loadCellDuzenlemeIndex: -1
86
87     property bool loadCellAraEkranGoster: false
88     property bool loadCellGoruntuleModu: false
89     property bool loadCellTamDuzenlemeModu: false
90     property bool loadCellYenidenOnayGoster: false
91     property bool loadCellTamamlandiOnayGoster: false
92
93     property bool loadCellBildirimGoster: false
94     property bool loadCellBildirimBasarili: true
95     property string loadCellBildirimMesaj: ""
96
97     property string mesafeSonTarih: ""
98
99     property bool mesafeOlcumKayitliVar: false
100    property int mesafeOlcumNoktaSayisi: 0
101    property var mesafeNoktalar: []
102    property int mesafeToplamNokta: 10
103    property bool mesafeSayiSecildi: false
104    property int mesafeDuzenlemeIndex: -1
105
106    property bool mesafeAraEkranGoster: false
107    property bool mesafeGoruntuleModu: false
108    property bool mesafeTamDuzenlemeModu: false
109    property bool mesafeYenidenOnayGoster: false
110    property bool mesafeTamamlandiOnayGoster: false
111
112    property bool mesafeBildirimGoster: false
113    property bool mesafeBildirimBasarili: true
114    property string mesafeBildirimMesaj: ""
115
116    function loadCellSiraliKopya(dizi) {
117        var kopya = dizi.slice()
118        kopya.sort(function(a, b) { return a.hamDeger - b.hamDeger })
119        return kopya
120    }
121
122    function loadCellKaydetVeBildir() {
123        var sirali = loadCellSiraliKopya(loadCellNoktalar)
124        var basarili = database.loadCellNoktalarKaydet(sirali)
125        loadCellBildirimBasarili = basarili
126        loadCellBildirimMesaj = basarili ? txt("kalibrasyonKaydedildiMesaj") : txt("kalibrasyonKaydedilemediMesaj")
127        loadCellBildirimGoster = true
128        loadCellBildirimTimer.restart()
129    }
130
131    Timer {
132        id: loadCellBildirimTimer
133        interval: loadCellBildirimBasarili ? 1800 : 2500
134        repeat: false
135        onTriggered: {
136            loadCellBildirimGoster = false
137            if (loadCellBildirimBasarili) {
138                seciliSensor = -1
139            }
140        }
141    }
142
143    function mesafeSiraliKopya(dizi) {
144        var kopya = dizi.slice()
145        kopya.sort(function(a, b) { return a.hamDeger - b.hamDeger })
146        return kopya
147    }
148

```

```

149 function mesafeKaydetVeBildir() {
150     var sirali = mesafeSiralikopya(mesafeNoktalar)
151     var basarili = database.mesafeNoktalarKaydet(sirali)
152     if (basarili) {
153         wifiManager.kalibrasyonYenidenYukle()
154     }
155     mesafeBildirimBasarili = basarili
156     mesafeBildirimMesaj = basarili ? txt("kalibrasyonKaydedildiMesaj") : txt("kalibrasyonKaydedilemediMesaj")
157     mesafeBildirimGoster = true
158     mesafeBildirimTimer.restart()
159 }
160
161 Timer {
162     id: mesafeBildirimTimer
163     interval: mesafeBildirimBasarili ? 1800 : 2500
164     repeat: false
165     onTriggered: {
166         mesafeBildirimGoster = false
167         if (mesafeBildirimBasarili) {
168             seciliSensor = -1
169         }
170     }
171 }
172
173 function egimSifirlaVeBildir() {
174     var biasX = sensorManager.hamAccelX
175     var biasY = sensorManager.hamAccelY
176     var basarili = database.egimKalibrasyonuKaydet(biasX, biasY, 0.0, 1.0, 1.0, 1.0)
177     if (basarili) {
178         wifiManager.kalibrasyonYenidenYukle()
179     }
180     egimBildirimBasarili = basarili
181     egimBildirimMesaj = basarili ? txt("kalibrasyonKaydedildiMesaj") : txt("kalibrasyonKaydedilemediMesaj")
182     egimBildirimGoster = true
183     egimBildirimTimer.restart()
184 }
185
186 Timer {
187     id: egimBildirimTimer
188     interval: egimBildirimBasarili ? 1800 : 2500
189     repeat: false
190     onTriggered: {
191         egimBildirimGoster = false
192         if (egimBildirimBasarili) {
193             seciliSensor = -1
194         }
195     }
196 }
197
198 // Sensör kartlarının "Kalibre Edildi" rozeti ve tarihi kalıcı depodan (SQLite)
199 // okunur; sadece o an kalibrasyon yapılıncaya değil, sayfa her açıldığında yeniden
200 // yüklenmeli ki uygulama yeniden başlatıldığında da rozet doğru görünsün.
201 function kalibrasyonDurumunuYukle() {
202     var egimKal = database.egimKalibrasyonuGetir()
203     egimKayitliVar = egimKal.mevcut
204     egimSonTarih = egimKal.mevcut ? egimKal.tarih : ""
205
206     var lcNoktalar = database.loadCellNoktalarGetir()
207     loadCellKayitliVar = lcNoktalar.length > 0
208     loadCellNoktaSayisi = lcNoktalar.length
209     loadCellSonTarih = loadCellKayitliVar ? database.kalibrasyonTarihiGetir("loadcell") : ""
210
211     var mNoktalar = database.mesafeNoktalarGetir()
212     mesafeOlcumKayitliVar = mNoktalar.length > 0
213     mesafeOlcumNoktaSayisi = mNoktalar.length
214     mesafeSonTarih = mesafeOlcumKayitliVar ? database.kalibrasyonTarihiGetir("mesafe_olcum") : ""
215 }
216
217 Component.onCompleted: kalibrasyonDurumunuYukle()
218
219 onVisibleChanged: {
220     if (visible) {
221         kalibrasyonDurumunuYukle()
222     }
223 }
224
225 onSeciliSensorChanged: {
226     if (seciliSensor === 0) {
227         var egimKal = database.egimKalibrasyonuGetir()
228         egimKayitliVar = egimKal.mevcut

```

```

229         egimSonTarih = egimKal.mevcut ? egimKal.tarih : ""
230         egimBildirimGoster = false
231     } else if (seciliSensor === 1) {
232         var noktalar = database.loadCellNoktalarGetir()
233         loadCellKayitliVar = noktalar.length > 0
234         loadCellNoktaSayisi = noktalar.length
235         loadCellSonTarih = loadCellKayitliVar ? database.kalibrasyonTarihiGetir("loadcell") : ""
236         loadCellNoktalar = []
237         loadCellSayiSecildi = false
238         loadCellDuzenlemeIndex = -1
239         loadCellTamDuzenlemeModu = false
240         loadCellGoruntuleModu = false
241         loadCellYenidenOnayGoster = false
242         loadCellTamamlandiOnayGoster = false
243         loadCellBildirimGoster = false
244         loadCellAraEkranGoster = loadCellKayitliVar
245     } else if (seciliSensor === 2) {
246         var mNoktalar = database.mesafeNoktalarGetir()
247         mesafeOlcumKayitliVar = mNoktalar.length > 0
248         mesafeOlcumNoktaSayisi = mNoktalar.length
249         mesafeSonTarih = mesafeOlcumKayitliVar ? database.kalibrasyonTarihiGetir("mesafe_olcum") : ""
250         mesafeNoktalar = []
251         mesafeSayiSecildi = false
252         mesafeDuzenlemeIndex = -1
253         mesafeTamDuzenlemeModu = false
254         mesafeGoruntuleModu = false
255         mesafeYenidenOnayGoster = false
256         mesafeTamamlandiOnayGoster = false
257         mesafeBildirimGoster = false
258         mesafeAraEkranGoster = mesafeOlcumKayitliVar
259     }
260 }

261
262 StackLayout {
263     anchors.fill: parent
264     currentIndex: seciliSensor === -1 ? 0 : 1
265
266     Item {
267         Text {
268             anchors.top: parent.top
269             anchors.left: parent.left
270             anchors.margins: 20
271             text: txt("baslikSensorSec")
272             color: "#6b7280"
273             font.family: "Segoe UI"
274             font.pixelSize: 12
275             font.bold: true
276             font.letterSpacing: 1
277         }
278
279         Row {
280             anchors.centerIn: parent
281             spacing: 24
282
283             Rectangle {
284                 id: egimKarti
285                 width: 350
286                 height: 450
287                 radius: 12
288                 color: "#0f1420"
289                 border.color: egimKarti.hovered ? "#14b8a6" : "#1e2a3f"
290                 border.width: 1
291                 property bool hovered: false
292                 scale: hovered ? 1.03 : 1.0
293                 Behavior on border.color { ColorAnimation { duration: 120 } }
294                 Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
295
296                 Column {
297                     anchors.centerIn: parent
298                     spacing: 14
299
300                     Item {
301                         anchors.horizontalCenter: parent.horizontalCenter
302                         width: 210
303                         height: 150
304
305                         Image {
306                             anchors.centerIn: parent
307                             width: parent.width
308                             height: parent.height

```

```

309         source: "../assets/egim_sensor.png"
310         fillMode: Image.PreserveAspectFit
311     }
312 }
313
314 Text { anchors.horizontalCenter: parent.horizontalCenter; text: txt("egimSensoru"); color: "#dce8f5"; font.family:
"Segoe UI"; font.pixelSize: 24; font.bold: true }
315 Text { anchors.horizontalCenter: parent.horizontalCenter; text: txt("egimAciklama"); color: "#6b7280";
font.family: "Segoe UI"; font.pixelSize: 14 }
316
317 Rectangle {
318     anchors.horizontalCenter: parent.horizontalCenter
319     visible: egimKayitliVar
320     width: kaliMetni1.implicitWidth + 16
321     height: 20
322     radius: 10
323     color: "#123321"
324
325     Text {
326         id: kaliMetni1
327         anchors.centerIn: parent
328         text: txt("kalibreEdildiOnEk") + egimSonTarih
329         color: "#4ade80"
330         font.pixelSize: 10
331         font.bold: true
332     }
333 }
334 }
335
336 MouseArea {
337     anchors.fill: parent
338     hoverEnabled: true
339     cursorShape: Qt.PointingHandCursor
340     onEntered: egimKarti.hovered = true
341     onExited: egimKarti.hovered = false
342     onClicked: seciliSensor = 0
343 }
344 }
345
346 Rectangle {
347     id: loadCellKarti
348     width: 350
349     height: 450
350     radius: 12
351     color: "#0f1420"
352     border.color: loadCellKarti.hovered ? "#3b82f6" : "#1e2a3f"
353     border.width: 1
354     property bool hovered: false
355     scale: hovered ? 1.03 : 1.0
356     Behavior on border.color { ColorAnimation { duration: 120 } }
357     Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
358
359     Column {
360         anchors.centerIn: parent
361         spacing: 14
362
363         Item {
364             anchors.horizontalCenter: parent.horizontalCenter
365             width: 210
366             height: 150
367
368             Image {
369                 anchors.centerIn: parent
370                 width: parent.width
371                 height: parent.height
372                 source: "../assets/loadcell.png"
373                 fillMode: Image.PreserveAspectFit
374             }
375         }
376
377         Text { anchors.horizontalCenter: parent.horizontalCenter; text: txt("loadCellAdi"); color: "#dce8f5"; font.family:
"Segoe UI"; font.pixelSize: 24; font.bold: true }
378         Text { anchors.horizontalCenter: parent.horizontalCenter; text: txt("loadCellAciklama"); color: "#6b7280";
font.family: "Segoe UI"; font.pixelSize: 14 }
379
380         Rectangle {
381             anchors.horizontalCenter: parent.horizontalCenter
382             visible: loadCellKayitliVar
383             width: kaliMetni2.implicitWidth + 16
384             height: 20

```

```

385         radius: 10
386         color: "#123321"
387
388         Text {
389             id: kaliMetni2
390             anchors.centerIn: parent
391             text: txt("kalibreEdildiOnEk") + loadCellSonTarih
392             color: "#4ade80"
393             font.pixelSize: 10
394             font.bold: true
395         }
396     }
397 }
398
399 MouseArea {
400     anchors.fill: parent
401     hoverEnabled: true
402     cursorShape: Qt.PointingHandCursor
403     onEntered: loadCellKarti.hovered = true
404     onExited: loadCellKarti.hovered = false
405     onClicked: seciliSensor = 1
406 }
407
408
409 Rectangle {
410     id: mesafeKarti
411     width: 350
412     height: 450
413     radius: 12
414     color: "#0f1420"
415     border.color: mesafeKarti.hovered ? "#9333ea" : "#1e2a3f"
416     border.width: 1
417     property bool hovered: false
418     scale: hovered ? 1.03 : 1.0
419     Behavior on border.color { ColorAnimation { duration: 120 } }
420     Behavior on scale { NumberAnimation { duration: 120; easing.type: Easing.OutQuad } }
421
422     Column {
423         anchors.centerIn: parent
424         spacing: 14
425
426         Item {
427             anchors.horizontalCenter: parent.horizontalCenter
428             width: 210
429             height: 150
430
431             Image {
432                 anchors.centerIn: parent
433                 width: parent.width
434                 height: parent.height
435                 source: "../assets/mesafe_sensor.png"
436                 fillMode: Image.PreserveAspectFit
437             }
438         }
439
440         Text { anchors.horizontalCenter: parent.horizontalCenter; text: txt("mesafeSensoru"); color: "#dce8f5";
font.family: "Segoe UI"; font.pixelSize: 24; font.bold: true }
441         Text { anchors.horizontalCenter: parent.horizontalCenter; text: txt("mesafeAciklama"); color: "#6b7280";
font.family: "Segoe UI"; font.pixelSize: 14 }
442
443         Rectangle {
444             anchors.horizontalCenter: parent.horizontalCenter
445             visible: mesafeOlcumKayitliVar
446             width: kaliMetni3.implicitWidth + 16
447             height: 20
448             radius: 10
449             color: "#123321"
450
451             Text {
452                 id: kaliMetni3
453                 anchors.centerIn: parent
454                 text: txt("kalibreEdildiOnEk") + mesafeSonTarih
455                 color: "#4ade80"
456                 font.pixelSize: 10
457                 font.bold: true
458             }
459         }
460     }
461
462     MouseArea {

```

```

463         anchors.fill: parent
464         hoverEnabled: true
465         cursorShape: Qt.PointingHandCursor
466         onEntered: mesafeKarti.hovered = true
467         onExited: mesafeKarti.hovered = false
468         onClicked: seciliSensor = 2
469     }
470 }
471 }
472 }
473
474 Item {
475     Row {
476         anchors.top: parent.top
477         anchors.left: parent.left
478         anchors.margins: 20
479         spacing: 12
480
481         Button {
482             text: txt("geriButon")
483             font.pixelSize: 13
484
485             background: Rectangle {
486                 radius: 6
487                 color: parent.hovered ? "#1e2a3f" : "#0f1420"
488                 border.color: "#1e2a3f"
489                 border.width: 1
490                 Behavior on color { ColorAnimation { duration: 120 } }
491             }
492
493             contentItem: Text {
494                 text: parent.text
495                 color: "#dce8f5"
496                 font: parent.font
497                 horizontalAlignment: Text.AlignHCenter
498                 verticalAlignment: Text.AlignVCenter
499                 leftPadding: 10
500                 rightPadding: 10
501             }
502
503             onClicked: seciliSensor = -1
504         }
505
506         Rectangle {
507             width: 8
508             height: 8
509             radius: 4
510             anchors.verticalCenter: parent.verticalCenter
511             color: {
512                 if (seciliSensor === 0) return "#14b8a6"
513                 if (seciliSensor === 1) return "#3b82f6"
514                 if (seciliSensor === 2) return "#9333ea"
515                 return "#6b7280"
516             }
517         }
518
519         Text {
520             anchors.verticalCenter: parent.verticalCenter
521             text: {
522                 if (seciliSensor === 0) return txt("egimBaslik")
523                 if (seciliSensor === 1) return txt("loadCellBaslik")
524                 if (seciliSensor === 2) return txt("mesafeBaslik")
525                 return ""
526             }
527             color: "#dce8f5"
528             font.family: "Segoe UI"
529             font.pixelSize: 16
530             font.bold: true
531         }
532     }
533
534     Item {
535         anchors.top: parent.top
536         anchors.left: parent.left
537         anchors.right: parent.right
538         anchors.bottom: parent.bottom
539         anchors.topMargin: 70
540         visible: seciliSensor === 0
541
542         Column {

```

```

543     anchors.centerIn: parent
544     spacing: 20
545     width: 360
546
547     Rectangle {
548         anchors.horizontalCenter: parent.horizontalCenter
549         visible: egimKayitliVar
550         width: egimSonKaliMetni.implicitWidth + 20
551         height: 26
552         radius: 13
553         color: "#0f1420"
554         border.color: "#1e2a3f"
555         border.width: 1
556
557         Text {
558             id: egimSonKaliMetni
559             anchors.centerIn: parent
560             text: txt("sonKalibrasyonOnEk") + egimSonTarih
561             color: "#6b7280"
562             font.pixelSize: 11
563         }
564     }
565
566     Text {
567         width: parent.width
568         horizontalAlignment: Text.AlignHCenter
569         wrapMode: Text.WordWrap
570         text: txt("cihazıVerlestin")
571         color: "#dce8f5"
572         font.family: "Segoe UI"
573         font.pixelSize: 14
574     }
575
576     SuTerazisi {
577         anchors.horizontalCenter: parent.horizontalCenter
578         egimX: sensorManager.egimX
579         egimY: sensorManager.egimY
580     }
581
582     Rectangle {
583         anchors.horizontalCenter: parent.horizontalCenter
584         width: 220
585         height: 72
586         radius: 12
587         color: "#0f2e2a"
588         border.color: "#14b8a6"
589         border.width: 1
590
591         Text {
592             anchors.centerIn: parent
593             text: "X: " + sensorManager.egimX.toFixed(1) + "°   Y: " + sensorManager.egimY.toFixed(1) + "°"
594             color: "#2dd4bf"
595             font.family: "Segoe UI"
596             font.pixelSize: 22
597             font.bold: true
598         }
599     }
600
601     Text {
602         width: parent.width
603         horizontalAlignment: Text.AlignHCenter
604         wrapMode: Text.WordWrap
605         visible: !wifiManager.baglandi
606         text: txt("cihazBagliDegilKalibrasyon")
607         color: "#f87171"
608         font.family: "Segoe UI"
609         font.pixelSize: 11
610     }
611
612     Button {
613         width: parent.width
614         height: 44
615         text: txt("sifirlaButon")
616         font.pixelSize: 14
617         font.bold: true
618         enabled: wifiManager.baglandi
619
620         onClicked: egimSifirlaVeBildir()
621
622         background: Rectangle {

```

```

623         radius: 8
624         color: parent.enabled ? (parent.hovered ? "#2dd4bf" : "#14b8a6") : "#1e2a3f"
625         Behavior on color { ColorAnimation { duration: 120 } }
626     }
627
628     contentItem: Text {
629         text: parent.text
630         color: "#ffffff"
631         font: parent.font
632         horizontalAlignment: Text.AlignHCenter
633         verticalAlignment: Text.AlignVCenter
634     }
635 }
636
637
638 Rectangle {
639     id: egimBildirimKutusu
640     anchors.top: parent.top
641     anchors.horizontalCenter: parent.horizontalCenter
642     anchors.topMargin: 12
643     z: 100
644     width: egimBildirimMetni.implicitWidth + 24
645     height: 32
646     radius: 16
647     color: egimBildirimBasarili ? "#123321" : "#3a1414"
648     opacity: egimBildirimGoster ? 1 : 0
649     visible: opacity > 0
650     Behavior on opacity { NumberAnimation { duration: 200 } }
651
652     Text {
653         id: egimBildirimMetni
654         anchors.centerIn: parent
655         text: egimBildirimMesaj
656         color: egimBildirimBasarili ? "#4ade80" : "#f87171"
657         font.family: "Segoe UI"
658         font.pixelSize: 12
659         font.bold: true
660     }
661 }
662
663
664 Item {
665     anchors.top: parent.top
666     anchors.left: parent.left
667     anchors.right: parent.right
668     anchors.bottom: parent.bottom
669     anchors.topMargin: 70
670     visible: seciliSensor === 1
671
672     Column {
673         anchors.centerIn: parent
674         spacing: 24
675         width: 420
676         visible: loadCellAraEkranGoster
677
678         Rectangle {
679             anchors.horizontalCenter: parent.horizontalCenter
680             width: araEkranBadge.implicitWidth + 20
681             height: 26
682             radius: 13
683             color: "#0f1420"
684             border.color: "#1e2a3f"
685             border.width: 1
686
687             Text {
688                 id: araEkranBadge
689                 anchors.centerIn: parent
690                 text: txt("kayitliOnEk") + loadCellNoktaSayisi + txt("noktaSonEk")
691                 color: "#6b7280"
692                 font.pixelSize: 11
693             }
694         }
695
696         Text {
697             width: parent.width
698             horizontalAlignment: Text.AlignHCenter
699             wrapMode: Text.WordWrap
700             text: txt("kayitliKalibrasyonBulundu")
701             color: "#dce8f5"
702             font.family: "Segoe UI"

```



```

703         font.pixelSize: 16
704         font.bold: true
705     }
706
707     Row {
708         anchors.horizontalCenter: parent.horizontalCenter
709         spacing: 20
710
711         Rectangle {
712             id: goruntuleKarti
713             width: 190
714             height: 180
715             radius: 14
716             color: "#0f1420"
717             border.color: goruntuleKarti.hovered ? "#3b82f6" : "#1e2a3f"
718             border.width: 1
719             property bool hovered: false
720             Behavior on border.color { ColorAnimation { duration: 120 } }
721
722             Column {
723                 anchors.centerIn: parent
724                 spacing: 12
725                 width: parent.width - 28
726
727                 Canvas {
728                     anchors.horizontalCenter: parent.horizontalCenter
729                     width: 32
730                     height: 32
731                     onPaint: {
732                         var ctx = getContext("2d")
733                         ctx.clearRect(0, 0, width, height)
734                         ctx.lineWidth = 1.8
735                         ctx.lineCap = "round"
736                         ctx.lineJoin = "round"
737                         ctx.strokeStyle = "#3b82f6"
738
739                         ctx.beginPath()
740                         ctx.moveTo(3, 16)
741                         ctx.bezierCurveTo(7.5, 8.5, 12.5, 6.8, 16, 6.8)
742                         ctx.bezierCurveTo(19.5, 6.8, 24.5, 8.5, 29, 16)
743                         ctx.bezierCurveTo(24.5, 23.5, 19.5, 25.2, 16, 25.2)
744                         ctx.bezierCurveTo(12.5, 25.2, 7.5, 23.5, 3, 16)
745                         ctx.stroke()
746
747                         ctx.beginPath()
748                         ctx.arc(16, 16, 4.4, 0, Math.PI * 2, false)
749                         ctx.stroke()
750                     }
751                 }
752
753                 Text {
754                     width: parent.width
755                     horizontalAlignment: Text.AlignHCenter
756                     wrapMode: Text.WordWrap
757                     text: txt("mevcutGoruntule")
758                     color: "#dce8f5"
759                     font.family: "Segoe UI"
760                     font.pixelSize: 13
761                     font.bold: true
762                 }
763
764                 Text {
765                     width: parent.width
766                     horizontalAlignment: Text.AlignHCenter
767                     wrapMode: Text.WordWrap
768                     text: txt("kayitliNoktalariIncele")
769                     color: "#6b7280"
770                     font.family: "Segoe UI"
771                     font.pixelSize: 11
772                 }
773             }
774
775             MouseArea {
776                 anchors.fill: parent
777                 hoverEnabled: true
778                 cursorShape: Qt.PointingHandCursor
779                 onEntered: goruntuleKarti.hovered = true
780                 onExited: goruntuleKarti.hovered = false
781                 onClicked: {
782                     loadCellNoktalari = database.loadCellNoktalariGetir()

```

```

783         loadCellAraEkranGoster = false
784         loadCellGoruntuleModu = true
785         loadCellDuzenlemeIndex = -1
786     }
787 }
788 }
789
790 Rectangle {
791     id: yenidenKarti
792     width: 190
793     height: 180
794     radius: 14
795     color: "#0f1420"
796     border.color: yenidenKarti.hovered ? "#9ca3af" : "#1e2a3f"
797     border.width: 1
798     property bool hovered: false
799     Behavior on border.color { ColorAnimation { duration: 120 } }
800
801     Column {
802         anchors.centerIn: parent
803         spacing: 12
804         width: parent.width - 28
805
806         Canvas {
807             anchors.horizontalCenter: parent.horizontalCenter
808             width: 32
809             height: 32
810             onPaint: {
811                 var ctx = getContext("2d")
812                 ctx.clearRect(0, 0, width, height)
813                 ctx.strokeStyle = "#9ca3af"
814                 ctx.fillStyle = "#9ca3af"
815                 ctx.lineWidth = 3.4
816                 ctx.lineCap = "butt"
817
818                 function yayOku(startDeg, endDeg) {
819                     var s = startDeg * Math.PI / 180
820                     var e = endDeg * Math.PI / 180
821                     var cx = 16, cy = 16, r = 10.5
822
823                     ctx.beginPath()
824                     ctx.arc(cx, cy, r, s, e, false)
825                     ctx.stroke()
826
827                     var ex = cx + r * Math.cos(e)
828                     var ey = cy + r * Math.sin(e)
829                     var tx = -Math.sin(e)
830                     var ty = Math.cos(e)
831                     var px = -ty
832                     var py = tx
833                     var geri = 6.2
834                     var yaricap = 3.6
835                     var bx = ex - tx * geri
836                     var by = ey - ty * geri
837
838                     ctx.beginPath()
839                     ctx.moveTo(ex, ey)
840                     ctx.lineTo(bx + px * yaricap, by + py * yaricap)
841                     ctx.lineTo(bx - px * yaricap, by - py * yaricap)
842                     ctx.closePath()
843                     ctx.fill()
844                 }
845
846                 yayOku(195, 345)
847                 yayOku(15, 165)
848             }
849         }
850
851         Text {
852             width: parent.width
853             horizontalAlignment: Text.AlignHCenter
854             wrapMode: Text.WordWrap
855             text: txt("yenidenKalibreEt")
856             color: "#dce8f5"
857             font.family: "Segoe UI"
858             font.pixelSize: 13
859             font.bold: true
860         }
861     }
862     Text {

```

```

863         width: parent.width
864         horizontalAlignment: Text.AlignHCenter
865         wrapMode: Text.WordWrap
866         text: txt("sifirdanYeniOlcum")
867         color: "#6b7280"
868         font.family: "Segoe UI"
869         font.pixelSize: 11
870     }
871 }
872
873 MouseArea {
874     anchors.fill: parent
875     hoverEnabled: true
876     cursorShape: Qt.PointingHandCursor
877     onEntered: yenidenKarti.hovered = true
878     onExited: yenidenKarti.hovered = false
879     onClicked: loadCellYenidenOnayGoster = true
880 }
881 }
882 }
883 }
884
885 Column {
886     anchors.centerIn: parent
887     spacing: 16
888     width: 340
889     visible: loadCellGoruntuleModu
890
891     Text {
892         text: txt("kayitliNoktalarBaslik")
893         color: "#6b7280"
894         font.family: "Segoe UI"
895         font.pixelSize: 11
896         font.bold: true
897         font.letterSpacing: 1
898     }
899
900     Rectangle {
901         width: parent.width
902         height: 320
903         radius: 14
904         color: "#0f1420"
905         border.color: "#1e2a3f"
906         border.width: 1
907
908         ListView {
909             anchors.fill: parent
910             anchors.margins: 16
911             clip: true
912             spacing: 8
913
914             model: loadCellNoktalari
915
916             delegate: Rectangle {
917                 width: ListView.view.width
918                 height: 48
919                 radius: 8
920                 color: "#0a0e17"
921                 border.color: "#1e2a3f"
922                 border.width: 1
923
924                 Row {
925                     anchors.fill: parent
926                     anchors.leftMargin: 12
927                     anchors.rightMargin: 12
928                     spacing: 8
929
930                     Rectangle {
931                         width: 22
932                         height: 22
933                         radius: 11
934                         anchors.verticalCenter: parent.verticalCenter
935                         color: "#182644"
936                         border.color: "#3b82f6"
937                         border.width: 1
938
939                         Text {
940                             anchors.centerIn: parent
941                             text: index + 1
942                             color: "#3b82f6"

```

```

943         font.pixelSize: 10
944         font.bold: true
945     }
946 }
947
948 Column {
949     anchors.verticalCenter: parent.verticalCenter
950     spacing: 1
951
952     Text {
953         text: modelData.hedefKg.toFixed(2) + " kg"
954         color: "#dce8f5"
955         font.family: "Segoe UI"
956         font.pixelSize: 13
957         font.bold: true
958     }
959
960     Text {
961         text: "ADC: " + modelData.hamDeger.toFixed(0)
962         color: "#6b7280"
963         font.family: "Segoe UI"
964         font.pixelSize: 10
965     }
966 }
967 }
968 }
969 }
970 }
971
972 Button {
973     width: parent.width
974     height: 46
975     text: txt("duzenleButon")
976     font.pixelSize: 14
977     font.bold: true
978
979     onClicked: {
980         loadCellToplamNokta = loadCellNoktalari.length
981         loadCellGoruntuleModu = false
982         loadCellTamDuzenlemeModu = true
983         loadCellSayiSecildi = true
984         loadCellDuzenlemeIndex = -1
985     }
986
987     background: Rectangle {
988         radius: 8
989         color: parent.hovered ? "#4f8cf7" : "#3b82f6"
990         Behavior on color { ColorAnimation { duration: 120 } }
991     }
992
993     contentItem: Text {
994         text: parent.text
995         color: "ffffff"
996         font: parent.font
997         horizontalAlignment: Text.AlignHCenter
998         verticalAlignment: Text.AlignVCenter
999     }
1000 }
1001 }
1002
1003 Rectangle {
1004     anchors.fill: parent
1005     color: "#000000b3"
1006     visible: loadCellYenidenOnayGoster
1007     z: 200
1008
1009     MouseArea { anchors.fill: parent }
1010
1011     Rectangle {
1012         anchors.centerIn: parent
1013         width: 360
1014         height: 200
1015         radius: 14
1016         color: "#0f1420"
1017         border.color: "#1e2a3f"
1018         border.width: 1
1019
1020         Column {
1021             anchors.centerIn: parent
1022             spacing: 18

```

```

1023         width: parent.width - 40
1024
1025         Text {
1026             width: parent.width
1027             horizontalAlignment: Text.AlignHCenter
1028             wrapMode: Text.WordWrap
1029             text: txt("silinecekOnay")
1030             color: "#dce8f5"
1031             font.family: "Segoe UI"
1032             font.pixelSize: 14
1033             font.bold: true
1034         }
1035
1036         Row {
1037             anchors.horizontalCenter: parent.horizontalCenter
1038             spacing: 12
1039
1040             Button {
1041                 width: 150
1042                 height: 42
1043                 text: txt("vazgecButon")
1044
1045                 onClicked: loadCellYenidenOnayGoster = false
1046
1047                 background: Rectangle {
1048                     radius: 8
1049                     color: parent.hovered ? "#1e2a3f" : "transparent"
1050                     border.color: "#1e2a3f"
1051                     border.width: 1
1052                     Behavior on color { ColorAnimation { duration: 120 } }
1053                 }
1054
1055                 contentItem: Text {
1056                     text: parent.text
1057                     color: "#9ca3af"
1058                     font: parent.font
1059                     horizontalAlignment: Text.AlignHCenter
1060                     verticalAlignment: Text.AlignVCenter
1061                 }
1062             }
1063
1064             Button {
1065                 width: 150
1066                 height: 42
1067                 text: txt("evetYenidenBaslaButon")
1068                 font.bold: true
1069
1070                 onClicked: {
1071                     loadCellYenidenOnayGoster = false
1072                     loadCellAraEkranGoster = false
1073                     loadCellGoruntuleModu = false
1074                     loadCellTamDuzenlemeModu = false
1075                     loadCellNoktalarlari = []
1076                     loadCellSayiSecildi = false
1077                     loadCellDuzenlemeIndex = -1
1078                 }
1079
1080                 background: Rectangle {
1081                     radius: 8
1082                     color: parent.hovered ? "#b91c1c" : "#991b1b"
1083                     Behavior on color { ColorAnimation { duration: 120 } }
1084                 }
1085
1086                 contentItem: Text {
1087                     text: parent.text
1088                     color: "#ffffff"
1089                     font: parent.font
1090                     horizontalAlignment: Text.AlignHCenter
1091                     verticalAlignment: Text.AlignVCenter
1092                 }
1093             }
1094         }
1095     }
1096 }
1097
1098
1099 Rectangle {
1100     anchors.fill: parent
1101     color: "#000000b3"
1102     visible: loadCellTamamlandiOnayGoster

```

```

1103         z: 200
1104
1105     MouseArea { anchors.fill: parent }
1106
1107     Rectangle {
1108         anchors.centerIn: parent
1109         width: 380
1110         height: 220
1111         radius: 14
1112         color: "#0f1420"
1113         border.color: "#16a34a"
1114         border.width: 1
1115
1116         Column {
1117             anchors.centerIn: parent
1118             spacing: 18
1119             width: parent.width - 40
1120
1121             Column {
1122                 width: parent.width
1123                 spacing: 6
1124
1125                 Text {
1126                     width: parent.width
1127                     horizontalAlignment: Text.AlignHCenter
1128                     text: txt("kalibrasyonTamamlandiBaslik")
1129                     color: "#4ade80"
1130                     font.family: "Segoe UI"
1131                     font.pixelSize: 17
1132                     font.bold: true
1133                 }
1134
1135                 Text {
1136                     width: parent.width
1137                     horizontalAlignment: Text.AlignHCenter
1138                     wrapMode: Text.WordWrap
1139                     text: txt("toplamlanEk") + loadCellToplamNokta + txt("noktaGirildiSonEk")
1140                     color: "#9ca3af"
1141                     font.family: "Segoe UI"
1142                     font.pixelSize: 13
1143                 }
1144             }
1145
1146             Row {
1147                 anchors.horizontalCenter: parent.horizontalCenter
1148                 spacing: 12
1149
1150                 Button {
1151                     width: 150
1152                     height: 42
1153                     text: txt("noktalarikontrolEt")
1154
1155                     onClicked: loadCellTamamlandiOnayGoster = false
1156
1157                     background: Rectangle {
1158                         radius: 8
1159                         color: parent.hovered ? "#1e2a3f" : "transparent"
1160                         border.color: "#1e2a3f"
1161                         border.width: 1
1162                         Behavior on color { ColorAnimation { duration: 120 } }
1163                     }
1164
1165                     contentItem: Text {
1166                         text: parent.text
1167                         color: "#9ca3af"
1168                         font: parent.font
1169                         horizontalAlignment: Text.AlignHCenter
1170                         verticalAlignment: Text.AlignVCenter
1171                     }
1172                 }
1173
1174                 Button {
1175                     width: 150
1176                     height: 42
1177                     text: txt("kaydetVeBitir")
1178                     font.bold: true
1179
1180                     onClicked: {
1181                         loadCellTamamlandiOnayGoster = false
1182                         loadCellKaydetVeBildir()

```

```

1183     }
1184
1185     background: Rectangle {
1186         radius: 8
1187         color: parent.hovered ? "#22c55e" : "#16a34a"
1188         Behavior on color { ColorAnimation { duration: 120 } }
1189     }
1190
1191     contentItem: Text {
1192         text: parent.text
1193         color: "#ffffff"
1194         font: parent.font
1195         horizontalAlignment: Text.AlignHCenter
1196         verticalAlignment: Text.AlignVCenter
1197     }
1198     }
1199 }
1200
1201 }
1202
1203 }
1204
1205 Rectangle {
1206     id: loadCellBildirirKutusu
1207     anchors.top: parent.top
1208     anchors.horizontalCenter: parent.horizontalCenter
1209     anchors.topMargin: 12
1210     z: 100
1211     width: loadCellBildirirMetni.implicitWidth + 24
1212     height: 32
1213     radius: 16
1214     color: loadCellBildirirBasarili ? "#123321" : "#3a1414"
1215     opacity: loadCellBildirirGoster ? 1 : 0
1216     visible: opacity > 0
1217     Behavior on opacity { NumberAnimation { duration: 200 } }
1218
1219     Text {
1220         id: loadCellBildirirMetni
1221         anchors.centerIn: parent
1222         text: loadCellBildirirMesaj
1223         color: loadCellBildirirBasarili ? "#4ade80" : "#f87171"
1224         font.family: "Segoe UI"
1225         font.pixelSize: 12
1226         font.bold: true
1227     }
1228 }
1229
1230 Column {
1231     anchors.centerIn: parent
1232     spacing: 24
1233     width: 360
1234     visible: !loadCellSayiSecildi && !loadCellAraEkranGoster && !loadCellGoruntuleModu
1235
1236     Text {
1237         width: parent.width
1238         horizontalAlignment: Text.AlignHCenter
1239         wrapMode: Text.WordWrap
1240         text: txt("kacNoktaSoru")
1241         color: "#dce8f5"
1242         font.family: "Segoe UI"
1243         font.pixelSize: 16
1244         font.bold: true
1245     }
1246
1247     Text {
1248         width: parent.width
1249         horizontalAlignment: Text.AlignHCenter
1250         wrapMode: Text.WordWrap
1251         text: txt("dahaFazlaNokta")
1252         color: "#6b7280"
1253         font.family: "Segoe UI"
1254         font.pixelSize: 12
1255     }
1256
1257     Grid {
1258         anchors.horizontalCenter: parent.horizontalCenter
1259         columns: 4
1260         spacing: 12
1261
1262         Repeater {
1263             model: [4, 6, 8, 10]

```

```

1263
1264         Rectangle {
1265             id: secKarti
1266             width: 72
1267             height: 72
1268             radius: 12
1269             color: secKarti.hovered ? "#182644" : "#0f1420"
1270             border.color: "#3b82f6"
1271             border.width: 1
1272             property bool hovered: false
1273             Behavior on color { ColorAnimation { duration: 120 } }
1274
1275             Text {
1276                 anchors.centerIn: parent
1277                 text: modelData
1278                 color: "#dce8f5"
1279                 font.family: "Segoe UI"
1280                 font.pixelSize: 26
1281                 font.bold: true
1282             }
1283
1284             MouseArea {
1285                 anchors.fill: parent
1286                 hoverEnabled: true
1287                 cursorShape: Qt.PointingHandCursor
1288                 onEntered: secKarti.hovered = true
1289                 onExited: secKarti.hovered = false
1290                 onClicked: {
1291                     loadCellToplamNokta = modelData
1292                     loadCellNoktaları = []
1293                     loadCellDuzenlemeIndex = -1
1294                     loadCellSayıSecildi = true
1295                 }
1296             }
1297         }
1298     }
1299 }
1300
1301
1302 Row {
1303     anchors.centerIn: parent
1304     spacing: 24
1305     visible: loadCellSayıSecildi
1306
1307     Rectangle {
1308         width: 340
1309         height: 500
1310         radius: 14
1311         color: "#0f1420"
1312         border.color: "#1e2a3f"
1313         border.width: 1
1314
1315         Column {
1316             anchors.centerIn: parent
1317             spacing: 16
1318             width: parent.width - 40
1319
1320             Rectangle {
1321                 anchors.horizontalCenter: parent.horizontalCenter
1322                 visible: loadCellKayıtlıVar && !loadCellTamDuzenlemeModu
1323                 width: sonKaliMetni2.implicitWidth + 20
1324                 height: 24
1325                 radius: 12
1326                 color: "#0a0e17"
1327                 border.color: "#1e2a3f"
1328                 border.width: 1
1329
1330                 Text {
1331                     id: sonKaliMetni2
1332                     anchors.centerIn: parent
1333                     text: txt("kayıtlıOnEk") + loadCellNoktaSayısı + txt("noktaSonEk")
1334                     color: "#6b7280"
1335                     font.pixelSize: 11
1336                 }
1337             }
1338
1339             Text {
1340                 anchors.horizontalCenter: parent.horizontalCenter
1341                 text: loadCellTamDuzenlemeModu
1342                     ? txt("kalibrasyonuDuzenle")

```



```

1343         : txt("noktaOnEk") + (loadCellNoktalari.length + 1) + " / " + loadCellToplamNokta
1344         color: "#3b82f6"
1345         font.family: "Segoe UI"
1346         font.pixelSize: 13
1347         font.bold: true
1348         font.letterSpacing: 1
1349     }
1350
1351     Rectangle {
1352         width: parent.width
1353         height: 6
1354         radius: 3
1355         color: "#1e2a3f"
1356
1357         Rectangle {
1358             width: parent.width * (loadCellNoktalari.length / loadCellToplamNokta)
1359             height: parent.height
1360             radius: 3
1361             color: "#3b82f6"
1362             Behavior on width { NumberAnimation { duration: 150 } }
1363         }
1364     }
1365
1366     Text {
1367         width: parent.width
1368         horizontalAlignment: Text.AlignHCenter
1369         wrapMode: Text.WordWrap
1370         text: loadCellDuzenlemeIndex >= 0
1371             ? txt("noktaOnEk") + (loadCellDuzenlemeIndex + 1) + txt("duzenleniyorSonEk")
1372             : (loadCellTamDuzenlemeModu
1373                 ? txt("duzenlemekIcinTikla")
1374                 : txt("hedefAgirlikYonerge"))
1375         color: loadCellDuzenlemeIndex >= 0 ? "#3b82f6" : "#9ca3af"
1376         font.family: "Segoe UI"
1377         font.bold: loadCellDuzenlemeIndex >= 0
1378         font.pixelSize: 13
1379     }
1380
1381     TextField {
1382         id: hedefKgKutusu
1383         anchors.horizontalCenter: parent.horizontalCenter
1384         width: 180
1385         height: 44
1386         placeholderText: "kg"
1387         placeholderTextColor: "#4b5563"
1388         color: "ffffff"
1389         font.pixelSize: 16
1390         horizontalAlignment: TextInput.AlignHCenter
1391         validator: DoubleValidator { bottom: 0; top: 500; decimals: 2 }
1392         background: Rectangle {
1393             color: "#0a0e17"
1394             radius: 10
1395             border.color: hedefKgKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
1396             border.width: 1
1397         }
1398     }
1399
1400     Rectangle {
1401         anchors.horizontalCenter: parent.horizontalCenter
1402         width: parent.width
1403         height: 64
1404         radius: 12
1405         color: "#132335"
1406         border.color: wifiManager.baglandi ? "#3b82f6" : "#dc2626"
1407         border.width: 1
1408         Behavior on border.color { ColorAnimation { duration: 120 } }
1409
1410         Text {
1411             anchors.centerIn: parent
1412             text: txt("hamDegerOnEk") + sensorManager.hamAgirlik.toFixed(0)
1413             color: wifiManager.baglandi ? "#3b82f6" : "#f87171"
1414             font.family: "Segoe UI"
1415             font.pixelSize: 20
1416             font.bold: true
1417         }
1418     }
1419
1420     Text {
1421         width: parent.width
1422         horizontalAlignment: Text.AlignHCenter

```

```

1423         wrapMode: Text.WordWrap
1424         visible: !wifiManager.baglandi
1425         text: txt("hamDegerSifirUyari")
1426         color: "#f87171"
1427         font.family: "Segoe UI"
1428         font.pixelSize: 11
1429     }
1430
1431     Button {
1432         id: noktaKaydetButonu
1433         width: parent.width
1434         height: 46
1435         property bool tumNoktalarGirildi: !loadCellTamDuzenlemeModu && loadCellDuzenlemeIndex < 0 &&
loadCellNoktalar.length >= loadCellToplamNokta
1436         text: loadCellDuzenlemeIndex >= 0
1437             ? txt("noktayiGuncelle")
1438             : (tumNoktalarGirildi
1439                 ? txt("kalibrasyonuTamamla")
1440                 : (loadCellNoktalar.length >= loadCellToplamNokta - 1 ? txt("sonNoktayiKaydet") :
txt("buNoktayiKaydet")))
1441         font.pixelSize: 14
1442         font.bold: true
1443         enabled: tumNoktalarGirildi || (hedefKgKutusu.text.length > 0 && (loadCellDuzenlemeIndex >= 0 ||
loadCellNoktalar.length < loadCellToplamNokta))
1444
1445         onClicked: {
1446             if (tumNoktalarGirildi) {
1447                 loadCellTamamlandiOnayGoster = true
1448                 return
1449             }
1450
1451             var yeniNokta = { hedefKg: parseFloat(hedefKgKutusu.text), hamDeger: sensorManager.hamAgirlik }
1452
1453             if (loadCellDuzenlemeIndex >= 0) {
1454                 // Var olan bir noktayı yerinde güncelliyoruz, diğer noktalar etkilenmiyor
1455                 var guncelDizi = loadCellNoktalar.slice()
1456                 guncelDizi[loadCellDuzenlemeIndex] = yeniNokta
1457                 loadCellNoktalar = guncelDizi
1458                 loadCellDuzenlemeIndex = -1
1459                 hedefKgKutusu.text = ""
1460                 return
1461             }
1462
1463             var yeniDizi = loadCellNoktalar.slice()
1464             yeniDizi.push(yeniNokta)
1465             loadCellNoktalar = yeniDizi
1466             hedefKgKutusu.text = ""
1467
1468             if (loadCellNoktalar.length >= loadCellToplamNokta) {
1469                 loadCellTamamlandiOnayGoster = true
1470             }
1471         }
1472
1473         background: Rectangle {
1474             radius: 10
1475             color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
1476             Behavior on color { ColorAnimation { duration: 120 } }
1477         }
1478
1479         contentItem: Text {
1480             text: parent.text
1481             color: "#ffffff"
1482             font: parent.font
1483             horizontalAlignment: Text.AlignHCenter
1484             verticalAlignment: Text.AlignVCenter
1485         }
1486     }
1487
1488     Button {
1489         width: parent.width
1490         height: 46
1491         visible: loadCellTamDuzenlemeModu
1492         text: txt("degisiklikleriKaydet")
1493         font.pixelSize: 14
1494         font.bold: true
1495
1496         onClicked: loadCellKaydetVeBildir()
1497
1498         background: Rectangle {
1499             radius: 10

```

```

1500         color: parent.hovered ? "#22c55e" : "#16a34a"
1501         Behavior on color { ColorAnimation { duration: 120 } }
1502     }
1503
1504     contentItem: Text {
1505         text: parent.text
1506         color: "#ffffff"
1507         font: parent.font
1508         horizontalAlignment: Text.AlignHCenter
1509         verticalAlignment: Text.AlignVCenter
1510     }
1511 }
1512
1513 Button {
1514     width: parent.width
1515     height: 38
1516     visible: loadCellDuzenlemeIndex >= 0
1517     text: txt("vazgecButon")
1518     font.pixelSize: 13
1519
1520     onClicked: {
1521         loadCellDuzenlemeIndex = -1
1522         hedefKgKutusu.text = ""
1523     }
1524
1525     background: Rectangle {
1526         radius: 8
1527         color: parent.hovered ? "#1e2a3f" : "transparent"
1528         border.color: "#1e2a3f"
1529         border.width: 1
1530         Behavior on color { ColorAnimation { duration: 120 } }
1531     }
1532
1533     contentItem: Text {
1534         text: parent.text
1535         color: "#9ca3af"
1536         font: parent.font
1537         horizontalAlignment: Text.AlignHCenter
1538         verticalAlignment: Text.AlignVCenter
1539     }
1540 }
1541 }
1542 }
1543
1544 Rectangle {
1545     width: 300
1546     height: 500
1547     radius: 14
1548     color: "#0f1420"
1549     border.color: "#1e2a3f"
1550     border.width: 1
1551
1552     Column {
1553         anchors.top: parent.top
1554         anchors.left: parent.left
1555         anchors.right: parent.right
1556         anchors.margins: 20
1557         spacing: 4
1558
1559         Text {
1560             text: txt("kaydedilenNoktalarBaslik")
1561             color: "#6b7280"
1562             font.family: "Segoe UI"
1563             font.pixelSize: 11
1564             font.bold: true
1565             font.letterSpacing: 1
1566         }
1567
1568         Text {
1569             text: txt("duzeltmekIcinTikla")
1570             color: "#4b5563"
1571             font.family: "Segoe UI"
1572             font.pixelSize: 10
1573         }
1574     }
1575
1576     ListView {
1577         anchors.top: parent.top
1578         anchors.left: parent.left
1579         anchors.right: parent.right

```

```

1580     anchors.bottom: parent.bottom
1581     anchors.margins: 16
1582     anchors.topMargin: 60
1583     clip: true
1584     spacing: 8
1585
1586     model: loadCellNoktaları
1587
1588     delegate: Rectangle {
1589         width: ListView.view.width
1590         height: 48
1591         radius: 8
1592         color: satirAlani.containsMouse ? "#182644" : "#0a0e17"
1593         border.color: index === loadCellDuzenlemeIndex ? "#3b82f6" : "#1e2a3f"
1594         border.width: index === loadCellDuzenlemeIndex ? 2 : 1
1595         Behavior on color { ColorAnimation { duration: 120 } }
1596
1597         Row {
1598             anchors.fill: parent
1599             anchors.leftMargin: 12
1600             anchors.rightMargin: 12
1601             spacing: 8
1602
1603             Rectangle {
1604                 width: 22
1605                 height: 22
1606                 radius: 11
1607                 anchors.verticalCenter: parent.verticalCenter
1608                 color: "#182644"
1609                 border.color: "#3b82f6"
1610                 border.width: 1
1611
1612                 Text {
1613                     anchors.centerIn: parent
1614                     text: index + 1
1615                     color: "#3b82f6"
1616                     font.pixelSize: 10
1617                     font.bold: true
1618                 }
1619             }
1620
1621             Column {
1622                 anchors.verticalCenter: parent.verticalCenter
1623                 spacing: 1
1624
1625                 Text {
1626                     text: modelData.hedefKg.toFixed(2) + " kg"
1627                     color: "#dce8f5"
1628                     font.family: "Segoe UI"
1629                     font.pixelSize: 13
1630                     font.bold: true
1631                 }
1632
1633                 Text {
1634                     text: "ADC: " + modelData.hamDeger.toFixed(0)
1635                     color: "#6b7280"
1636                     font.family: "Segoe UI"
1637                     font.pixelSize: 10
1638                 }
1639             }
1640         }
1641
1642         Text {
1643             anchors.right: parent.right
1644             anchors.verticalCenter: parent.verticalCenter
1645             anchors.rightMargin: 12
1646             text: "↖"
1647             color: "#6b7280"
1648             font.pixelSize: 13
1649             visible: satirAlani.containsMouse
1650         }
1651
1652         MouseArea {
1653             id: satirAlani
1654             anchors.fill: parent
1655             hoverEnabled: true
1656             cursorShape: Qt.PointingHandCursor
1657             onClicked: {
1658                 hedefKgKutusu.text = modelData.hedefKg.toString()
1659                 loadCellDuzenlemeIndex = index

```

```

1660     }
1661     }
1662     }
1663     }
1664     }
1665     }
1666 }
1667
1668 Item {
1669     anchors.top: parent.top
1670     anchors.left: parent.left
1671     anchors.right: parent.right
1672     anchors.bottom: parent.bottom
1673     anchors.topMargin: 70
1674     visible: seciliSensor === 2
1675
1676     // =====
1677     // OLCUM KALIBRASYONU – cok nokta (load cell birebir kopyasi)
1678     // =====
1679     Item {
1680         anchors.fill: parent
1681
1682         Column {
1683             anchors.centerIn: parent
1684             spacing: 24
1685             width: 420
1686             visible: mesafeAraEkranGoster
1687
1688             Rectangle {
1689                 anchors.horizontalCenter: parent.horizontalCenter
1690                 width: mesafeAraEkranBadge.implicitWidth + 20
1691                 height: 26
1692                 radius: 13
1693                 color: "#0f1420"
1694                 border.color: "#1e2a3f"
1695                 border.width: 1
1696
1697                 Text {
1698                     id: mesafeAraEkranBadge
1699                     anchors.centerIn: parent
1700                     text: txt("kayitliOnEk") + mesafeOlcumNoktaSayisi + txt("noktaSonEk")
1701                     color: "#6b7280"
1702                     font.pixelSize: 11
1703                 }
1704             }
1705
1706             Text {
1707                 width: parent.width
1708                 horizontalAlignment: Text.AlignHCenter
1709                 wrapMode: Text.WordWrap
1710                 text: txt("kayitliKalibrasyonBulundu")
1711                 color: "#dce8f5"
1712                 font.family: "Segoe UI"
1713                 font.pixelSize: 16
1714                 font.bold: true
1715             }
1716
1717             Row {
1718                 anchors.horizontalCenter: parent.horizontalCenter
1719                 spacing: 20
1720
1721                 Rectangle {
1722                     id: mesafeGoruntuleKarti
1723                     width: 190
1724                     height: 180
1725                     radius: 14
1726                     color: "#0f1420"
1727                     border.color: mesafeGoruntuleKarti.hovered ? "#9333ea" : "#1e2a3f"
1728                     border.width: 1
1729                     property bool hovered: false
1730                     Behavior on border.color { ColorAnimation { duration: 120 } }
1731
1732                     Column {
1733                         anchors.centerIn: parent
1734                         spacing: 12
1735                         width: parent.width - 28
1736
1737                         Canvas {
1738                             anchors.horizontalCenter: parent.horizontalCenter
1739                             width: 32

```

```

1740         height: 32
1741         onPaint: {
1742             var ctx = getContext("2d")
1743             ctx.clearRect(0, 0, width, height)
1744             ctx.lineWidth = 1.8
1745             ctx.lineCap = "round"
1746             ctx.lineJoin = "round"
1747             ctx.strokeStyle = "#9333ea"
1748
1749             ctx.beginPath()
1750             ctx.moveTo(3, 16)
1751             ctx.bezierCurveTo(7.5, 8.5, 12.5, 6.8, 16, 6.8)
1752             ctx.bezierCurveTo(19.5, 6.8, 24.5, 8.5, 29, 16)
1753             ctx.bezierCurveTo(24.5, 23.5, 19.5, 25.2, 16, 25.2)
1754             ctx.bezierCurveTo(12.5, 25.2, 7.5, 23.5, 3, 16)
1755             ctx.stroke()
1756
1757             ctx.beginPath()
1758             ctx.arc(16, 16, 4.4, 0, Math.PI * 2, false)
1759             ctx.stroke()
1760         }
1761     }
1762
1763     Text {
1764         width: parent.width
1765         horizontalAlignment: Text.AlignHCenter
1766         wrapMode: Text.WordWrap
1767         text: txt("mevcutGoruntule")
1768         color: "#dce8f5"
1769         font.family: "Segoe UI"
1770         font.pixelSize: 13
1771         font.bold: true
1772     }
1773
1774     Text {
1775         width: parent.width
1776         horizontalAlignment: Text.AlignHCenter
1777         wrapMode: Text.WordWrap
1778         text: txt("kayitliNoktalariIncele")
1779         color: "#6b7280"
1780         font.family: "Segoe UI"
1781         font.pixelSize: 11
1782     }
1783 }
1784
1785 MouseArea {
1786     anchors.fill: parent
1787     hoverEnabled: true
1788     cursorShape: Qt.PointingHandCursor
1789     onEntered: mesafeGoruntuleKarti.hovered = true
1790     onExited: mesafeGoruntuleKarti.hovered = false
1791     onClicked: {
1792         mesafeNoktalari = database.mesafeNoktalariGetir()
1793         mesafeAraEkranGoster = false
1794         mesafeGoruntuleModu = true
1795         mesafeDuzenlemeIndex = -1
1796     }
1797 }
1798
1799 Rectangle {
1800     id: mesafeYenidenKarti
1801     width: 190
1802     height: 180
1803     radius: 14
1804     color: "#0f1420"
1805     border.color: mesafeYenidenKarti.hovered ? "#9ca3af" : "#1e2a3f"
1806     border.width: 1
1807     property bool hovered: false
1808     Behavior on border.color { ColorAnimation { duration: 120 } }
1809
1810     Column {
1811         anchors.centerIn: parent
1812         spacing: 12
1813         width: parent.width - 28
1814
1815         Canvas {
1816             anchors.horizontalCenter: parent.horizontalCenter
1817             width: 32
1818             height: 32

```

```

1820         onPaint: {
1821             var ctx = getContext("2d")
1822             ctx.clearRect(0, 0, width, height)
1823             ctx.strokeStyle = "#9ca3af"
1824             ctx.fillStyle = "#9ca3af"
1825             ctx.lineWidth = 3.4
1826             ctx.lineCap = "butt"
1827
1828             function yayOku(startDeg, endDeg) {
1829                 var s = startDeg * Math.PI / 180
1830                 var e = endDeg * Math.PI / 180
1831                 var cx = 16, cy = 16, r = 10.5
1832
1833                 ctx.beginPath()
1834                 ctx.arc(cx, cy, r, s, e, false)
1835                 ctx.stroke()
1836
1837                 var ex = cx + r * Math.cos(e)
1838                 var ey = cy + r * Math.sin(e)
1839                 var tx = -Math.sin(e)
1840                 var ty = Math.cos(e)
1841                 var px = -ty
1842                 var py = tx
1843                 var geri = 6.2
1844                 var yaricap = 3.6
1845                 var bx = ex - tx * geri
1846                 var by = ey - ty * geri
1847
1848                 ctx.beginPath()
1849                 ctx.moveTo(ex, ey)
1850                 ctx.lineTo(bx + px * yaricap, by + py * yaricap)
1851                 ctx.lineTo(bx - px * yaricap, by - py * yaricap)
1852                 ctx.closePath()
1853                 ctx.fill()
1854             }
1855
1856             yayOku(195, 345)
1857             yayOku(15, 165)
1858         }
1859     }
1860
1861     Text {
1862         width: parent.width
1863         horizontalAlignment: Text.AlignHCenter
1864         wrapMode: Text.WordWrap
1865         text: txt("yenidenKalibreEt")
1866         color: "#dce8f5"
1867         font.family: "Segoe UI"
1868         font.pixelSize: 13
1869         font.bold: true
1870     }
1871
1872     Text {
1873         width: parent.width
1874         horizontalAlignment: Text.AlignHCenter
1875         wrapMode: Text.WordWrap
1876         text: txt("sifirdanYeniOlcum")
1877         color: "#6b7280"
1878         font.family: "Segoe UI"
1879         font.pixelSize: 11
1880     }
1881 }
1882
1883 MouseArea {
1884     anchors.fill: parent
1885     hoverEnabled: true
1886     cursorShape: Qt.PointingHandCursor
1887     onEntered: mesafeYenidenKarti.hovered = true
1888     onExited: mesafeYenidenKarti.hovered = false
1889     onClicked: mesafeYenidenOnayGoster = true
1890 }
1891
1892 }
1893
1894
1895 Column {
1896     anchors.centerIn: parent
1897     spacing: 16
1898     width: 340
1899     visible: mesafeGoruntuleModu

```

```

1900
1901 Text {
1902     text: txt("kayitliNoktalarBaslik")
1903     color: "#6b7280"
1904     font.family: "Segoe UI"
1905     font.pixelSize: 11
1906     font.bold: true
1907     font.letterSpacing: 1
1908 }
1909
1910 Rectangle {
1911     width: parent.width
1912     height: 320
1913     radius: 14
1914     color: "#0f1420"
1915     border.color: "#1e2a3f"
1916     border.width: 1
1917
1918     ListView {
1919         anchors.fill: parent
1920         anchors.margins: 16
1921         clip: true
1922         spacing: 8
1923
1924         model: mesafeNoktalarIari
1925
1926         delegate: Rectangle {
1927             width: ListView.view.width
1928             height: 48
1929             radius: 8
1930             color: "#0a0e17"
1931             border.color: "#1e2a3f"
1932             border.width: 1
1933
1934             Row {
1935                 anchors.fill: parent
1936                 anchors.leftMargin: 12
1937                 anchors.rightMargin: 12
1938                 spacing: 8
1939
1940                 Rectangle {
1941                     width: 22
1942                     height: 22
1943                     radius: 11
1944                     anchors.verticalCenter: parent.verticalCenter
1945                     color: "#241a38"
1946                     border.color: "#9333ea"
1947                     border.width: 1
1948
1949                     Text {
1950                         anchors.centerIn: parent
1951                         text: index + 1
1952                         color: "#c084fc"
1953                         font.pixelSize: 10
1954                         font.bold: true
1955                     }
1956                 }
1957
1958                 Column {
1959                     anchors.verticalCenter: parent.verticalCenter
1960                     spacing: 1
1961
1962                     Text {
1963                         text: (modelData.hedefMm / 10).toFixed(1) + " cm"
1964                         color: "#dce8f5"
1965                         font.family: "Segoe UI"
1966                         font.pixelSize: 13
1967                         font.bold: true
1968                     }
1969
1970                     Text {
1971                         text: "ADC: " + modelData.hamDeger.toFixed(0)
1972                         color: "#6b7280"
1973                         font.family: "Segoe UI"
1974                         font.pixelSize: 10
1975                     }
1976                 }
1977             }
1978         }
1979     }

```



```

1980     }
1981
1982     Button {
1983         width: parent.width
1984         height: 46
1985         text: txt("duzenleButon")
1986         font.pixelSize: 14
1987         font.bold: true
1988
1989         onClicked: {
1990             mesafeToplamNokta = mesafeNoktalari.length
1991             mesafeGoruntuleModu = false
1992             mesafeTamDuzenlemeModu = true
1993             mesafeSayiSecildi = true
1994             mesafeDuzenlemeIndex = -1
1995         }
1996
1997         background: Rectangle {
1998             radius: 8
1999             color: parent.hovered ? "#a855f7" : "#9333ea"
2000             Behavior on color { ColorAnimation { duration: 120 } }
2001         }
2002
2003         contentItem: Text {
2004             text: parent.text
2005             color: "#ffffff"
2006             font: parent.font
2007             horizontalAlignment: Text.AlignHCenter
2008             verticalAlignment: Text.AlignVCenter
2009         }
2010     }
2011 }
2012
2013 Rectangle {
2014     anchors.fill: parent
2015     color: "#000000b3"
2016     visible: mesafeYenidenOnayGoster
2017     z: 200
2018
2019     MouseArea { anchors.fill: parent }
2020
2021     Rectangle {
2022         anchors.centerIn: parent
2023         width: 360
2024         height: 200
2025         radius: 14
2026         color: "#0f1420"
2027         border.color: "#1e2a3f"
2028         border.width: 1
2029
2030         Column {
2031             anchors.centerIn: parent
2032             spacing: 18
2033             width: parent.width - 40
2034
2035             Text {
2036                 width: parent.width
2037                 horizontalAlignment: Text.AlignHCenter
2038                 wrapMode: Text.WordWrap
2039                 text: txt("silinecekOnay")
2040                 color: "#dce8f5"
2041                 font.family: "Segoe UI"
2042                 font.pixelSize: 14
2043                 font.bold: true
2044             }
2045
2046             Row {
2047                 anchors.horizontalCenter: parent.horizontalCenter
2048                 spacing: 12
2049
2050                 Button {
2051                     width: 150
2052                     height: 42
2053                     text: txt("vazgecButon")
2054
2055                     onClicked: mesafeYenidenOnayGoster = false
2056
2057                     background: Rectangle {
2058                         radius: 8
2059                         color: parent.hovered ? "#1e2a3f" : "transparent"

```

```

2060         border.color: "#1e2a3f"
2061         border.width: 1
2062         Behavior on color { ColorAnimation { duration: 120 } }
2063     }
2064
2065     contentItem: Text {
2066         text: parent.text
2067         color: "#9ca3af"
2068         font: parent.font
2069         horizontalAlignment: Text.AlignHCenter
2070         verticalAlignment: Text.AlignVCenter
2071     }
2072 }
2073
2074 Button {
2075     width: 150
2076     height: 42
2077     text: txt("evetYenidenBaslaButon")
2078     font.bold: true
2079
2080     onClicked: {
2081         mesafeYenidenOnayGoster = false
2082         mesafeAraEkranGoster = false
2083         mesafeGoruntuleModu = false
2084         mesafeTamDuzenlemeModu = false
2085         mesafeNoktalari = []
2086         mesafeSayiSecildi = false
2087         mesafeDuzenlemeIndex = -1
2088     }
2089
2090     background: Rectangle {
2091         radius: 8
2092         color: parent.hovered ? "#b91c1c" : "#991b1b"
2093         Behavior on color { ColorAnimation { duration: 120 } }
2094     }
2095
2096     contentItem: Text {
2097         text: parent.text
2098         color: "#ffffff"
2099         font: parent.font
2100         horizontalAlignment: Text.AlignHCenter
2101         verticalAlignment: Text.AlignVCenter
2102     }
2103 }
2104 }
2105 }
2106 }
2107 }
2108
2109 Rectangle {
2110     anchors.fill: parent
2111     color: "#000000b3"
2112     visible: mesafeTamamlandiOnayGoster
2113     z: 200
2114
2115     MouseArea { anchors.fill: parent }
2116
2117     Rectangle {
2118         anchors.centerIn: parent
2119         width: 380
2120         height: 220
2121         radius: 14
2122         color: "#0f1420"
2123         border.color: "#16a34a"
2124         border.width: 1
2125
2126         Column {
2127             anchors.centerIn: parent
2128             spacing: 18
2129             width: parent.width - 40
2130
2131             Column {
2132                 width: parent.width
2133                 spacing: 6
2134
2135                 Text {
2136                     width: parent.width
2137                     horizontalAlignment: Text.AlignHCenter
2138                     text: txt("kalibrasyonTamamlandiBaslik")
2139                     color: "#4ade80"

```

```

2140         font.family: "Segoe UI"
2141         font.pixelSize: 17
2142         font.bold: true
2143     }
2144
2145     Text {
2146         width: parent.width
2147         horizontalAlignment: Text.AlignHCenter
2148         wrapMode: Text.WordWrap
2149         text: txt("toplamlOnEk") + mesafeToplamNokta + txt("noktaGirildiSonEk")
2150         color: "#9ca3af"
2151         font.family: "Segoe UI"
2152         font.pixelSize: 13
2153     }
2154 }
2155
2156 Row {
2157     anchors.horizontalCenter: parent.horizontalCenter
2158     spacing: 12
2159
2160     Button {
2161         width: 150
2162         height: 42
2163         text: txt("noktalarlKontrolEt")
2164
2165         onClicked: mesafeTamamlandiOnayGoster = false
2166
2167         background: Rectangle {
2168             radius: 8
2169             color: parent.hovered ? "#1e2a3f" : "transparent"
2170             border.color: "#1e2a3f"
2171             border.width: 1
2172             Behavior on color { ColorAnimation { duration: 120 } }
2173         }
2174
2175         contentItem: Text {
2176             text: parent.text
2177             color: "#9ca3af"
2178             font: parent.font
2179             horizontalAlignment: Text.AlignHCenter
2180             verticalAlignment: Text.AlignVCenter
2181         }
2182     }
2183
2184     Button {
2185         width: 150
2186         height: 42
2187         text: txt("kaydetVeBitir")
2188         font.bold: true
2189
2190         onClicked: {
2191             mesafeTamamlandiOnayGoster = false
2192             mesafeKaydetVeBildir()
2193         }
2194
2195         background: Rectangle {
2196             radius: 8
2197             color: parent.hovered ? "#22c55e" : "#16a34a"
2198             Behavior on color { ColorAnimation { duration: 120 } }
2199         }
2200
2201         contentItem: Text {
2202             text: parent.text
2203             color: "ffffff"
2204             font: parent.font
2205             horizontalAlignment: Text.AlignHCenter
2206             verticalAlignment: Text.AlignVCenter
2207         }
2208     }
2209 }
2210
2211 }
2212
2213
2214 Column {
2215     anchors.centerIn: parent
2216     spacing: 24
2217     width: 360
2218     visible: !mesafeSayiSecildi && !mesafeAraEkranGoster && !mesafeGoruntuleModu
2219

```

```

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: txt("kacNoktaSoru")
    color: "#dce8f5"
    font.family: "Segoe UI"
    font.pixelSize: 16
    font.bold: true
}

Text {
    width: parent.width
    horizontalAlignment: Text.AlignHCenter
    wrapMode: Text.WordWrap
    text: txt("dahaFazlaNokta")
    color: "#6b7280"
    font.family: "Segoe UI"
    font.pixelSize: 12
}

Grid {
    anchors.horizontalCenter: parent.horizontalCenter
    columns: 4
    spacing: 12

    Repeater {
        model: [4, 6, 8, 10]

        Rectangle {
            id: mesafeSecKarti
            width: 72
            height: 72
            radius: 12
            color: mesafeSecKarti.hovered ? "#241a38" : "#0f1420"
            border.color: "#9333ea"
            border.width: 1
            property bool hovered: false
            Behavior on color { ColorAnimation { duration: 120 } }

            Text {
                anchors.centerIn: parent
                text: modelData
                color: "#dce8f5"
                font.family: "Segoe UI"
                font.pixelSize: 26
                font.bold: true
            }

            MouseArea {
                anchors.fill: parent
                hoverEnabled: true
                cursorShape: Qt.PointingHandCursor
                onEntered: mesafeSecKarti.hovered = true
                onExited: mesafeSecKarti.hovered = false
                onClicked: {
                    mesafeToplamNokta = modelData
                    mesafeNoktalari = []
                    mesafeDuzenlemeIndex = -1
                    mesafeSayiSecildi = true
                }
            }
        }
    }
}

Row {
    anchors.centerIn: parent
    spacing: 24
    visible: mesafeSayiSecildi

    Rectangle {
        width: 340
        height: 500
        radius: 14
        color: "#0f1420"
        border.color: "#1e2a3f"
        border.width: 1
    }
}

```

```

2300 Column {
2301     anchors.centerIn: parent
2302     spacing: 16
2303     width: parent.width - 40
2304
2305     Rectangle {
2306         anchors.horizontalCenter: parent.horizontalCenter
2307         visible: mesafeOlcumKayitliVar && !mesafeTamDuzenlemeModu
2308         width: mesafeSonKaliMetni.implicitWidth + 20
2309         height: 24
2310         radius: 12
2311         color: "#0a0e17"
2312         border.color: "#1e2a3f"
2313         border.width: 1
2314
2315         Text {
2316             id: mesafeSonKaliMetni
2317             anchors.centerIn: parent
2318             text: txt("kayitliOnEk") + mesafeOlcumNoktaSayisi + txt("noktaSonEk")
2319             color: "#6b7280"
2320             font.pixelSize: 11
2321         }
2322     }
2323
2324     Text {
2325         anchors.horizontalCenter: parent.horizontalCenter
2326         text: mesafeTamDuzenlemeModu
2327             ? txt("kalibrasyonuDuzenle")
2328             : txt("noktaOnEk") + (mesafeNoktalari.length + 1) + " / " + mesafeToplamNokta
2329         color: "#9333ea"
2330         font.family: "Segoe UI"
2331         font.pixelSize: 13
2332         font.bold: true
2333         font.letterSpacing: 1
2334     }
2335
2336     Rectangle {
2337         width: parent.width
2338         height: 6
2339         radius: 3
2340         color: "#1e2a3f"
2341
2342         Rectangle {
2343             width: parent.width * (mesafeNoktalari.length / mesafeToplamNokta)
2344             height: parent.height
2345             radius: 3
2346             color: "#9333ea"
2347             Behavior on width { NumberAnimation { duration: 150 } }
2348         }
2349     }
2350
2351     Text {
2352         width: parent.width
2353         horizontalAlignment: Text.AlignHCenter
2354         wrapMode: Text.WordWrap
2355         text: mesafeDuzenlemeIndex >= 0
2356             ? txt("noktaOnEk") + (mesafeDuzenlemeIndex + 1) + txt("duzenleniyorSonEk")
2357             : (mesafeTamDuzenlemeModu
2358                 ? txt("duzenlemekIcinTikla")
2359                 : txt("hedefMesafeYonerge"))
2360         color: mesafeDuzenlemeIndex >= 0 ? "#9333ea" : "#9ca3af"
2361         font.family: "Segoe UI"
2362         font.bold: mesafeDuzenlemeIndex >= 0
2363         font.pixelSize: 13
2364     }
2365
2366     TextField {
2367         id: hedefCmKutusu
2368         anchors.horizontalCenter: parent.horizontalCenter
2369         width: 180
2370         height: 44
2371         placeholderText: "cm"
2372         placeholderTextColor: "#4b5563"
2373         color: "#ffffff"
2374         font.pixelSize: 16
2375         horizontalAlignment: TextInput.AlignHCenter
2376         validator: DoubleValidator { bottom: 0; top: 300; decimals: 1 }
2377         background: Rectangle {
2378             color: "#0a0e17"
2379             radius: 10

```

```

2380         border.color: hedefCmKutusu.activeFocus ? "#9333ea" : "#1e2a3f"
2381         border.width: 1
2382     }
2383 }
2384
2385 Rectangle {
2386     anchors.horizontalCenter: parent.horizontalCenter
2387     width: parent.width
2388     height: 64
2389     radius: 12
2390     color: "#132335"
2391     border.color: wifiManager.baglandi ? "#9333ea" : "#dc2626"
2392     border.width: 1
2393     Behavior on border.color { ColorAnimation { duration: 120 } }
2394
2395     Text {
2396         anchors.centerIn: parent
2397         text: txt("hamDegerOnEk") + sensorManager.hamMesafe.toFixed(0)
2398         color: wifiManager.baglandi ? "#c084fc" : "#f87171"
2399         font.family: "Segoe UI"
2400         font.pixelSize: 20
2401         font.bold: true
2402     }
2403 }
2404
2405 Text {
2406     width: parent.width
2407     horizontalAlignment: Text.AlignHCenter
2408     wrapMode: Text.WordWrap
2409     visible: !wifiManager.baglandi
2410     text: txt("hamDegerSifirUyari")
2411     color: "#f87171"
2412     font.family: "Segoe UI"
2413     font.pixelSize: 11
2414 }
2415
2416 Button {
2417     id: mesafeNoktaKaydetButonu
2418     width: parent.width
2419     height: 46
2420     property bool tumNoktalarGirildi: !mesafeTamDuzenlemeModu && mesafeDuzenlemeIndex < 0 &&
mesafeNoktalari.length >= mesafeToplamNokta
2421     text: mesafeDuzenlemeIndex >= 0
2422         ? txt("noktayiGuncelle")
2423         : (tumNoktalarGirildi
2424             ? txt("kalibrasyonuTamamla")
2425             : (mesafeNoktalari.length >= mesafeToplamNokta - 1 ? txt("sonNoktayiKaydet") :
txt("buNoktayiKaydet")))
2426     font.pixelSize: 14
2427     font.bold: true
2428     enabled: tumNoktalarGirildi || (hedefCmKutusu.text.length > 0 && (mesafeDuzenlemeIndex >= 0 ||
mesafeNoktalari.length < mesafeToplamNokta))
2429
2430     onClicked: {
2431         if (tumNoktalarGirildi) {
2432             mesafeTamamlandiOnayGoster = true
2433             return
2434         }
2435
2436         var yeniNokta = { hedefMm: parseFloat(hedefCmKutusu.text) * 10, hamDeger: sensorManager.hamMesafe
}
2437
2438         if (mesafeDuzenlemeIndex >= 0) {
2439             // Var olan bir noktayı yerinde güncelliyoruz, diğer noktalar etkilenmiyor
2440             var guncelDizi = mesafeNoktalari.slice()
2441             guncelDizi[mesafeDuzenlemeIndex] = yeniNokta
2442             mesafeNoktalari = guncelDizi
2443             mesafeDuzenlemeIndex = -1
2444             hedefCmKutusu.text = ""
2445             return
2446         }
2447
2448         var yeniDizi = mesafeNoktalari.slice()
2449         yeniDizi.push(yeniNokta)
2450         mesafeNoktalari = yeniDizi
2451         hedefCmKutusu.text = ""
2452
2453         if (mesafeNoktalari.length >= mesafeToplamNokta) {
2454             mesafeTamamlandiOnayGoster = true
2455         }

```

```

2456     }
2457
2458     background: Rectangle {
2459         radius: 10
2460         color: parent.enabled ? (parent.hovered ? "#a855f7" : "#9333ea") : "#1e2a3f"
2461         Behavior on color { ColorAnimation { duration: 120 } }
2462     }
2463
2464     contentItem: Text {
2465         text: parent.text
2466         color: "#ffffff"
2467         font: parent.font
2468         horizontalAlignment: Text.AlignHCenter
2469         verticalAlignment: Text.AlignVCenter
2470     }
2471 }
2472
2473 Button {
2474     width: parent.width
2475     height: 46
2476     visible: mesafeTamDuzenlemeModu
2477     text: txt("degisiklikleriKaydet")
2478     font.pixelSize: 14
2479     font.bold: true
2480
2481     onClicked: mesafeKaydetVeBildir()
2482
2483     background: Rectangle {
2484         radius: 10
2485         color: parent.hovered ? "#22c55e" : "#16a34a"
2486         Behavior on color { ColorAnimation { duration: 120 } }
2487     }
2488
2489     contentItem: Text {
2490         text: parent.text
2491         color: "#ffffff"
2492         font: parent.font
2493         horizontalAlignment: Text.AlignHCenter
2494         verticalAlignment: Text.AlignVCenter
2495     }
2496 }
2497
2498 Button {
2499     width: parent.width
2500     height: 38
2501     visible: mesafeDuzenlemeIndex >= 0
2502     text: txt("vazgecButon")
2503     font.pixelSize: 13
2504
2505     onClicked: {
2506         mesafeDuzenlemeIndex = -1
2507         hedefCmKutusu.text = ""
2508     }
2509
2510     background: Rectangle {
2511         radius: 8
2512         color: parent.hovered ? "#1e2a3f" : "transparent"
2513         border.color: "#1e2a3f"
2514         border.width: 1
2515         Behavior on color { ColorAnimation { duration: 120 } }
2516     }
2517
2518     contentItem: Text {
2519         text: parent.text
2520         color: "#9ca3af"
2521         font: parent.font
2522         horizontalAlignment: Text.AlignHCenter
2523         verticalAlignment: Text.AlignVCenter
2524     }
2525 }
2526 }
2527
2528
2529 Rectangle {
2530     width: 300
2531     height: 500
2532     radius: 14
2533     color: "#0f1420"
2534     border.color: "#1e2a3f"
2535     border.width: 1

```

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2613
2614
2615

Column {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.margins: 20
    spacing: 4

    Text {
        text: txt("kaydedilenNoktalarBaslik")
        color: "#6b7280"
        font.family: "Segoe UI"
        font.pixelSize: 11
        font.bold: true
        font.letterSpacing: 1
    }

    Text {
        text: txt("duzeltmekIcinTikla")
        color: "#4b5563"
        font.family: "Segoe UI"
        font.pixelSize: 10
    }
}

ListView {
    anchors.top: parent.top
    anchors.left: parent.left
    anchors.right: parent.right
    anchors.bottom: parent.bottom
    anchors.margins: 16
    anchors.topMargin: 60
    clip: true
    spacing: 8

    model: mesafeNoktalarI

    delegate: Rectangle {
        width: ListView.view.width
        height: 48
        radius: 8
        color: mesafeSatirAlani.containsMouse ? "#241a38" : "#0a0e17"
        border.color: index === mesafeDuzenlemeIndex ? "#9333ea" : "#1e2a3f"
        border.width: index === mesafeDuzenlemeIndex ? 2 : 1
        Behavior on color { ColorAnimation { duration: 120 } }

        Row {
            anchors.fill: parent
            anchors.leftMargin: 12
            anchors.rightMargin: 12
            spacing: 8

            Rectangle {
                width: 22
                height: 22
                radius: 11
                anchors.verticalCenter: parent.verticalCenter
                color: "#241a38"
                border.color: "#9333ea"
                border.width: 1

                Text {
                    anchors.centerIn: parent
                    text: index + 1
                    color: "#c084fc"
                    font.pixelSize: 10
                    font.bold: true
                }
            }
        }

        Column {
            anchors.verticalCenter: parent.verticalCenter
            spacing: 1

            Text {
                text: (modelData.hedefMm / 10).toFixed(1) + " cm"
                color: "#dce8f5"
                font.family: "Segoe UI"
                font.pixelSize: 13
                font.bold: true
            }
        }
    }
}

```



```

2616         }
2617
2618         Text {
2619             text: "ADC: " + modelData.hamDeger.toFixed(0)
2620             color: "#6b7280"
2621             font.family: "Segoe UI"
2622             font.pixelSize: 10
2623         }
2624     }
2625 }
2626
2627 Text {
2628     anchors.right: parent.right
2629     anchors.verticalCenter: parent.verticalCenter
2630     anchors.rightMargin: 12
2631     text: "↖"
2632     color: "#6b7280"
2633     font.pixelSize: 13
2634     visible: mesafeSatirAlani.containsMouse
2635 }
2636
2637 MouseArea {
2638     id: mesafeSatirAlani
2639     anchors.fill: parent
2640     hoverEnabled: true
2641     cursorShape: Qt.PointingHandCursor
2642     onClicked: {
2643         hedefCmKutusu.text = (modelData.hedefMm / 10).toString()
2644         mesafeDuzenlemeIndex = index
2645     }
2646 }
2647
2648     }
2649 }
2650
2651 }
2652
2653 Rectangle {
2654     id: mesafeBildirimKutusu
2655     anchors.top: parent.top
2656     anchors.horizontalCenter: parent.horizontalCenter
2657     anchors.topMargin: 12
2658     z: 100
2659     width: mesafeBildirimMetni.implicitWidth + 24
2660     height: 32
2661     radius: 16
2662     color: mesafeBildirimBasarili ? "#123321" : "#3a1414"
2663     opacity: mesafeBildirimGoster ? 1 : 0
2664     visible: opacity > 0
2665     Behavior on opacity { NumberAnimation { duration: 200 } }
2666
2667     Text {
2668         id: mesafeBildirimMetni
2669         anchors.centerIn: parent
2670         text: mesafeBildirimMesaj
2671         color: mesafeBildirimBasarili ? "#4ade80" : "#f87171"
2672         font.family: "Segoe UI"
2673         font.pixelSize: 12
2674         font.bold: true
2675     }
2676 }
2677
2678     }
2679 }
2680 }

```

```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import sliper
4
5 Rectangle {
6     id: sayfa
7
8     property bool sifreGorunur: false
9     property bool beniHatirla: false
10
11     signal girisBasarili()
12
13     readonly property var metinler: ({
14         girisBasligi: { tr: "Sisteme Giriş", en: "System Login" },
15         girisAltBaslik: { tr: "Yetkili hesabınızla devam edin", en: "Continue with your authorized account" },
16         kullanıcıAdiEtiket: { tr: "KULLANICI ADI", en: "USERNAME" },
17         kullanıcıAdiPlaceholder: { tr: "kullanici_adi", en: "username" },
18         sifreEtiket: { tr: "ŞİFRE", en: "PASSWORD" },
19         girisYapButon: { tr: "Giriş Yap", en: "Log In" },
20         telifHakki: { tr: "© 2026 Liya Test A.Ş. SLIPER Analiz Yazılımı", en: "© 2026 Liya Test A.Ş. SLIPER Analysis Software" }
21     })
22
23     function txt(anahtar) {
24         return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
25     }
26
27     gradient: Gradient {
28         GradientStop { position: 0.0; color: "#101c33" }
29         GradientStop { position: 1.0; color: "#0a1220" }
30     }
31
32     Row {
33         anchors.top: parent.top
34         anchors.right: parent.right
35         anchors.margins: 24
36         spacing: 8
37
38         Rectangle {
39             width: 44
40             height: 30
41             radius: 6
42             color: "#0f1420"
43             border.color: Translations.turkish ? "#3b82f6" : "#1e2a3f"
44             border.width: 1
45
46             Text {
47                 anchors.centerIn: parent
48                 text: "TR"
49                 color: Translations.turkish ? "ffffff" : "#6b7280"
50                 font.pixelSize: 12
51                 font.bold: Translations.turkish
52             }
53
54             MouseArea {
55                 anchors.fill: parent
56                 cursorShape: Qt.PointingHandCursor
57                 onClicked: Translations.turkish = true
58             }
59         }
60
61         Rectangle {
62             width: 44
63             height: 30
64             radius: 6
65             color: "#0f1420"
66             border.color: !Translations.turkish ? "#3b82f6" : "#1e2a3f"
67             border.width: 1
68
69             Text {
70                 anchors.centerIn: parent
71                 text: "EN"
72                 color: !Translations.turkish ? "ffffff" : "#6b7280"
73                 font.pixelSize: 12
74                 font.bold: !Translations.turkish
75             }
76
77             MouseArea {

```

```

78         anchors.fill: parent
79         cursorShape: Qt.PointingHandCursor
80         onClicked: Translations.turkish = false
81     }
82 }
83 }
84
85 Rectangle {
86     anchors.centerIn: kart
87     width: kart.width + 48
88     height: kart.height + 48
89     radius: 28
90     color: "#18000000"
91 }
92
93 Rectangle {
94     anchors.centerIn: kart
95     width: kart.width + 32
96     height: kart.height + 32
97     radius: 24
98     color: "#25000000"
99 }
100
101 Rectangle {
102     anchors.centerIn: kart
103     width: kart.width + 18
104     height: kart.height + 18
105     radius: 20
106     color: "#35000000"
107 }
108
109 Rectangle {
110     anchors.centerIn: kart
111     width: kart.width + 8
112     height: kart.height + 8
113     radius: 18
114     color: "#45000000"
115 }
116
117 Rectangle {
118     id: kart
119     anchors.centerIn: parent
120     width: 400
121     color: "#0f1420"
122     radius: 16
123     border.color: "#4b5563"
124     border.width: 1
125     height: govde.implicitHeight + 48
126
127     Column {
128         id: govde
129         anchors.top: parent.top
130         anchors.horizontalCenter: parent.horizontalCenter
131         anchors.topMargin: 32
132         width: parent.width - 48
133         spacing: 20
134
135         Image {
136             anchors.horizontalCenter: parent.horizontalCenter
137             source: "../assets/logo.png"
138             width: 144
139             height: 144
140             fillMode: Image.PreserveAspectFit
141         }
142
143         Rectangle {
144             width: parent.width
145             height: 1
146             color: "#1e2a3f"
147         }
148
149         Column {
150             width: parent.width
151             spacing: 4
152
153             Text {
154                 anchors.horizontalCenter: parent.horizontalCenter
155                 text: txt("girisBasligi")
156                 color: "#ffffff"
157                 font.family: "Segoe UI"

```

```

158         font.pixelSize: 22
159         font.bold: true
160     }
161
162     Text {
163         anchors.horizontalCenter: parent.horizontalCenter
164         text: txt("girisAltBaslik")
165         color: "#6b7280"
166         font.family: "Segoe UI"
167         font.pixelSize: 12
168     }
169 }
170
171 Column {
172     width: parent.width
173     spacing: 6
174
175     Text {
176         text: txt("kullaniciAdiEtiket")
177         color: "#6b7280"
178         font.family: "Segoe UI"
179         font.pixelSize: 10
180         font.letterSpacing: 1
181     }
182
183     TextField {
184         id: kullaniciAdiKutusu
185         width: parent.width
186         height: 42
187         placeholderText: txt("kullaniciAdiPlaceholder")
188         placeholderTextColor: "#4b5563"
189         color: "ffffff"
190         font.pixelSize: 13
191         leftPadding: 12
192         verticalAlignment: TextInput.AlignVCenter
193         background: Rectangle {
194             color: "#0a0e17"
195             radius: 6
196             border.color: kullaniciAdiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
197             border.width: 1
198         }
199     }
200 }
201
202 Column {
203     width: parent.width
204     spacing: 6
205
206     Text {
207         text: txt("sifreEtiket")
208         color: "#6b7280"
209         font.family: "Segoe UI"
210         font.pixelSize: 10
211         font.letterSpacing: 1
212     }
213
214     TextField {
215         id: sifreKutusu
216         width: parent.width
217         height: 42
218         placeholderText: "*****"
219         placeholderTextColor: "#4b5563"
220         echoMode: sayfa.sifreGorunur ? TextInput.Normal : TextInput.Password
221         color: "ffffff"
222         font.pixelSize: 13
223         leftPadding: 12
224         rightPadding: 48
225         verticalAlignment: TextInput.AlignVCenter
226         background: Rectangle {
227             color: "#0a0e17"
228             radius: 6
229             border.color: sifreKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
230             border.width: 1
231         }
232     }
233
234     Rectangle {
235         id: sifreGosterButonu
236         anchors.right: parent.right
237         anchors.verticalCenter: parent.verticalCenter
238         anchors.rightMargin: 8

```

```

width: 28
height: 28
radius: 6
color: sifreGosterMouse.pressed ? "#162033" : (sifreGosterMouse.containsMouse ? "#111827" : "transparent")
border.color: sayfa.sifreGorunur ? "#334155" : "transparent"
border.width: 1

property color ikonRengi: sifreGosterMouse.containsMouse || sayfa.sifreGorunur ? "#cbd5e1" : "#6b7280"

Canvas {
    id: sifreIkonu
    anchors.centerIn: parent
    width: 18
    height: 18

    onPaint: {
        var ctx = getContext("2d")
        ctx.clearRect(0, 0, width, height)
        ctx.lineWidth = 1.7
        ctx.lineCap = "round"
        ctx.lineJoin = "round"
        ctx.strokeStyle = sifreGosterButonu.ikonRengi

        ctx.beginPath()
        ctx.moveTo(1.5, 9)
        ctx.bezierCurveTo(4.2, 4.8, 7.0, 3.8, 9, 3.8)
        ctx.bezierCurveTo(11.0, 3.8, 13.8, 4.8, 16.5, 9)
        ctx.bezierCurveTo(13.8, 13.2, 11.0, 14.2, 9, 14.2)
        ctx.bezierCurveTo(7.0, 14.2, 4.2, 13.2, 1.5, 9)
        ctx.stroke()

        ctx.beginPath()
        ctx.arc(9, 9, 2.5, 0, Math.PI * 2, false)
        ctx.stroke()

        if (sayfa.sifreGorunur) {
            ctx.beginPath()
            ctx.moveTo(3, 15)
            ctx.lineTo(15, 3)
            ctx.stroke()
        }
    }

    Connections {
        target: sayfa
        function onSifreGorunurChanged() {
            sifreIkonu.requestPaint()
        }
    }

    Connections {
        target: sifreGosterButonu
        function onIkonRengiChanged() {
            sifreIkonu.requestPaint()
        }
    }
}

MouseArea {
    id: sifreGosterMouse
    anchors.fill: parent
    hoverEnabled: true
    cursorShape: Qt.PointingHandCursor
    onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
}

Text {
    anchors.right: parent.right
    anchors.verticalCenter: parent.verticalCenter
    anchors.rightMargin: 12
    text: sayfa.sifreGorunur ? "👁" : "👁"
    color: "#6b7280"
    font.pixelSize: 14
    visible: false

    MouseArea {
        anchors.fill: parent
        anchors.margins: -8
        cursorShape: Qt.PointingHandCursor
    }
}

```

```

318         onClicked: sayfa.sifreGorunur = !sayfa.sifreGorunur
319     }
320 }
321 }
322 }
323
324 Button {
325     width: parent.width
326     height: 44
327     text: txt("girisYapButon")
328     font.family: "Segoe UI"
329     font.pixelSize: 14
330     font.bold: true
331
332     onClicked: {
333         var sonuc = backend.girisYap(kullaniciAdiKutusu.text, sifreKutusu.text)
334
335         if (sonuc) {
336             sayfa.girisBasarili()
337         } else {
338             console.log("Kullanici adi veya sifre hatali")
339         }
340     }
341
342     background: Rectangle {
343         radius: 8
344         color: parent.pressed ? "#2563eb" : (parent.hovered ? "#4f8cf7" : "#3b82f6")
345         Behavior on color { ColorAnimation { duration: 120 } }
346     }
347
348     contentItem: Text {
349         text: parent.text
350         color: "#ffffff"
351         font: parent.font
352         horizontalAlignment: Text.AlignHCenter
353         verticalAlignment: Text.AlignVCenter
354     }
355 }
356
357 Rectangle {
358     width: parent.width
359     height: 1
360     color: "#1e2a3f"
361 }
362
363 Text {
364     anchors.horizontalCenter: parent.horizontalCenter
365     text: txt("telifHakki")
366     color: "#4b5563"
367     font.family: "Segoe UI"
368     font.pixelSize: 10
369 }
370 }
371 }
372 }
373

```

```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtCharts 6.7
4 import sliper
5
6 Rectangle {
7     color: "#0a0e17"
8
9     readonly property var metinler: ({
10         testBilgileri: { tr: "TEST BİLGİLERİ", en: "TEST INFORMATION" },
11         musteri: { tr: "Müşteri", en: "Customer" },
12         musteriPlaceholder: { tr: "Musteri adi girin...", en: "Enter customer name..." },
13         betonRecetesi: { tr: "Beton Reçetesi", en: "Concrete Mix Design" },
14         receteSeciniz: { tr: "Reçete Seçin...", en: "Select Mix Design..." },
15         cimentoEtiket: { tr: "Çimento ", en: "Cement " },
16         suCimentoEtiket: { tr: " • Su/Çimento ", en: " • Water/Cement " },
17         eklenenAgirlik: { tr: "Eklenen Ağırlık (kg)", en: "Added Weight (kg)" },
18         geriButon: { tr: "- Geri", en: "- Undo" },
19         sifirlaButon: { tr: "Ü Sıfırla", en: "Ü Reset" },
20         olcumuBaslatButon: { tr: "► Ölçümü Başlat", en: "► Start Measurement" },
21         devamEtButon: { tr: "► Devam Et", en: "► Resume" },
22         duraklatButon: { tr: "|| Duraklat", en: "|| Pause" },
23         bitirButon: { tr: "■ Bitir", en: "■ Finish" },
24         veriYok: { tr: "VERİ YOK", en: "NO DATA" },
25         duraklatildiDurum: { tr: "DURAKLATILDI", en: "PAUSED" },
26         anlikDegerler: { tr: "ANLIK DEĞERLER", en: "CURRENT VALUES" },
27         basincEtiket: { tr: "BASINÇ", en: "PRESSURE" },
28         konumEtiket: { tr: "KONUM", en: "POSITION" },
29         hizEtiket: { tr: "HIZ", en: "SPEED" },
30         debiEtiket: { tr: "DEBİ", en: "FLOW RATE" },
31         basincZaman: { tr: "Basınç - Zaman", en: "Pressure - Time" },
32         konumZaman: { tr: "Konum - Zaman", en: "Position - Time" },
33         hizZaman: { tr: "Hız - Zaman", en: "Speed - Time" },
34         debiZaman: { tr: "Debi - Zaman", en: "Flow Rate - Time" },
35         sesliDinleniyor: { tr: "🔊 Dinleniyor – \"Başlat / Durdur / Devam Et / Bitir / Ekle\"", en: "🔊 Listening – \"Start / Stop / Resume / Finish / Add\"" },
36         sesliHazirlaniyor: { tr: "🔊 Sesli komut hazırlanıyor...", en: "🔊 Preparing voice commands..." },
37         duyulanEtiket: { tr: "Duyulan: ", en: "Heard: " },
38         bitirOnaySorusu: { tr: "Ölçümü bitirmek istediğinize emin misiniz?", en: "Are you sure you want to finish the measurement?" },
39         kaydetVeBitir: { tr: "💾 Kaydet ve Bitir", en: "💾 Save and Finish" },
40         testiSilVeBitir: { tr: "🗑️ Testi Sil ve Bitir", en: "🗑️ Delete Test and Finish" },
41         vazgecTesteDevam: { tr: "Vazgeç, Teste Devam Et", en: "Cancel, Continue Testing" },
42         uyariMusteriAdi: { tr: "Lütfen müşteri adını girin.", en: "Please enter the customer name." },
43         uyariReceteSec: { tr: "Lütfen bir beton reçetesi seçin.", en: "Please select a concrete mix design." },
44         uyariCihazBagliDegil: { tr: "Cihaz bağlı değil. Lütfen önce sol alttan SLIPER-ESP32'ye bağlanın.", en: "Device not connected. Please connect to SLIPER-ESP32 from the bottom left first." },
45         uyariAktifOlcumYok: { tr: "Aktif bir ölçüm yok. Önce ölçümü başlatın.", en: "No active measurement. Please start a measurement first." }
46     })
47
48     function txt(anahtar) {
49         return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
50     }
51
52     property real zamanSayaci: 0
53     property int aktifOlcumId: -1
54     property string uyariMesaji: ""
55     property bool bitirIcinDuraklatildi: false
56
57     property var basincGecmis: []
58     property var konumGecmis: []
59     property var hizGecmis: []
60     property var debiGecmis: []
61
62     readonly property real agirlikBirim: 1.6
63     property int agirlikAdedi: 0
64     readonly property real agirlikToplam: agirlikAdedi * agirlikBirim
65
66     // TS EN 206 / TS 13515 normal beton dayanım sınıfları (C sınıfı) – çimento dozajı
67     // ve azami su/çimento oranı, ilgili sınıfın minimum bağlayıcı gereksinimine göre.
68     readonly property var receteListesi: [
69         { sinifTr: "Reçete Seçin...", sinifEn: "Select Mix Design...", cimentoTr: "", cimentoEn: "", suCimentoTr: "", suCimentoEn: "",
70           aciklamaTr: "", aciklamaEn: "" },
71         { sinifTr: "C16/20", sinifEn: "C16/20", cimentoTr: "min. 260 kg/m³", cimentoEn: "min. 260 kg/m³", suCimentoTr: "maks. 0.65",
72           suCimentoEn: "max. 0.65", aciklamaTr: "Hafif yükte yalın/donatılı beton", aciklamaEn: "Plain/reinforced concrete for light loads" },
73         { sinifTr: "C20/25", sinifEn: "C20/25", cimentoTr: "min. 280 kg/m³", cimentoEn: "min. 280 kg/m³", suCimentoTr: "maks. 0.60",
74           suCimentoEn: "max. 0.60", aciklamaTr: "Standart betonarme", aciklamaEn: "Standard reinforced concrete" },
75         { sinifTr: "C25/30", sinifEn: "C25/30", cimentoTr: "min. 300 kg/m³", cimentoEn: "min. 300 kg/m³", suCimentoTr: "maks. 0.55",

```

```

suCimentoEn: "max. 0.55", aciklamaTr: "Standart betonarme (TS EN 206)", aciklamaEn: "Standard reinforced concrete (TS EN 206)" },
72     { sinifTr: "C30/37", sinifEn: "C30/37", cimentoTr: "min. 320 kg/m³", cimentoEn: "min. 320 kg/m³", suCimentoTr: "maks. 0.50",
suCimentoEn: "max. 0.50", aciklamaTr: "Standart betonarme", aciklamaEn: "Standard reinforced concrete" },
73     { sinifTr: "C30/37 SCC", sinifEn: "C30/37 SCC", cimentoTr: "min. 380 kg/m³", cimentoEn: "min. 380 kg/m³", suCimentoTr: "maks.
0.45", suCimentoEn: "max. 0.45", aciklamaTr: "Kendiliğinden yerleşen beton (SF2)", aciklamaEn: "Self-compacting concrete (SF2)" },
74     { sinifTr: "C35/45", sinifEn: "C35/45", cimentoTr: "min. 340 kg/m³", cimentoEn: "min. 340 kg/m³", suCimentoTr: "maks. 0.45",
suCimentoEn: "max. 0.45", aciklamaTr: "Orta-yüksek dayanım", aciklamaEn: "Medium-high strength" },
75     { sinifTr: "C40/50", sinifEn: "C40/50", cimentoTr: "min. 360 kg/m³", cimentoEn: "min. 360 kg/m³", suCimentoTr: "maks. 0.40",
suCimentoEn: "max. 0.40", aciklamaTr: "Yüksek dayanım", aciklamaEn: "High strength" },
76     { sinifTr: "C45/55", sinifEn: "C45/55", cimentoTr: "min. 380 kg/m³", cimentoEn: "min. 380 kg/m³", suCimentoTr: "maks. 0.38",
suCimentoEn: "max. 0.38", aciklamaTr: "Yüksek dayanım", aciklamaEn: "High strength" },
77     { sinifTr: "C50/60", sinifEn: "C50/60", cimentoTr: "min. 400 kg/m³", cimentoEn: "min. 400 kg/m³", suCimentoTr: "maks. 0.35",
suCimentoEn: "max. 0.35", aciklamaTr: "Yüksek dayanım (TS EN 206 normal beton sınırı)", aciklamaEn: "High strength (TS EN 206 normal concrete
limit)" }
78     ]
79
80     signal olcumTamamlandi(int id)
81     signal agirlikEklendi()
82
83     function resetMeasurementScreen() {
84         aktifOlcumId = -1
85         uyariMesaji = ""
86         bitirIcinDuraklatildi = false
87         musterikutusu.text = ""
88         receteKutusu.currentIndex = 0
89         agirlikAdedi = 0
90         calculator.sifirla()
91         grafikleriSifirla()
92     }
93
94     function baslatOlcumu() {
95         if (aktifOlcumId > 0) return
96
97         if (musterikutusu.text.trim().length === 0) {
98             uyariMesaji = txt("uyariMusteriAdi")
99             return
100         }
101
102         if (receteKutusu.currentIndex <= 0) {
103             uyariMesaji = txt("uyariReceteSec")
104             return
105         }
106
107         if (!sensorManager.veriGecerli) {
108             console.warn("Gerçek sensor verisi yok, olcum baslatilmadi.")
109             uyariMesaji = txt("uyariCihazBagliDegil")
110             return
111         }
112
113         uyariMesaji = ""
114         grafikleriSifirla()
115
116         calculator.sifirla()
117         aktifOlcumId = database.olcumBaslat(
118             musterikutusu.text,
119             receteKutusu.currentText,
120             agirlikToplam
121         )
122         console.log("Aktif olcum id:", aktifOlcumId)
123     }
124
125     function bitirTalebiGoster() {
126         if (aktifOlcumId > 0) {
127             if (!calculator.duraklatildi) {
128                 calculator.duraklat()
129                 bitirIcinDuraklatildi = true
130             }
131             bitirOnayPopup.open()
132         } else {
133             uyariMesaji = txt("uyariAktifOlcumYok")
134         }
135     }
136
137     function grafikleriSifirla() {
138         zamanSayaci = 0
139         basincSerisi.clear()
140         konumSerisi.clear()
141         hizSerisi.clear()
142         debiSerisi.clear()
143         basincGecmis = []

```



```

144     konumGecmis = []
145     hizGecmis = []
146     debiGecmis = []
147     xEkseni.min = 0
148     xEkseni.max = 60
149     yEkseni.min = 0
150     yEkseni.max = 1100
151     konumXEkseni.min = 0
152     konumXEkseni.max = 60
153     konumYEkseni.max = 500
154     hizXEkseni.min = 0
155     hizXEkseni.max = 60
156     hizYEkseni.max = 4
157     debiXEkseni.min = 0
158     debiXEkseni.max = 60
159     debiYEkseni.max = 180
160 }
161
162 function eksenTavaniHesapla(dizi, tavan, taban) {
163     if (dizi.length === 0) return taban
164     var maxDeger = Math.max.apply(null, dizi)
165     var hedef = maxDeger * 1.25
166     if (hedef < taban) hedef = taban
167     if (hedef > tavan) hedef = tavan
168     return hedef
169 }
170
171 // Basınc (mbar) atmosferik seviyede sabit bir taban etrafında dalgalanır,
172 // bu yüzden sabit tavanlı ölçekleme yerine veriye göre kayan min/max kullanılır.
173 function basincEkseniHesapla(dizi) {
174     if (dizi.length === 0) return { min: 0, max: 1100 }
175     var maxDeger = Math.max.apply(null, dizi)
176     var minDeger = Math.min.apply(null, dizi)
177     var pad = Math.max((maxDeger - minDeger) * 0.2, maxDeger * 0.02, 5)
178     return { min: Math.max(0, minDeger - pad), max: maxDeger + pad }
179 }
180
181 Connections {
182     target: sensorManager
183     function onVeriGuncellendi() {
184         if (!sensorManager.veriGecerli || aktifOlcumId <= 0 || calculator.duraklatildi) return
185
186         zamanSayaci += 0.2
187
188         basincSerisi.append(zamanSayaci, sensorManager.basinc)
189         konumSerisi.append(zamanSayaci, sensorManager.konum)
190         hizSerisi.append(zamanSayaci, sensorManager.hiz)
191         debiSerisi.append(zamanSayaci, sensorManager.debi)
192
193         basincGecmis.push(sensorManager.basinc)
194         konumGecmis.push(sensorManager.konum)
195         hizGecmis.push(sensorManager.hiz)
196         debiGecmis.push(sensorManager.debi)
197
198         if (zamanSayaci > 60) {
199             basincSerisi.remove(0)
200             konumSerisi.remove(0)
201             hizSerisi.remove(0)
202             debiSerisi.remove(0)
203
204             basincGecmis.shift()
205             konumGecmis.shift()
206             hizGecmis.shift()
207             debiGecmis.shift()
208
209             xEkseni.min = zamanSayaci - 60
210             xEkseni.max = zamanSayaci
211             konumXEkseni.min = zamanSayaci - 60
212             konumXEkseni.max = zamanSayaci
213             hizXEkseni.min = zamanSayaci - 60
214             hizXEkseni.max = zamanSayaci
215             debiXEkseni.min = zamanSayaci - 60
216             debiXEkseni.max = zamanSayaci
217         }
218
219         var basincAralik = basincEkseniHesapla(basincGecmis)
220         yEkseni.min = basincAralik.min
221         yEkseni.max = basincAralik.max
222         konumYEkseni.max = eksenTavaniHesapla(konumGecmis, 500, 50)
223         hizYEkseni.max = eksenTavaniHesapla(hizGecmis, 4, 0.5)

```

```

224         debiYEksenini.max = eksenTavaniHesapla(debiGecmis, 180, 10)
225
226         calculator.konumGuncelle(sensorManager.konum, sensorManager.hiz)
227     }
228 }
229
230 // Sesli komutlar: "Başlat / Durdur / Bitir". Bu sayfa arka planda
231 // (StackLayout icinde) da yasadigi icin, komutlar sadece Ölçüm sayfası
232 // ekranda goruntulenirken (visible) uygulanir.
233 Connections {
234     target: voiceCommandManager
235
236     function onBaslatKomutu() {
237         if (!visible) return
238         baslatOlcumu()
239     }
240
241     function onDurdurKomutu() {
242         if (!visible || aktifOlcumId <= 0) return
243         if (!calculator.duraklatildi) calculator.duraklat()
244     }
245
246     function onDevamKomutu() {
247         if (!visible || aktifOlcumId <= 0) return
248         if (calculator.duraklatildi) calculator.devamEt()
249     }
250
251     function onBitirKomutu() {
252         if (!visible) return
253         bitirTalebiGoster()
254     }
255
256     function onEkleKomutu() {
257         if (!visible) return
258         agirlikAdedi += 1
259         agirlikEklendi()
260     }
261 }
262
263 Row {
264     anchors.fill: parent
265     spacing: 0
266
267     Rectangle {
268         id: testBilgileriPaneli
269         width: 280
270         height: parent.height
271         color: "#0f1420"
272
273         Text {
274             anchors.top: parent.top
275             anchors.left: parent.left
276             anchors.margins: 20
277             text: txt("testBilgileri")
278             color: "#6b7280"
279             font.family: "Segoe UI"
280             font.pixelSize: 12
281             font.bold: true
282             font.letterSpacing: 1
283         }
284
285         Column {
286             id: formAlanlari
287             anchors.top: parent.top
288             anchors.left: parent.left
289             anchors.right: parent.right
290             anchors.margins: 20
291             anchors.topMargin: 60
292             spacing: 6
293
294             Text {
295                 text: txt("musteri")
296                 color: "#9ca3af"
297                 font.family: "Segoe UI"
298                 font.pixelSize: 12
299             }
300
301             TextField {
302                 id: musterikutusu
303                 width: parent.width

```

```

304         height: 38
305         placeholderText: txt("musteriPlaceholder")
306         placeholderTextColor: "#4b5563"
307         color: "#ffffff"
308         font.pixelSize: 13
309         leftPadding: 10
310         verticalAlignment: TextInput.AlignVCenter
311         background: Rectangle {
312             color: "#0a0e17"
313             radius: 6
314             border.color: musterikutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
315             border.width: 1
316         }
317     }
318
319     Text {
320         text: txt("betonRecetesi")
321         color: "#9ca3af"
322         font.family: "Segoe UI"
323         font.pixelSize: 12
324     }
325
326     ComboBox {
327         id: receteKutusu
328         width: parent.width
329         height: 38
330         model: receteListesi
331         textRole: Translations.turkish ? "sinifTr" : "sinifEn"
332
333         background: Rectangle {
334             color: "#0a0e17"
335             radius: 6
336             border.color: receteKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
337             border.width: 1
338         }
339
340         contentItem: Text {
341             text: receteKutusu.displayText
342             color: "#9ca3af"
343             font.pixelSize: 13
344             leftPadding: 10
345             verticalAlignment: Text.AlignVCenter
346         }
347
348         delegate: ItemDelegate {
349             id: receteDelege
350             width: receteKutusu.width
351             height: modelData.cimentoTr.length > 0 ? 52 : 36
352             highlighted: receteKutusu.highlightedIndex === index
353
354             background: Rectangle {
355                 color: receteDelege.highlighted ? "#182644" : "#0a0e17"
356             }
357
358             contentItem: Column {
359                 anchors.verticalCenter: parent.verticalCenter
360                 anchors.left: parent.left
361                 anchors.leftMargin: 10
362                 anchors.right: parent.right
363                 anchors.rightMargin: 10
364                 spacing: 2
365
366                 Text {
367                     text: Translations.turkish ? modelData.sinifTr : modelData.sinifEn
368                     color: "#dce8f5"
369                     font.family: "Segoe UI"
370                     font.pixelSize: 13
371                     font.bold: true
372                 }
373
374                 Text {
375                     visible: modelData.cimentoTr.length > 0
376                     text: Translations.turkish
377                         ? (txt("cimentoEtiket") + modelData.cimentoTr + txt("suCimentoEtiket") + modelData.suCimentoTr)
378                         : (txt("cimentoEtiket") + modelData.cimentoEn + txt("suCimentoEtiket") + modelData.suCimentoEn)
379                     color: "#6b7280"
380                     font.family: "Segoe UI"
381                     font.pixelSize: 10
382                 }
383             }

```

```

384     }
385 }
386
387 Text {
388     text: txt("eklenenAgirlik")
389     color: "#9ca3af"
390     font.family: "Segoe UI"
391     font.pixelSize: 12
392 }
393
394 Column {
395     width: parent.width
396     spacing: 8
397
398     Row {
399         width: parent.width
400         spacing: 8
401
402         Button {
403             id: agirlikEkleButonu
404             width: (parent.width - 2 * parent.spacing) / 3
405             height: 38
406             text: "+ 1.6"
407             font.family: "Segoe UI"
408             font.pixelSize: 12
409
410             onClicked: {
411                 agirlikAdedi += 1
412                 agirlikEklendi()
413             }
414
415             background: Rectangle {
416                 radius: 6
417                 color: agirlikEkleButonu.pressed ? "#1e3a8a" : (agirlikEkleButonu.hovered ? "#2563eb" : "#1d4ed8")
418                 border.color: "#3b82f6"
419                 border.width: 1
420             }
421
422             contentItem: Text {
423                 text: agirlikEkleButonu.text
424                 color: "#ffffff"
425                 font: agirlikEkleButonu.font
426                 horizontalAlignment: Text.AlignHCenter
427                 verticalAlignment: Text.AlignVCenter
428             }
429         }
430
431         Button {
432             id: agirlikAzaltButonu
433             width: (parent.width - 2 * parent.spacing) / 3
434             height: 38
435             text: txt("geriButon")
436             font.family: "Segoe UI"
437             font.pixelSize: 12
438             enabled: agirlikAdedi > 0
439
440             onClicked: agirlikAdedi -= 1
441
442             background: Rectangle {
443                 radius: 6
444                 color: agirlikAzaltButonu.pressed ? "#78350f" : (agirlikAzaltButonu.hovered ? "#a16207" : "#92400e")
445                 opacity: agirlikAzaltButonu.enabled ? 1.0 : 0.4
446                 border.color: "#d97706"
447                 border.width: 1
448             }
449
450             contentItem: Text {
451                 text: agirlikAzaltButonu.text
452                 color: "#ffffff"
453                 opacity: agirlikAzaltButonu.enabled ? 1.0 : 0.6
454                 font: agirlikAzaltButonu.font
455                 horizontalAlignment: Text.AlignHCenter
456                 verticalAlignment: Text.AlignVCenter
457             }
458         }
459
460         Button {
461             id: sifirlaButonu
462             width: (parent.width - 2 * parent.spacing) / 3
463             height: 38

```

```

464         text: txt("sifirlaButon")
465         font.family: "Segoe UI"
466         font.pixelSize: 12
467         enabled: agirlikAdedi > 0
468
469         onClicked: agirlikAdedi = 0
470
471         background: Rectangle {
472             radius: 6
473             color: sifirlaButonu.pressed ? "#7f1d1d" : (sifirlaButonu.hovered ? "#b91c1c" : "#991b1b")
474             opacity: sifirlaButonu.enabled ? 1.0 : 0.4
475             border.color: "#dc2626"
476             border.width: 1
477         }
478
479         contentItem: Text {
480             text: sifirlaButonu.text
481             color: "#ffffff"
482             opacity: sifirlaButonu.enabled ? 1.0 : 0.6
483             font: sifirlaButonu.font
484             horizontalAlignment: Text.AlignHCenter
485             verticalAlignment: Text.AlignVCenter
486         }
487     }
488 }
489
490 Rectangle {
491     width: parent.width
492     height: 34
493     radius: 6
494     color: "#0a0e17"
495     border.color: "#1e2a3f"
496     border.width: 1
497
498     Text {
499         anchors.centerIn: parent
500         text: agirlikAdedi + " x " + agirlikBirim.toFixed(1) + " = " + agirlikToplam.toFixed(1) + " kg"
501         color: agirlikAdedi > 0 ? "#dce8f5" : "#4b5563"
502         font.family: "Segoe UI"
503         font.pixelSize: 13
504         font.bold: true
505     }
506 }
507
508 Item { width: parent.width; height: 10 }
509
510 Button {
511     id: baslatButonu
512     width: parent.width
513     height: 44
514     text: txt("olcumuBaslatButon")
515     font.family: "Segoe UI"
516     font.pixelSize: 14
517     font.bold: true
518     enabled: aktifOlcumId <= 0
519
520     onClicked: baslatOlcumu()
521
522     background: Rectangle {
523         radius: 8
524         color: !baslatButonu.enabled ? "#14532d" : (baslatButonu.pressed ? "#15803d" : (baslatButonu.hovered ? "#22c55e" :
525 "#16a34a"))
526         opacity: baslatButonu.enabled ? 1.0 : 0.5
527         Behavior on color { ColorAnimation { duration: 120 } }
528     }
529
530     contentItem: Text {
531         text: baslatButonu.text
532         color: "#ffffff"
533         opacity: baslatButonu.enabled ? 1.0 : 0.6
534         font: baslatButonu.font
535         horizontalAlignment: Text.AlignHCenter
536         verticalAlignment: Text.AlignVCenter
537     }
538 }
539
540 Text {
541     width: parent.width
542     visible: uyariMesaji.length > 0

```

```

543         text: uyariMesaji
544         color: "#f87171"
545         font.family: "Segoe UI"
546         font.pixelSize: 11
547         wrapMode: Text.WordWrap
548     }
549
550     Row {
551         width: parent.width
552         spacing: 10
553
554         Button {
555             id: duraklatButonu
556             width: (parent.width - parent.spacing) / 2
557             height: 40
558             text: calculator.duraklatildi ? txt("devamEtButon") : txt("duraklatButon")
559             font.family: "Segoe UI"
560             font.pixelSize: 13
561
562             onClicked: {
563                 if (calculator.duraklatildi) {
564                     calculator.devamEt()
565                 } else {
566                     calculator.duraklat()
567                 }
568             }
569
570             background: Rectangle {
571                 radius: 8
572                 color: duraklatButonu.pressed ? "#78350f" : (duraklatButonu.hovered ? "#a16207" : "#92400e")
573                 border.color: "#d97706"
574                 border.width: 1
575             }
576
577             contentItem: Text {
578                 text: duraklatButonu.text
579                 color: "ffffff"
580                 font: duraklatButonu.font
581                 horizontalAlignment: Text.AlignHCenter
582                 verticalAlignment: Text.AlignVCenter
583             }
584         }
585
586         Button {
587             id: bitirButonu
588             width: (parent.width - parent.spacing) / 2
589             height: 40
590             text: txt("bitirButon")
591             font.family: "Segoe UI"
592             font.pixelSize: 13
593
594             onClicked: bitirTalebiGoster()
595
596             background: Rectangle {
597                 radius: 8
598                 color: bitirButonu.pressed ? "#7f1d1d" : (bitirButonu.hovered ? "#b91c1c" : "#991b1b")
599                 border.color: "#dc2626"
600                 border.width: 1
601             }
602
603             contentItem: Text {
604                 text: bitirButonu.text
605                 color: "ffffff"
606                 font: bitirButonu.font
607                 horizontalAlignment: Text.AlignHCenter
608                 verticalAlignment: Text.AlignVCenter
609             }
610         }
611     }
612 }
613
614 Rectangle {
615     id: durumKarti
616     anchors.top: formAlanlari.bottom
617     anchors.left: parent.left
618     anchors.right: parent.right
619     anchors.topMargin: 20
620     anchors.leftMargin: 20
621     anchors.rightMargin: 20
622     height: 52

```

```

623     radius: 10
624     color: "#0a0e17"
625     border.color: "#1e2a3f"
626     border.width: 1
627
628     Row {
629         anchors.left: parent.left
630         anchors.verticalCenter: parent.verticalCenter
631         anchors.leftMargin: 14
632         spacing: 10
633
634         Rectangle {
635             id: trafikIsigi
636             width: 12
637             height: 12
638             radius: 6
639             anchors.verticalCenter: parent.verticalCenter
640             color: {
641                 if (!sensorManager.veriGecerli) return "#dc2626"
642                 if (calculator.duraklatildi) return "#6b7280"
643                 if (calculator.durum == "YUKARIDA") return "#16a34a"
644                 if (calculator.durum == "INIYOR") return "#f59e0b"
645                 return "#dc2626"
646             }
647
648             Rectangle {
649                 anchors.centerIn: parent
650                 width: parent.width + 8
651                 height: parent.height + 8
652                 radius: width / 2
653                 color: "transparent"
654                 border.color: parent.color
655                 border.width: 1
656                 opacity: 0.35
657             }
658         }
659
660         Text {
661             anchors.verticalCenter: parent.verticalCenter
662             text: {
663                 if (!sensorManager.veriGecerli) return txt("veriYok")
664                 if (calculator.duraklatildi) return txt("duraklatildiDurum")
665                 return calculator.durum
666             }
667             color: "#dce8f5"
668             font.family: "Segoe UI"
669             font.pixelSize: 13
670             font.bold: true
671         }
672     }
673
674     Row {
675         anchors.right: parent.right
676         anchors.verticalCenter: parent.verticalCenter
677         anchors.rightMargin: 14
678         spacing: 6
679
680         Text {
681             anchors.verticalCenter: parent.verticalCenter
682             text: "STROKE"
683             color: "#6b7280"
684             font.family: "Segoe UI"
685             font.pixelSize: 10
686             font.letterSpacing: 1
687         }
688
689         Text {
690             anchors.verticalCenter: parent.verticalCenter
691             text: calculator.strokeSayisi
692             color: "#e8a020"
693             font.family: "Segoe UI"
694             font.pixelSize: 16
695             font.bold: true
696         }
697     }
698 }
699
700 Text {
701     id: anlikDegerlerBaslik
702     anchors.top: durumKarti.bottom

```

```

703         anchors.left: parent.left
704         anchors.topMargin: 24
705         anchors.leftMargin: 20
706         text: txt("anlikDegerler")
707         color: "#6b7280"
708         font.family: "Segoe UI"
709         font.pixelSize: 12
710         font.bold: true
711         font.letterSpacing: 1
712     }
713
714     Grid {
715         anchors.top: anlikDegerlerBaslik.bottom
716         anchors.left: parent.left
717         anchors.topMargin: 12
718         anchors.leftMargin: 20
719         columns: 2
720         spacing: 10
721
722         Rectangle {
723             id: basincKarti
724             width: 115
725             height: 90
726             radius: 10
727             color: "#0a0e17"
728             border.color: "#1e2a3f"
729             border.width: 1
730             clip: true
731
732             Rectangle { width: 4; height: parent.height; color: "#3b82f6" }
733
734             Column {
735                 anchors.top: parent.top
736                 anchors.right: parent.right
737                 anchors.bottom: parent.bottom
738                 anchors.left: parent.left
739                 anchors.topMargin: 10
740                 anchors.rightMargin: 10
741                 anchors.bottomMargin: 10
742                 anchors.leftMargin: 16
743                 spacing: 6
744
745                 Text {
746                     text: txt("basincEtiket")
747                     color: "#6b7280"
748                     font.family: "Segoe UI"
749                     font.pixelSize: 10
750                     font.letterSpacing: 1
751                 }
752
753                 Text {
754                     text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) : "--"
755                     color: "#3b82f6"
756                     font.family: "Segoe UI"
757                     font.pixelSize: 22
758                     font.bold: true
759                 }
760
761                 Text {
762                     text: "mbar"
763                     color: "#4b5563"
764                     font.family: "Segoe UI"
765                     font.pixelSize: 10
766                 }
767             }
768         }
769
770         Rectangle {
771             width: 115
772             height: 90
773             radius: 10
774             color: "#0a0e17"
775             border.color: "#1e2a3f"
776             border.width: 1
777             clip: true
778
779             Rectangle { width: 4; height: parent.height; color: "#9333ea" }
780
781             Column {
782                 anchors.top: parent.top

```



```

783         anchors.right: parent.right
784         anchors.bottom: parent.bottom
785         anchors.left: parent.left
786         anchors.topMargin: 10
787         anchors.rightMargin: 10
788         anchors.bottomMargin: 10
789         anchors.leftMargin: 16
790         spacing: 6
791
792         Text {
793             text: txt("konumEtiket")
794             color: "#6b7280"
795             font.family: "Segoe UI"
796             font.pixelSize: 10
797             font.letterSpacing: 1
798         }
799
800         Text {
801             text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) : "--"
802             color: "#9333ea"
803             font.family: "Segoe UI"
804             font.pixelSize: 22
805             font.bold: true
806         }
807
808         Text {
809             text: "mm"
810             color: "#4b5563"
811             font.family: "Segoe UI"
812             font.pixelSize: 10
813         }
814     }
815 }
816
817 Rectangle {
818     width: 115
819     height: 90
820     radius: 10
821     color: "#0a0e17"
822     border.color: "#1e2a3f"
823     border.width: 1
824     clip: true
825
826     Rectangle { width: 4; height: parent.height; color: "#f59e0b" }
827
828     Column {
829         anchors.top: parent.top
830         anchors.right: parent.right
831         anchors.bottom: parent.bottom
832         anchors.left: parent.left
833         anchors.topMargin: 10
834         anchors.rightMargin: 10
835         anchors.bottomMargin: 10
836         anchors.leftMargin: 16
837         spacing: 6
838
839         Text {
840             text: txt("hizEtiket")
841             color: "#6b7280"
842             font.family: "Segoe UI"
843             font.pixelSize: 10
844             font.letterSpacing: 1
845         }
846
847         Text {
848             text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) : "--"
849             color: "#f59e0b"
850             font.family: "Segoe UI"
851             font.pixelSize: 22
852             font.bold: true
853         }
854
855         Text {
856             text: "m/s"
857             color: "#4b5563"
858             font.family: "Segoe UI"
859             font.pixelSize: 10
860         }
861     }
862 }

```

```

863
864     Rectangle {
865         width: 115
866         height: 90
867         radius: 10
868         color: "#0a0e17"
869         border.color: "#1e2a3f"
870         border.width: 1
871         clip: true
872
873         Rectangle { width: 4; height: parent.height; color: "#16a34a" }
874
875         Column {
876             anchors.top: parent.top
877             anchors.right: parent.right
878             anchors.bottom: parent.bottom
879             anchors.left: parent.left
880             anchors.topMargin: 10
881             anchors.rightMargin: 10
882             anchors.bottomMargin: 10
883             anchors.leftMargin: 16
884             spacing: 6
885
886             Text {
887                 text: txt("debiEtiket")
888                 color: "#6b7280"
889                 font.family: "Segoe UI"
890                 font.pixelSize: 10
891                 font.letterSpacing: 1
892             }
893
894             Text {
895                 text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) : "--"
896                 color: "#16a34a"
897                 font.family: "Segoe UI"
898                 font.pixelSize: 22
899                 font.bold: true
900             }
901
902             Text {
903                 text: "m3/h"
904                 color: "#4b5563"
905                 font.family: "Segoe UI"
906                 font.pixelSize: 10
907             }
908         }
909     }
910 }
911
912
913 Rectangle {
914     id: panelAyraci
915     width: 3
916     height: parent.height
917     color: "#05070b"
918
919     Rectangle {
920         anchors.right: parent.right
921         anchors.top: parent.top
922         anchors.bottom: parent.bottom
923         width: 1
924         color: "#403b82f6"
925     }
926 }
927
928 Item {
929     width: parent.width - testBilgileriPaneli.width - panelAyraci.width
930     height: parent.height
931
932     Grid {
933         anchors.fill: parent
934         anchors.margins: 16
935         columns: 2
936         spacing: 16
937
938         Rectangle {
939             width: (parent.width - parent.spacing) / 2
940             height: (parent.height - parent.spacing) / 2
941             radius: 10
942             color: "#0f1420"

```

```

943 border.color: "#17263d"
944 border.width: 1
945
946 Item {
947     id: basincBaslik
948     anchors.top: parent.top
949     anchors.left: parent.left
950     anchors.right: parent.right
951     anchors.margins: 14
952     height: 20
953
954     Rectangle {
955         id: basincNoktasi
956         width: 8
957         height: 8
958         radius: 4
959         color: "#3b82f6"
960         anchors.left: parent.left
961         anchors.verticalCenter: parent.verticalCenter
962     }
963
964     Text {
965         anchors.left: basincNoktasi.right
966         anchors.leftMargin: 8
967         anchors.verticalCenter: parent.verticalCenter
968         text: txt("basincZaman")
969         color: "#9ca3af"
970         font.family: "Segoe UI"
971         font.pixelSize: 13
972         font.bold: true
973     }
974
975     Text {
976         anchors.right: parent.right
977         anchors.verticalCenter: parent.verticalCenter
978         text: sensorManager.veriGecerli ? sensorManager.basinc.toFixed(1) + " mbar" : "-- mbar"
979         color: "#3b82f6"
980         font.family: "Segoe UI"
981         font.pixelSize: 13
982         font.bold: true
983     }
984 }
985
986 ChartView {
987     anchors.top: basincBaslik.bottom
988     anchors.left: parent.left
989     anchors.right: parent.right
990     anchors.bottom: parent.bottom
991     anchors.topMargin: 4
992     anchors.bottomMargin: 4
993     backgroundColor: "transparent"
994     legend.visible: false
995     antialiasing: true
996
997     ValueAxis {
998         id: xEkseni
999         min: 0
1000         max: 60
1001         gridLineColor: "#182131"
1002         labelsColor: "#4b5563"
1003         labelsFont.pixelSize: 9
1004         lineVisible: false
1005         minorGridVisible: false
1006     }
1007
1008     ValueAxis {
1009         id: yEkseni
1010         min: 0
1011         max: 1100
1012         gridLineColor: "#182131"
1013         labelsColor: "#4b5563"
1014         labelsFont.pixelSize: 9
1015         lineVisible: false
1016         minorGridVisible: false
1017     }
1018
1019     LineSeries {
1020         id: basincSerisi
1021         axisX: xEkseni
1022         axisY: yEkseni

```

```

1023         color: "#3b82f6"
1024         width: 2
1025     }
1026 }
1027 }
1028
1029 Rectangle {
1030     width: (parent.width - parent.spacing) / 2
1031     height: (parent.height - parent.spacing) / 2
1032     radius: 10
1033     color: "#0f1420"
1034     border.color: "#241a38"
1035     border.width: 1
1036
1037     Item {
1038         id: konumBaslik
1039         anchors.top: parent.top
1040         anchors.left: parent.left
1041         anchors.right: parent.right
1042         anchors.margins: 14
1043         height: 20
1044
1045         Rectangle {
1046             id: konumNoktasi
1047             width: 8
1048             height: 8
1049             radius: 4
1050             color: "#9333ea"
1051             anchors.left: parent.left
1052             anchors.verticalCenter: parent.verticalCenter
1053         }
1054
1055         Text {
1056             anchors.left: konumNoktasi.right
1057             anchors.leftMargin: 8
1058             anchors.verticalCenter: parent.verticalCenter
1059             text: txt("konumZaman")
1060             color: "#9ca3af"
1061             font.family: "Segoe UI"
1062             font.pixelSize: 13
1063             font.bold: true
1064         }
1065
1066         Text {
1067             anchors.right: parent.right
1068             anchors.verticalCenter: parent.verticalCenter
1069             text: sensorManager.veriGecerli ? sensorManager.konum.toFixed(1) + " mm" : "-- mm"
1070             color: "#9333ea"
1071             font.family: "Segoe UI"
1072             font.pixelSize: 13
1073             font.bold: true
1074         }
1075     }
1076
1077     ChartView {
1078         anchors.top: konumBaslik.bottom
1079         anchors.left: parent.left
1080         anchors.right: parent.right
1081         anchors.bottom: parent.bottom
1082         anchors.topMargin: 4
1083         anchors.bottomMargin: 4
1084         backgroundColor: "transparent"
1085         legend.visible: false
1086         antialiasing: true
1087
1088         ValueAxis {
1089             id: konumXEkseni
1090             min: 0
1091             max: 60
1092             gridLineColor: "#182131"
1093             labelsColor: "#4b5563"
1094             labelsFont.pixelSize: 9
1095             lineVisible: false
1096             minorGridVisible: false
1097         }
1098
1099         ValueAxis {
1100             id: konumYEkseni
1101             min: 0
1102             max: 500

```

```

1103         gridLineColor: "#182131"
1104         labelsColor: "#4b5563"
1105         labelsFont.pixelSize: 9
1106         lineVisible: false
1107         minorGridVisible: false
1108     }
1109
1110     LineSeries {
1111         id: konumSerisi
1112         axisX: konumXEkseni
1113         axisY: konumYEkseni
1114         color: "#93333ea"
1115         width: 2
1116     }
1117
1118     Connections {
1119         target: calculator
1120         function onStrokeSayisiChanged() {
1121             if (aktifOlcumId > 0 && sensorManager.veriGecerli) {
1122                 // gecerli parametresi artik Calculator'un hiz>0 kontrolunden geliyor
1123                 // (orijinal SLIPER'daki "invalid stroke" mantigi), sabit "true" degil.
1124                 database.strokeKaydet(aktifOlcumId, sensorManager.basinc, sensorManager.konum, sensorManager.debi,
calculator.sonStrokeGecerli)
1125             }
1126         }
1127     }
1128 }
1129
1130
1131 Rectangle {
1132     width: (parent.width - parent.spacing) / 2
1133     height: (parent.height - parent.spacing) / 2
1134     radius: 10
1135     color: "#0f1420"
1136     border.color: "#3a2a14"
1137     border.width: 1
1138
1139     Item {
1140         id: hizBaslik
1141         anchors.top: parent.top
1142         anchors.left: parent.left
1143         anchors.right: parent.right
1144         anchors.margins: 14
1145         height: 20
1146
1147         Rectangle {
1148             id: hizNoktasi
1149             width: 8
1150             height: 8
1151             radius: 4
1152             color: "#f59e0b"
1153             anchors.left: parent.left
1154             anchors.verticalCenter: parent.verticalCenter
1155         }
1156
1157         Text {
1158             anchors.left: hizNoktasi.right
1159             anchors.leftMargin: 8
1160             anchors.verticalCenter: parent.verticalCenter
1161             text: txt("hizZaman")
1162             color: "#9ca3af"
1163             font.family: "Segoe UI"
1164             font.pixelSize: 13
1165             font.bold: true
1166         }
1167
1168         Text {
1169             anchors.right: parent.right
1170             anchors.verticalCenter: parent.verticalCenter
1171             text: sensorManager.veriGecerli ? sensorManager.hiz.toFixed(1) + " m/s" : "-- m/s"
1172             color: "#f59e0b"
1173             font.family: "Segoe UI"
1174             font.pixelSize: 13
1175             font.bold: true
1176         }
1177     }
1178
1179 ChartView {
1180     anchors.top: hizBaslik.bottom
1181     anchors.left: parent.left

```

```

1182     anchors.right: parent.right
1183     anchors.bottom: parent.bottom
1184     anchors.topMargin: 4
1185     anchors.bottomMargin: 4
1186     backgroundColor: "transparent"
1187     legend.visible: false
1188     antialiasing: true
1189
1190     ValueAxis {
1191         id: hizXEkseni
1192         min: 0
1193         max: 60
1194         gridLineColor: "#182131"
1195         labelsColor: "#4b5563"
1196         labelsFont.pixelSize: 9
1197         lineVisible: false
1198         minorGridVisible: false
1199     }
1200
1201     ValueAxis {
1202         id: hizYEkseni
1203         min: 0
1204         max: 4
1205         gridLineColor: "#182131"
1206         labelsColor: "#4b5563"
1207         labelsFont.pixelSize: 9
1208         lineVisible: false
1209         minorGridVisible: false
1210     }
1211
1212     LineSeries {
1213         id: hizSerisi
1214         axisX: hizXEkseni
1215         axisY: hizYEkseni
1216         color: "#f59e0b"
1217         width: 2
1218     }
1219 }
1220
1221
1222 Rectangle {
1223     width: (parent.width - parent.spacing) / 2
1224     height: (parent.height - parent.spacing) / 2
1225     radius: 10
1226     color: "#0f1420"
1227     border.color: "#163321"
1228     border.width: 1
1229
1230     Item {
1231         id: debiBaslik
1232         anchors.top: parent.top
1233         anchors.left: parent.left
1234         anchors.right: parent.right
1235         anchors.margins: 14
1236         height: 20
1237
1238         Rectangle {
1239             id: debiNoktasi
1240             width: 8
1241             height: 8
1242             radius: 4
1243             color: "#16a34a"
1244             anchors.left: parent.left
1245             anchors.verticalCenter: parent.verticalCenter
1246         }
1247
1248         Text {
1249             anchors.left: debiNoktasi.right
1250             anchors.leftMargin: 8
1251             anchors.verticalCenter: parent.verticalCenter
1252             text: txt("debiZaman")
1253             color: "#9ca3af"
1254             font.family: "Segoe UI"
1255             font.pixelSize: 13
1256             font.bold: true
1257         }
1258
1259         Text {
1260             anchors.right: parent.right
1261             anchors.verticalCenter: parent.verticalCenter

```

```

1262         text: sensorManager.veriGecerli ? sensorManager.debi.toFixed(1) + " m3/h" : "-- m3/h"
1263         color: "#16a34a"
1264         font.family: "Segoe UI"
1265         font.pixelSize: 13
1266         font.bold: true
1267     }
1268 }
1269
1270 ChartView {
1271     anchors.top: debiBaslik.bottom
1272     anchors.left: parent.left
1273     anchors.right: parent.right
1274     anchors.bottom: parent.bottom
1275     anchors.topMargin: 4
1276     anchors.bottomMargin: 4
1277     backgroundColor: "transparent"
1278     legend.visible: false
1279     antialiasing: true
1280
1281     ValueAxis {
1282         id: debiXEkseni
1283         min: 0
1284         max: 60
1285         gridLineColor: "#182131"
1286         labelsColor: "#4b5563"
1287         labelsFont.pixelSize: 9
1288         lineVisible: false
1289         minorGridVisible: false
1290     }
1291
1292     ValueAxis {
1293         id: debiYEkseni
1294         min: 0
1295         max: 180
1296         gridLineColor: "#182131"
1297         labelsColor: "#4b5563"
1298         labelsFont.pixelSize: 9
1299         lineVisible: false
1300         minorGridVisible: false
1301     }
1302
1303     LineSeries {
1304         id: debiSerisi
1305         axisX: debiXEkseni
1306         axisY: debiYEkseni
1307         color: "#16a34a"
1308         width: 2
1309     }
1310 }
1311 }
1312 }
1313 }
1314 }
1315
1316 // Sesli komut geri bildirimi: eller kirliiyken dokunmadan "Başlat /
1317 // Durdur / Bitir" denildiğinde ne duyulduğunu göstererek kullanıcıya
1318 // güven verir.
1319 Rectangle {
1320     id: sesliKomutRozeti
1321     visible: voiceCommandManager.etkin
1322     anchors.top: parent.top
1323     anchors.right: parent.right
1324     anchors.margins: 16
1325     width: sesliKomutIcerik.implicitWidth + 24
1326     height: sesliKomutIcerik.implicitHeight + 16
1327     radius: 10
1328     color: "#0a0e17"
1329     border.color: voiceCommandManager.dinliyor ? "#16a34a" : "#1e2a3f"
1330     border.width: 1
1331     z: 10
1332
1333     Column {
1334         id: sesliKomutIcerik
1335         anchors.centerIn: parent
1336         spacing: 4
1337
1338         Row {
1339             spacing: 6
1340             Rectangle {
1341                 width: 8

```

```

1342         height: 8
1343         radius: 4
1344         anchors.verticalCenter: parent.verticalCenter
1345         color: voiceCommandManager.dinliyor ? "#16a34a" : "#6b7280"
1346     }
1347     Text {
1348         text: voiceCommandManager.dinliyor ? txt("sesliDinleniyor") : txt("sesliHazirlaniyor")
1349         color: "#9ca3af"
1350         font.family: "Segoe UI"
1351         font.pixelSize: 11
1352     }
1353 }
1354
1355 Text {
1356     visible: voiceCommandManager.anlikMetin.length > 0
1357     text: txt("duyulanEtiket") + voiceCommandManager.anlikMetin
1358     color: "#4b5563"
1359     font.family: "Segoe UI"
1360     font.pixelSize: 10
1361     font.italic: true
1362 }
1363 }
1364 }
1365
1366 Popup {
1367     id: bitirOnayPopup
1368     modal: true
1369     focus: true
1370     anchors.centerIn: parent
1371     width: 340
1372     padding: 24
1373     closePolicy: Popup.CloseOnEscape | Popup.CloseOnPressOutside
1374
1375     background: Rectangle {
1376         color: "#0f1420"
1377         radius: 12
1378         border.color: "#1e2a3f"
1379         border.width: 1
1380     }
1381
1382     Overlay.modal: Rectangle {
1383         color: "#a6000000"
1384     }
1385
1386     contentItem: Column {
1387         spacing: 20
1388
1389         Text {
1390             width: 292
1391             text: txt("bitirOnaySorusu")
1392             color: "#dce8f5"
1393             font.family: "Segoe UI"
1394             font.pixelSize: 14
1395             font.bold: true
1396             wrapMode: Text.WordWrap
1397         }
1398
1399         Column {
1400             width: 292
1401             spacing: 10
1402
1403             Button {
1404                 id: kaydetBitirButonu
1405                 width: parent.width
1406                 height: 40
1407                 text: txt("kaydetVeBitir")
1408                 font.family: "Segoe UI"
1409                 font.pixelSize: 13
1410
1411                 onClicked: {
1412                     bitirOnayPopup.close()
1413                     var bitenId = aktifOlcumId
1414                     console.log("Olcum kaydedilip sonlandırıldı, id:", bitenId)
1415                     resetMeasurementScreen()
1416                     olcumTamamlandi(bitenId)
1417                 }
1418
1419                 background: Rectangle {
1420                     radius: 8
1421                     color: kaydetBitirButonu.pressed ? "#14532d" : (kaydetBitirButonu.hovered ? "#15803d" : "#16a34a")

```



```

1422         border.color: "#22c55e"
1423         border.width: 1
1424     }
1425
1426     contentItem: Text {
1427         text: kaydetBitirButonu.text
1428         color: "ffffff"
1429         font: kaydetBitirButonu.font
1430         horizontalAlignment: Text.AlignHCenter
1431         verticalAlignment: Text.AlignVCenter
1432     }
1433 }
1434
1435 Button {
1436     id: silBitirButonu
1437     width: parent.width
1438     height: 40
1439     text: txt("testiSilVeBitir")
1440     font.family: "Segoe UI"
1441     font.pixelSize: 13
1442
1443     onClicked: {
1444         bitirOnayPopup.close()
1445         var silinecekId = aktifOlcumId
1446         var basarili = database.olcumSil(silinecekId)
1447         console.log("Olcum silinip sonlandirildi, id:", silinecekId, "basarili:", basarili)
1448         resetMeasurementScreen()
1449     }
1450
1451     background: Rectangle {
1452         radius: 8
1453         color: silBitirButonu.pressed ? "#7f1d1d" : (silBitirButonu.hovered ? "#b91c1c" : "#991b1b")
1454         border.color: "#dc2626"
1455         border.width: 1
1456     }
1457
1458     contentItem: Text {
1459         text: silBitirButonu.text
1460         color: "ffffff"
1461         font: silBitirButonu.font
1462         horizontalAlignment: Text.AlignHCenter
1463         verticalAlignment: Text.AlignVCenter
1464     }
1465 }
1466
1467 Button {
1468     id: vazgecButonu
1469     width: parent.width
1470     height: 40
1471     text: txt("vazgecTesteDevam")
1472     font.family: "Segoe UI"
1473     font.pixelSize: 13
1474
1475     onClicked: {
1476         bitirOnayPopup.close()
1477         if (bitirIcinDuraklatildi) {
1478             calculator.devamEt()
1479             bitirIcinDuraklatildi = false
1480         }
1481     }
1482
1483     background: Rectangle {
1484         radius: 8
1485         color: vazgecButonu.pressed ? "#1e2a3f" : (vazgecButonu.hovered ? "#243349" : "#182131")
1486         border.color: "#2c3b52"
1487         border.width: 1
1488     }
1489
1490     contentItem: Text {
1491         text: vazgecButonu.text
1492         color: "#dce8f5"
1493         font: vazgecButonu.font
1494         horizontalAlignment: Text.AlignHCenter
1495         verticalAlignment: Text.AlignVCenter
1496     }
1497 }
1498 }
1499 }
1500 }
1501 }

```

```

1 import QtQuick 6.7
2 import QtQuick.Controls 6.7
3 import QtCharts 6.7
4 import sliper
5
6 Rectangle {
7     color: "#0a0e17"
8
9     property int olcumId: -1
10    property real hesaplananTau0: 0
11    property real hesaplananMu: 0
12    property int playbackAdimi: -1
13    property bool oynatiliyor: false
14    property int playbackHizi: 1
15    property stringmusteriAdi: ""
16    property stringreceteAdi: ""
17    property real agirlikDegeri: 0
18    property string olcumTarihi: ""
19
20    readonly property var metinler: ({
21        sonucOzeti: { tr: "SONUÇ ÖZETİ", en: "RESULT SUMMARY" },
22        akmaGerilmesi: { tr: "AKMA GERİLMESİ ( $\tau_0$ )", en: "YIELD STRESS ( $\tau_0$ )" },
23        plastikViskozite: { tr: "PLASTİK VİSKOZİTE ( $\mu$ )", en: "PLASTIC VISCOSITY ( $\mu$ )" },
24        uyumKalitesi: { tr: "UYUM KALİTESİ ( $R^2$ )", en: "FIT QUALITY ( $R^2$ )" },
25        boruHattiTahmini: { tr: "BORU HATTI TAHMİNİ", en: "PIPELINE ESTIMATE" },
26        hedefBoruCapi: { tr: "Hedef Boru Çapı (mm)", en: "Target Pipe Diameter (mm)" },
27        boruUzunlugu: { tr: "Boru Uzunluğu (m) - karşılaştırma referansı", en: "Pipe Length (m) - comparison reference" },
28        hedefDebi: { tr: "Hedef Debi ( $m^3/h$ )", en: "Target Flow Rate ( $m^3/h$ )" },
29        hataPayi: { tr: "Hata Payı:", en: "Margin of Error:" },
30        tahminiHesapla: { tr: "Tahmini Hesapla", en: "Calculate Estimate" },
31        gecerliDegerGirin: { tr: "Geçerli değer girin", en: "Enter a valid value" },
32        hesaplanamadi: { tr: "Hesaplanamadı", en: "Could not calculate" },
33        pompalanabilir: { tr: "POMPALANABİLİR", en: "PUMPABLE" },
34        pompalamaSorunu: { tr: "POMPALAMA SORUNU", en: "PUMPING ISSUE" },
35        durumHerIkisiYuksek: { tr: "Hem akma gerilmesi hem plastik viskozite yüksek. Su/çimento oranını artırın ve agrega gradasyonunu
gözden geçirin.", en: "Both yield stress and plastic viscosity are high. Increase the water/cement ratio and review the aggregate gradation." },
36        durumTau0Yuksek: { tr: "Akma gerilmesi yüksek. Su/çimento oranını artırmayı veya süperakışkanlaştırıcı dozajını yükseltmeyi
değerlendirin.", en: "Yield stress is high. Consider increasing the water/cement ratio or raising the superplasticizer dosage." },
37        durumMuYuksek: { tr: "Plastik viskozite yüksek. İnce agrega oranını azaltmayı veya su azaltıcı katkı eklemeyi değerlendirin.", en:
"Plastic viscosity is high. Consider reducing the fine aggregate ratio or adding a water-reducing admixture." },
38        durumYetersizVeri: { tr: "Yeterli stroke verisi bulunamadı.", en: "Not enough stroke data found." },
39        pdfRaporOnizle: { tr: "PDF Rapor Önizle", en: "Preview PDF Report" },
40        excelCsvOnizle: { tr: "Excel (CSV) Önizle", en: "Preview Excel (CSV)" },
41        excelXmlDisaAktor: { tr: "Excel (XML) Dışa Aktar", en: "Export Excel (XML)" },
42        disaAktorildi: { tr: "✓ Dışa aktarıldı: ", en: "✓ Exported: " },
43        disaAktorilamadi: { tr: "✗ Dışa aktarılamadı", en: "✗ Export failed" },
44        oynatmaPlayback: { tr: "OYNATMA (PLAYBACK)", en: "PLAYBACK" },
45        strokeEtiket: { tr: "Stroke", en: "Stroke" },
46        oynatilacakStrokeYok: { tr: "Oynatılacak stroke yok", en: "No stroke to play back" },
47        pqDagilimGrafigi: { tr: "P-Q Dağılım Grafiği", en: "P-Q Distribution Chart" },
48        olcumNoktaları: { tr: "Ölçüm Noktaları", en: "Measurement Points" },
49        regresyonDogrusu: { tr: "Regresyon Doğrusu", en: "Regression Line" },
50        oynatmaEtiket: { tr: "Oynatma", en: "Playback" },
51        strokeTablosu: { tr: "Stroke Tablosu", en: "Stroke Table" },
52        strokeBaslik: { tr: "STROKE", en: "STROKE" },
53        durumBaslik: { tr: "DURUM", en: "STATUS" },
54        gecerliEtiket: { tr: "Geçerli", en: "Valid" },
55        hataliEtiket: { tr: "Hatalı", en: "Invalid" },
56        pdfRaporOnizlemeBaslik: { tr: "PDF Rapor Önizleme", en: "PDF Report Preview" },
57        indirPdf: { tr: "İndir (PDF)", en: "Download (PDF)" },
58        pdfKaydedildi: { tr: "PDF kaydedildi: ", en: "PDF saved: " },
59        pdfOlusturulamadi: { tr: "PDF oluşturulamadı.", en: "Could not generate PDF." },
60        kapat: { tr: "Kapat", en: "Close" },
61        excelCsvOnizlemeBaslik: { tr: "Excel (CSV) Önizleme", en: "Excel (CSV) Preview" },
62        olcumIdEtiket: { tr: "Ölçüm ID: ", en: "Measurement ID: " },
63        tarihEtiket: { tr: "Tarih: ", en: "Date: " },
64       musteriEtiket: { tr: "Müşteri: ", en: "Customer: " },
65        receteEtiket: { tr: "Reçete: ", en: "Recipe: " },
66        agirlikEtiket: { tr: "Ağırlık (kg): ", en: "Weight (kg): " },
67        basincBaslik: { tr: "BASINÇ (mbar)", en: "PRESSURE (mbar)" },
68        konumBaslik: { tr: "KONUM (mm)", en: "POSITION (mm)" },
69        debiBuyukBaslik: { tr: "DEBİ ( $m^3/h$ )", en: "FLOW ( $m^3/h$ )" },
70        gecerliBuyukBaslik: { tr: "GEÇERLİ", en: "VALID" },
71        evet: { tr: "Evet", en: "Yes" },
72        hayir: { tr: "Hayır", en: "No" },
73        indirCsv: { tr: "İndir (CSV)", en: "Download (CSV)" },
74        csvKaydedildi: { tr: "CSV kaydedildi: ", en: "CSV saved: " },

```

```

75     csvOlusturulamadi: { tr: "CSV oluşturulamadı.", en: "Could not generate CSV." },
76     bilinmiyor: { tr: "Bilinmiyor", en: "Unknown" }
77 })
78
79 function txt(anahtar) {
80     return Translations.turkish ? metinler[anahtar].tr : metinler[anahtar].en
81 }
82
83 onOlcumIdChanged: verileriYukle()
84
85 function verileriYukle() {
86     if (olcumId <= 0) {
87         return
88     }
89
90     playbackAdimi = -1
91     oynatiliyor = false
92     playbackTimer.stop()
93
94     var bilgi = database.olcumBilgisiGetir(olcumId)
95     musteriAdi = bilgi.bulundu ? bilgi.musteri : txt("bilinmiyor")
96     receteAdi = bilgi.bulundu ? bilgi.recete : txt("bilinmiyor")
97     agirlikDegeri = bilgi.bulundu ? bilgi.agirlik : 0
98     olcumTarihi = bilgi.bulundu ? bilgi.tarih : ""
99
100     var sonuc = database.binghamHesapla(olcumId)
101
102     hesaplananTau0 = sonuc.tau0
103     hesaplananMu = sonuc.mu
104
105     tau0Metni.text = sonuc.tau0.toFixed(2) + " mbar"
106     muMetni.text = sonuc.mu.toFixed(2) + " mbar·h/m³"
107     r2Metni.text = sonuc.r2.toFixed(2)
108
109     durumKutusu.durumIyiMi = sonuc.yeterliVeri && sonuc.tau0 < 5 && sonuc.mu < 10
110     durumKutusu.tau0Yuksek = sonuc.tau0 >= 5
111     durumKutusu.muYuksek = sonuc.mu >= 10
112
113     pqSerisi.clear()
114     regresyonCizgisi.clear()
115     oynatmaSerisi.clear()
116     strokeModeli.clear()
117
118     var strokeListesi = database.strokeVerileriGetir(olcumId)
119     var pMaks = 0
120     var qMaks = 0
121     for (var i = 0; i < strokeListesi.length; i++) {
122         var s = strokeListesi[i]
123         pqSerisi.append(s.debi, s.basinc)
124         strokeModeli.append({ stroke: s.stroke, p: s.basinc, q: s.debi, gecerli: s.gecerli })
125         if (s.basinc > pMaks) pMaks = s.basinc
126         if (s.debi > qMaks) qMaks = s.debi
127     }
128
129     // Eksenler gerçek ölçüm verisine göre ayarlanır; sabit aralık kullanılırsa
130     // basınc/debi bu aralığı aştığında noktalar (ve oynatma imleci) grafik
131     // dışına taşıp görünmez olur.
132     qMaks = Math.max(qMaks, 5)
133     regresyonCizgisi.append(0, sonuc.tau0)
134     regresyonCizgisi.append(qMaks, sonuc.tau0 + sonuc.mu * qMaks)
135
136     pMaks = Math.max(pMaks, sonuc.tau0 + sonuc.mu * qMaks, 10)
137     pEkseni.max = pMaks * 1.15
138     qEkseni.max = qMaks * 1.15
139 }
140
141 Timer {
142     id: playbackTimer
143     interval: 1000 / playbackHizi
144     repeat: true
145     running: false
146
147     onTriggered: {
148         if (playbackAdimi < strokeModeli.count - 1) {
149             playbackAdimi++
150             var satir = strokeModeli.get(playbackAdimi)
151             oynatmaSerisi.clear()
152             oynatmaSerisi.append(satir.q, satir.p)
153         } else {
154             oynatiliyor = false

```

```

155         playbackTimer.stop()
156     }
157 }
158 }
159
160 Row {
161     anchors.fill: parent
162     spacing: 0
163
164     Rectangle {
165         id: ozetPaneli
166         width: 280
167         height: parent.height
168         color: "#0f1420"
169
170         Text {
171             anchors.top: parent.top
172             anchors.left: parent.left
173             anchors.margins: 20
174             text: txt("sonucOzeti")
175             color: "#6b7280"
176             font.family: "Segoe UI"
177             font.pixelSize: 12
178             font.bold: true
179             font.letterSpacing: 1
180         }
181
182         Flickable {
183             anchors.top: parent.top
184             anchors.left: parent.left
185             anchors.right: parent.right
186             anchors.bottom: parent.bottom
187             anchors.margins: 20
188             anchors.topMargin: 60
189             contentWidth: width
190             contentHeight: sonucKartlari.height
191             clip: true
192
193             Column {
194                 id: sonucKartlari
195                 width: parent.width
196                 spacing: 10
197
198                 Rectangle {
199                     width: parent.width
200                     height: 70
201                     radius: 10
202                     color: "#0a0e17"
203                     border.color: "#17263d"
204                     border.width: 1
205                     clip: true
206
207                     Rectangle { width: 4; height: parent.height; color: "#3b82f6" }
208
209                     Column {
210                         anchors.left: parent.left
211                         anchors.verticalCenter: parent.verticalCenter
212                         anchors.leftMargin: 20
213                         spacing: 2
214
215                         Text {
216                             text: txt("akmaGerilmesi")
217                             color: "#6b7280"
218                             font.family: "Segoe UI"
219                             font.pixelSize: 10
220                             font.letterSpacing: 1
221                         }
222
223                         Text {
224                             id: tau0Metni
225                             text: "-"
226                             color: "#3b82f6"
227                             font.family: "Segoe UI"
228                             font.pixelSize: 20
229                             font.bold: true
230                         }
231                     }
232                 }
233
234                 Rectangle {

```

```

235         width: parent.width
236         height: 70
237         radius: 10
238         color: "#0a0e17"
239         border.color: "#241a38"
240         border.width: 1
241         clip: true
242
243     Rectangle { width: 4; height: parent.height; color: "#9333ea" }
244
245     Column {
246         anchors.left: parent.left
247         anchors.verticalCenter: parent.verticalCenter
248         anchors.leftMargin: 20
249         spacing: 2
250
251         Text {
252             text: txt("plastikViskozite")
253             color: "#6b7280"
254             font.family: "Segoe UI"
255             font.pixelSize: 10
256             font.letterSpacing: 1
257         }
258
259         Text {
260             id: muMetni
261             text: "-"
262             color: "#9333ea"
263             font.family: "Segoe UI"
264             font.pixelSize: 20
265             font.bold: true
266         }
267     }
268 }
269
270 Rectangle {
271     width: parent.width
272     height: 70
273     radius: 10
274     color: "#0a0e17"
275     border.color: "#163321"
276     border.width: 1
277     clip: true
278
279     Rectangle { width: 4; height: parent.height; color: "#16a34a" }
280
281     Column {
282         anchors.left: parent.left
283         anchors.verticalCenter: parent.verticalCenter
284         anchors.leftMargin: 20
285         spacing: 2
286
287         Text {
288             text: txt("uyumKalitesi")
289             color: "#6b7280"
290             font.family: "Segoe UI"
291             font.pixelSize: 10
292             font.letterSpacing: 1
293         }
294
295         Text {
296             id: r2Metni
297             text: "-"
298             color: "#16a34a"
299             font.family: "Segoe UI"
300             font.pixelSize: 20
301             font.bold: true
302         }
303     }
304 }
305
306 Rectangle {
307     width: parent.width
308     radius: 10
309     color: "#0a0e17"
310     border.color: "#1e2a3f"
311     border.width: 1
312     height: tahminIcerik.implicitHeight + 24
313
314     Column {

```

```

315         id: tahminIcerik
316         anchors.top: parent.top
317         anchors.left: parent.left
318         anchors.right: parent.right
319         anchors.margins: 12
320         spacing: 8
321
322         Text {
323             text: txt("boruHattiTahmini")
324             color: "#6b7280"
325             font.family: "Segoe UI"
326             font.pixelSize: 10
327             font.letterSpacing: 1
328         }
329
330         TextField {
331             id: capKutusu
332             width: parent.width
333             height: 34
334             placeholderText: txt("hedefBoruCapi")
335             placeholderTextColor: "#4b5563"
336             color: "ffffff"
337             font.pixelSize: 12
338             leftPadding: 10
339             text: "125"
340             verticalAlignment: TextInput.AlignVCenter
341             validator: DoubleValidator { bottom: 10; top: 500; decimals: 0 }
342             background: Rectangle {
343                 color: "#060d17"
344                 radius: 6
345                 border.color: capKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
346                 border.width: 1
347             }
348         }
349
350         TextField {
351             id: uzunlukKutusu
352             width: parent.width
353             height: 34
354             placeholderText: txt("boruUzunlugu")
355             placeholderTextColor: "#4b5563"
356             color: "ffffff"
357             font.pixelSize: 12
358             leftPadding: 10
359             verticalAlignment: TextInput.AlignVCenter
360             validator: DoubleValidator { bottom: 0; top: 2000; decimals: 0 }
361             background: Rectangle {
362                 color: "#060d17"
363                 radius: 6
364                 border.color: uzunlukKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
365                 border.width: 1
366             }
367         }
368
369         TextField {
370             id: debiKutusu
371             width: parent.width
372             height: 34
373             placeholderText: txt("hedefDebi")
374             placeholderTextColor: "#4b5563"
375             color: "ffffff"
376             font.pixelSize: 12
377             leftPadding: 10
378             verticalAlignment: TextInput.AlignVCenter
379             validator: DoubleValidator { bottom: 0; top: 200; decimals: 1 }
380             background: Rectangle {
381                 color: "#060d17"
382                 radius: 6
383                 border.color: debiKutusu.activeFocus ? "#3b82f6" : "#1e2a3f"
384                 border.width: 1
385             }
386         }
387
388         Row {
389             width: parent.width
390             spacing: 8
391
392             Text {
393                 text: txt("hataPayi")
394                 color: "#9ca3af"

```

```

395         font.family: "Segoe UI"
396         font.pixelSize: 11
397         anchors.verticalCenter: parent.verticalCenter
398     }
399
400     ComboBox {
401         id: hataPayiSecici
402         width: 90
403         height: 28
404         model: ["0", "10", "20"]
405         currentIndex: 1
406
407         background: Rectangle {
408             color: "#060d17"
409             radius: 6
410             border.color: hataPayiSecici.activeFocus ? "#3b82f6" : "#1e2a3f"
411             border.width: 1
412         }
413
414         contentItem: Text {
415             text: hataPayiSecici.displayText
416             color: "#dce8f5"
417             font.family: "Segoe UI"
418             font.pixelSize: 11
419             leftPadding: 10
420             verticalAlignment: Text.AlignVCenter
421         }
422
423         indicator: Text {
424             x: hataPayiSecici.width - width - 8
425             anchors.verticalCenter: parent.verticalCenter
426             text: "▼"
427             color: "#6b7280"
428             font.pixelSize: 11
429         }
430
431         popup: Popup {
432             y: hataPayiSecici.height + 2
433             width: hataPayiSecici.width
434             implicitHeight: contentItem.implicitHeight
435             padding: 1
436
437             background: Rectangle {
438                 color: "#0a0e17"
439                 radius: 6
440                 border.color: "#1e2a3f"
441                 border.width: 1
442             }
443
444             contentItem: ListView {
445                 implicitHeight: contentHeight
446                 model: hataPayiSecici.popup.visible ? hataPayiSecici.delegateModel : null
447                 currentIndex: hataPayiSecici.highlightedIndex
448                 clip: true
449             }
450         }
451
452         delegate: ItemDelegate {
453             id: hataPayiDelege
454             width: hataPayiSecici.width
455             height: 28
456             highlighted: hataPayiSecici.highlightedIndex === index
457
458             background: Rectangle {
459                 color: hataPayiDelege.highlighted ? "#182644" : "#0a0e17"
460             }
461
462             contentItem: Text {
463                 text: modelData
464                 color: "#dce8f5"
465                 font.family: "Segoe UI"
466                 font.pixelSize: 11
467                 leftPadding: 10
468                 verticalAlignment: Text.AlignVCenter
469             }
470         }
471     }
472
473     Text {
474         text: "%"

```

```

475         color: "#9ca3af"
476         font.family: "Segoe UI"
477         font.pixelSize: 11
478         anchors.verticalCenter: parent.verticalCenter
479     }
480 }
481
482 Button {
483     width: parent.width
484     height: 36
485     text: txt("tahminiHesapla")
486     font.pixelSize: 12
487     font.bold: true
488
489     background: Rectangle {
490         radius: 6
491         color: parent.hovered ? "#4f8cf7" : "#3b82f6"
492         Behavior on color { ColorAnimation { duration: 120 } }
493     }
494
495     contentItem: Text {
496         text: parent.text
497         color: "#ffffff"
498         font: parent.font
499         horizontalAlignment: Text.AlignHCenter
500         verticalAlignment: Text.AlignVCenter
501     }
502
503     onClicked: {
504         var D = parseFloat(capKutusu.text)
505         var L = parseFloat(uzunlukKutusu.text)
506         var Q = parseFloat(debiKutusu.text)
507         var tolerans = parseFloat(hataPayiSecici.currentText)
508
509         if (isNaN(D) || isNaN(L) || isNaN(Q) || D <= 0 || L <= 0) {
510             tahminSonucu.text = txt("gecerliDegerGirin")
511             tahminKarsilastirma.text = ""
512             return
513         }
514
515         var sonuc = calculator.boruHattiTahminHesapla(
516             hesaplananTau0, hesaplananMu, Q, D, L, tolerans)
517
518         if (!sonuc.gecerliGiris) {
519             tahminSonucu.text = txt("hesaplanamadi")
520             tahminKarsilastirma.text = ""
521             return
522         }
523
524         tahminSonucu.text = sonuc.basincBar.toFixed(2) + " bar"
525         if (tolerans > 0) {
526             tahminSonucu.text += " (±1: %2 - %3 bar)".arg(tolerans.toFixed(0))
527                 .arg(sonuc.altSinirBar.toFixed(2)).arg(sonuc.ustSinirBar.toFixed(2))
528         }
529
530         // Aynı D/Q için farkli boru uzunluklarında karsilastirma
531         // (orijinal SLIPER'daki "birden fazla boru uzunlugu" karsilastirmasi)
532         var uzunluklar = [L * 0.5, L, L * 2]
533         var satirlar = []
534         for (var i = 0; i < uzunluklar.length; i++) {
535             var s = calculator.boruHattiTahminHesapla(
536                 hesaplananTau0, hesaplananMu, Q, D, uzunluklar[i], 0)
537             satirlar.push(uzunluklar[i].toFixed(0) + " m: " + s.basincBar.toFixed(2) + " bar")
538         }
539         tahminKarsilastirma.text = satirlar.join(" | ")
540     }
541 }
542
543 Text {
544     id: tahminSonucu
545     width: parent.width
546     text: "-"
547     color: "#e8a020"
548     font.family: "Segoe UI"
549     font.pixelSize: 18
550     font.bold: true
551     wrapMode: Text.WordWrap
552     horizontalAlignment: Text.AlignHCenter
553 }
554

```



```

555         Text {
556             id: tahminKarsilastirma
557             width: parent.width
558             text: ""
559             color: "#6b7280"
560             font.family: "Segoe UI"
561             font.pixelSize: 10
562             wrapMode: Text.WordWrap
563             horizontalAlignment: Text.AlignHCenter
564         }
565     }
566 }
567
568 Rectangle {
569     id: durumKutusu
570     width: parent.width
571     height: durumIcerik.implicitHeight + 24
572     radius: 10
573     color: durumIyiMi ? "#0f2417" : "#2a1414"
574     border.color: durumIyiMi ? "#16a34a" : "#dc2626"
575     border.width: 1
576
577     property bool durumIyiMi: true
578     property bool tau0Yuksek: false
579     property bool muYuksek: false
580
581     Column {
582         id: durumIcerik
583         anchors.top: parent.top
584         anchors.horizontalCenter: parent.horizontalCenter
585         anchors.topMargin: 12
586         spacing: 8
587         width: parent.width - 24
588
589         Row {
590             anchors.horizontalCenter: parent.horizontalCenter
591             spacing: 10
592
593             Rectangle {
594                 width: 24
595                 height: 24
596                 radius: 12
597                 anchors.verticalCenter: parent.verticalCenter
598                 color: durumKutusu.durumIyiMi ? "#16a34a" : "#dc2626"
599
600                 Text {
601                     anchors.centerIn: parent
602                     text: durumKutusu.durumIyiMi ? "✓" : "X"
603                     color: "ffffff"
604                     font.pixelSize: 13
605                     font.bold: true
606                 }
607             }
608
609             Text {
610                 anchors.verticalCenter: parent.verticalCenter
611                 text: durumKutusu.durumIyiMi ? txt("pompalanabilir") : txt("pompalamaSorunu")
612                 color: durumKutusu.durumIyiMi ? "#4ade80" : "#f87171"
613                 font.family: "Segoe UI"
614                 font.pixelSize: 15
615                 font.bold: true
616             }
617         }
618
619         Text {
620             width: parent.width
621             horizontalAlignment: Text.AlignHCenter
622             visible: !durumKutusu.durumIyiMi
623             text: {
624                 if (durumKutusu.tau0Yuksek && durumKutusu.muYuksek) {
625                     return txt("durumHerIkisiYuksek")
626                 } else if (durumKutusu.tau0Yuksek) {
627                     return txt("durumTau0Yuksek")
628                 } else if (durumKutusu.muYuksek) {
629                     return txt("durumMuYuksek")
630                 }
631                 return txt("durumYetersizVeri")
632             }
633             color: "#9ca3af"
634             font.family: "Segoe UI"

```

```

635         font.pixelSize: 11
636         wrapMode: Text.WordWrap
637     }
638 }
639 }
640
641 Button {
642     width: parent.width
643     height: 40
644     text: txt("pdfRaporOnizle")
645     font.pixelSize: 13
646     font.bold: true
647     enabled: olcumId > 0
648
649     onClicked: {
650         pdfOnizlemePopup.onizlemeHtml = reportManager.pdfOnizlemeHtml(
651             olcumId,
652             musteriAdi,
653             receteAdi,
654             hesaplananTau0,
655             hesaplananMu,
656             parseFloat(r2Metni.text)
657         )
658         pdfOnizlemePopup.open()
659     }
660
661     background: Rectangle {
662         radius: 8
663         color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
664         Behavior on color { ColorAnimation { duration: 120 } }
665     }
666
667     contentItem: Text {
668         text: parent.text
669         color: "#ffffff"
670         font: parent.font
671         horizontalAlignment: Text.AlignHCenter
672         verticalAlignment: Text.AlignVCenter
673     }
674 }
675
676 Button {
677     width: parent.width
678     height: 40
679     text: txt("excelCsvOnizle")
680     font.pixelSize: 13
681     font.bold: true
682     enabled: olcumId > 0
683
684     onClicked: {
685         csvOnizlemePopup.satirlar = database.strokeVerileriGetir(olcumId)
686         csvOnizlemePopup.open()
687     }
688
689     background: Rectangle {
690         radius: 8
691         color: parent.enabled ? (parent.hovered ? "#22c55e" : "#16a34a") : "#1e2a3f"
692         Behavior on color { ColorAnimation { duration: 120 } }
693     }
694
695     contentItem: Text {
696         text: parent.text
697         color: "#ffffff"
698         font: parent.font
699         horizontalAlignment: Text.AlignHCenter
700         verticalAlignment: Text.AlignVCenter
701     }
702 }
703
704 Button {
705     width: parent.width
706     height: 40
707     text: txt("excelXmlDisaAktar")
708     font.pixelSize: 13
709     font.bold: true
710     enabled: olcumId > 0
711
712     onClicked: {
713         var yol = database.xmlDisaAktar(olcumId)
714         xmlDisaAktarBildirim.text = yol.length > 0

```

```

715         ? txt("disaAktarildi") + yol
716         : txt("disaAktarilamadi")
717         xmlDisaAktarBildirim.visible = true
718         xmlDisaAktarTimer.restart()
719     }
720
721     background: Rectangle {
722         radius: 8
723         color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
724         Behavior on color { ColorAnimation { duration: 120 } }
725     }
726
727     contentItem: Text {
728         text: parent.text
729         color: "#ffffff"
730         font: parent.font
731         horizontalAlignment: Text.AlignHCenter
732         verticalAlignment: Text.AlignVCenter
733     }
734 }
735
736 Text {
737     id: xmlDisaAktarBildirim
738     width: parent.width
739     visible: false
740     text: ""
741     color: "#9ca3af"
742     font.family: "Segoe UI"
743     font.pixelSize: 10
744     wrapMode: Text.WrapAnywhere
745     horizontalAlignment: Text.AlignHCenter
746
747     Timer {
748         id: xmlDisaAktarTimer
749         interval: 4000
750         onTriggered: xmlDisaAktarBildirim.visible = false
751     }
752 }
753
754 Rectangle {
755     width: parent.width
756     radius: 10
757     color: "#0a0e17"
758     border.color: "#1e2a3f"
759     border.width: 1
760     height: playbackIcerik.implicitHeight + 24
761     visible: olcumId > 0
762
763     Column {
764         id: playbackIcerik
765         anchors.top: parent.top
766         anchors.left: parent.left
767         anchors.right: parent.right
768         anchors.margins: 12
769         spacing: 8
770
771         Text {
772             text: txt("oynatmaPlayback")
773             color: "#6b7280"
774             font.family: "Segoe UI"
775             font.pixelSize: 10
776             font.letterSpacing: 1
777         }
778
779         Text {
780             text: strokeModeli.count > 0
781                 ? txt("strokeEtiket") + " " + (playbackAdimi + 1) + " / " + strokeModeli.count
782                 : txt("oynatilacakStrokeYok")
783             color: "#dce8f5"
784             font.family: "Segoe UI"
785             font.pixelSize: 12
786         }
787
788         Flow {
789             width: parent.width
790             spacing: 5
791
792             Button {
793                 width: 40
794                 height: 34

```

```

795         text: oynatiliyor ? "||" : "▶"
796         font.pixelSize: 14
797         enabled: strokeModeli.count > 0
798
799         onClicked: {
800             if (oynatiliyor) {
801                 oynatiliyor = false
802                 playbackTimer.stop()
803             } else {
804                 if (playbackAdimi >= strokeModeli.count - 1) {
805                     playbackAdimi = -1
806                 }
807                 oynatiliyor = true
808                 playbackTimer.start()
809             }
810         }
811
812         background: Rectangle {
813             radius: 6
814             color: parent.enabled ? (parent.hovered ? "#4f8cf7" : "#3b82f6") : "#1e2a3f"
815             Behavior on color { ColorAnimation { duration: 120 } }
816         }
817
818         contentItem: Text {
819             text: parent.text
820             color: "#ffffff"
821             font: parent.font
822             horizontalAlignment: Text.AlignHCenter
823             verticalAlignment: Text.AlignVCenter
824         }
825     }
826
827     Repeater {
828         model: [1, 2, 5, 10]
829
830         Button {
831             width: 38
832             height: 34
833             text: modelData + "x"
834             font.pixelSize: 12
835
836             onClicked: playbackHizi = modelData
837
838             background: Rectangle {
839                 radius: 6
840                 color: playbackHizi === modelData ? "#3b82f6" : "#0f1420"
841                 border.color: "#1e2a3f"
842                 border.width: 1
843             }
844
845             contentItem: Text {
846                 text: parent.text
847                 color: playbackHizi === modelData ? "#ffffff" : "#9ca3af"
848                 font: parent.font
849                 horizontalAlignment: Text.AlignHCenter
850                 verticalAlignment: Text.AlignVCenter
851             }
852         }
853     }
854 }
855
856     }
857 }
858
859 }
860
861 Rectangle {
862     id: panelAyraci
863     width: 3
864     height: parent.height
865     color: "#05070b"
866
867     Rectangle {
868         anchors.right: parent.right
869         anchors.top: parent.top
870         anchors.bottom: parent.bottom
871         width: 1
872         color: "#403b82f6"
873     }
874 }

```

```

875
876 Item {
877     width: parent.width - ozetPaneli.width - panelAyraci.width
878     height: parent.height
879
880     Rectangle {
881         id: grafikKutusu
882         anchors.top: parent.top
883         anchors.left: parent.left
884         anchors.right: parent.right
885         anchors.margins: 16
886         height: parent.height * 0.55
887         radius: 10
888         color: "#0f1420"
889         border.color: "#1e2a3f"
890         border.width: 1
891
892         Item {
893             id: grafikBaslik
894             anchors.top: parent.top
895             anchors.left: parent.left
896             anchors.right: parent.right
897             anchors.margins: 14
898             height: 20
899
900             Text {
901                 anchors.left: parent.left
902                 anchors.verticalCenter: parent.verticalCenter
903                 text: txt("pqDagilimGrafigi")
904                 color: "#9ca3af"
905                 font.family: "Segoe UI"
906                 font.pixelSize: 13
907                 font.bold: true
908             }
909
910             Row {
911                 anchors.right: parent.right
912                 anchors.verticalCenter: parent.verticalCenter
913                 spacing: 16
914
915                 Row {
916                     spacing: 6
917                     anchors.verticalCenter: parent.verticalCenter
918
919                     Rectangle {
920                         width: 8
921                         height: 8
922                         radius: 4
923                         color: "#3b82f6"
924                         anchors.verticalCenter: parent.verticalCenter
925                     }
926
927                     Text {
928                         text: txt("olcumNoktalar")
929                         color: "#6b7280"
930                         font.family: "Segoe UI"
931                         font.pixelSize: 11
932                         anchors.verticalCenter: parent.verticalCenter
933                     }
934                 }
935
936                 Row {
937                     spacing: 6
938                     anchors.verticalCenter: parent.verticalCenter
939
940                     Rectangle {
941                         width: 12
942                         height: 2
943                         color: "#e8a020"
944                         anchors.verticalCenter: parent.verticalCenter
945                     }
946
947                     Text {
948                         text: txt("regresyonDogrusu")
949                         color: "#6b7280"
950                         font.family: "Segoe UI"
951                         font.pixelSize: 11
952                         anchors.verticalCenter: parent.verticalCenter
953                     }
954                 }
955             }
956         }
957     }
958 }

```

```

955
956         Row {
957             spacing: 6
958             anchors.verticalCenter: parent.verticalCenter
959
960             Rectangle {
961                 width: 8
962                 height: 8
963                 radius: 4
964                 color: "#e8a020"
965                 anchors.verticalCenter: parent.verticalCenter
966             }
967
968             Text {
969                 text: txt("oynatmaEtiket")
970                 color: "#6b7280"
971                 font.family: "Segoe UI"
972                 font.pixelSize: 11
973                 anchors.verticalCenter: parent.verticalCenter
974             }
975         }
976     }
977 }
978
979 ChartView {
980     anchors.top: grafikBaslik.bottom
981     anchors.left: parent.left
982     anchors.right: parent.right
983     anchors.bottom: parent.bottom
984     anchors.topMargin: 4
985     anchors.margins: 8
986     backgroundColor: "transparent"
987     legend.visible: false
988     antialiasing: true
989
990     ValueAxis {
991         id: qEkseni
992         min: 0
993         max: 20
994         titleText: Translations.turkish ? "Debi (Q) m³/h" : "Flow Rate (Q) m³/h"
995         gridLineColor: "#182131"
996         labelsColor: "#4b5563"
997         labelsFont.pixelSize: 9
998         lineVisible: false
999         minorGridVisible: false
1000     }
1001
1002     ValueAxis {
1003         id: pEkseni
1004         min: 0
1005         max: 100
1006         titleText: Translations.turkish ? "Basınç (P) mbar" : "Pressure (P) mbar"
1007         gridLineColor: "#182131"
1008         labelsColor: "#4b5563"
1009         labelsFont.pixelSize: 9
1010         lineVisible: false
1011         minorGridVisible: false
1012     }
1013
1014     ScatterSeries {
1015         id: pqSerisi
1016         axisX: qEkseni
1017         axisY: pEkseni
1018         color: "#3b82f6"
1019         markerSize: 10
1020     }
1021
1022     LineSeries {
1023         id: regresyonCizgisi
1024         axisX: qEkseni
1025         axisY: pEkseni
1026         color: "#e8a020"
1027         width: 2
1028     }
1029
1030     ScatterSeries {
1031         id: oynatmaSerisi
1032         axisX: qEkseni
1033         axisY: pEkseni
1034         color: "#e8a020"

```

```

1035         markerSize: 16
1036     }
1037 }
1038 }
1039
1040 Rectangle {
1041     anchors.top: grafikKutusu.bottom
1042     anchors.left: parent.left
1043     anchors.right: parent.right
1044     anchors.bottom: parent.bottom
1045     anchors.margins: 16
1046     anchors.topMargin: 8
1047     radius: 10
1048     color: "#0f1420"
1049     border.color: "#1e2a3f"
1050     border.width: 1
1051
1052     Text {
1053         anchors.top: parent.top
1054         anchors.left: parent.left
1055         anchors.margins: 14
1056         text: txt("strokeTablosu")
1057         color: "#9ca3af"
1058         font.family: "Segoe UI"
1059         font.pixelSize: 13
1060         font.bold: true
1061     }
1062
1063     Row {
1064         id: tabloBasligi
1065         anchors.top: parent.top
1066         anchors.left: parent.left
1067         anchors.right: parent.right
1068         anchors.topMargin: 36
1069         anchors.leftMargin: 12
1070         anchors.rightMargin: 12
1071
1072         Text { width: parent.width * 0.25; text: txt("strokeBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold: true;
font.letterSpacing: 1 }
1073         Text { width: parent.width * 0.25; text: "P (mbar)"; color: "#6b7280"; font.pixelSize: 11; font.bold: true;
font.letterSpacing: 1 }
1074         Text { width: parent.width * 0.25; text: "Q (m³/h)"; color: "#6b7280"; font.pixelSize: 11; font.bold: true;
font.letterSpacing: 1 }
1075         Text { width: parent.width * 0.25; text: txt("durumBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold: true;
font.letterSpacing: 1 }
1076     }
1077
1078     Rectangle {
1079         anchors.top: tabloBasligi.bottom
1080         anchors.left: parent.left
1081         anchors.right: parent.right
1082         anchors.topMargin: 8
1083         anchors.leftMargin: 12
1084         anchors.rightMargin: 12
1085         height: 1
1086         color: "#1e2a3f"
1087     }
1088
1089     ListView {
1090         anchors.top: tabloBasligi.bottom
1091         anchors.left: parent.left
1092         anchors.right: parent.right
1093         anchors.bottom: parent.bottom
1094         anchors.margins: 12
1095         anchors.topMargin: 14
1096         clip: true
1097
1098         model: ListModel {
1099             id: strokeModeli
1100         }
1101
1102         delegate: Rectangle {
1103             width: parent.width
1104             height: 32
1105             radius: 6
1106             color: index === playbackAdimi ? "#1e2a3f" : (index % 2 === 0 ? "transparent" : "#0a0e17")
1107
1108             Row {
1109                 anchors.fill: parent
1110                 anchors.leftMargin: 6

```

```

1111
1112 Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter; text: stroke; color:
"#dce8f5"; font.pixelSize: 12 }
1113 Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter; text: p.toFixed(1); color:
"#dce8f5"; font.pixelSize: 12 }
1114 Text { width: parent.width * 0.25; anchors.verticalCenter: parent.verticalCenter; text: q.toFixed(2); color:
"#dce8f5"; font.pixelSize: 12 }
1115
1116 Item {
1117     width: parent.width * 0.25
1118     height: parent.height
1119
1120     Rectangle {
1121         anchors.verticalCenter: parent.verticalCenter
1122         width: durumMetni.implicitWidth + 16
1123         height: 20
1124         radius: 10
1125         color: gecerli ? "#123321" : "#2a1414"
1126
1127         Text {
1128             id: durumMetni
1129             anchors.centerIn: parent
1130             text: gecerli ? txt("gecerliEtiket") : txt("hataliEtiket")
1131             color: gecerli ? "#4ade80" : "#f87171"
1132             font.pixelSize: 11
1133             font.bold: true
1134         }
1135     }
1136 }
1137 }
1138 }
1139 }
1140 }
1141 }
1142 }
1143
1144 Rectangle {
1145     id: bildirimKutusu
1146     property bool hataMi: false
1147
1148     function goster(mesaj, hata) {
1149         bildirimMetni.text = mesaj
1150         hataMi = hata
1151         opacity = 1
1152         bildirimTimer.restart()
1153     }
1154
1155     visible: opacity > 0
1156     opacity: 0
1157     z: 100
1158     anchors.bottom: parent.bottom
1159     anchors.horizontalCenter: parent.horizontalCenter
1160     anchors.bottomMargin: 24
1161     width: bildirimMetni.implicitWidth + 40
1162     height: 44
1163     radius: 10
1164     color: hataMi ? "#7f1d1d" : "#14532d"
1165     border.color: hataMi ? "#dc2626" : "#22c55e"
1166     border.width: 1
1167
1168     Behavior on opacity { NumberAnimation { duration: 200 } }
1169
1170     Text {
1171         id: bildirimMetni
1172         anchors.centerIn: parent
1173         text: ""
1174         color: "#ffffff"
1175         font.family: "Segoe UI"
1176         font.pixelSize: 13
1177         font.bold: true
1178     }
1179
1180     Timer {
1181         id: bildirimTimer
1182         interval: 3000
1183         onTriggered: bildirimKutusu.opacity = 0
1184     }
1185 }
1186
1187 Popup {

```



```

1188     id: pdfOnizlemePopup
1189     modal: true
1190     focus: true
1191     anchors.centerIn: parent
1192     width: 640
1193     height: Math.min(720, parent.height - 40)
1194     padding: 0
1195     closePolicy: Popup.CloseOnEscape
1196
1197     property string onizlemeHtml: ""
1198
1199     background: Rectangle {
1200         color: "#0f1420"
1201         radius: 12
1202         border.color: "#1e2a3f"
1203         border.width: 1
1204     }
1205
1206     Overlay.modal: Rectangle {
1207         color: "#a6000000"
1208     }
1209
1210     contentItem: Column {
1211         width: pdfOnizlemePopup.width
1212         spacing: 0
1213
1214         Rectangle {
1215             width: parent.width
1216             height: 52
1217             color: "#0a0e17"
1218
1219             Text {
1220                 anchors.left: parent.left
1221                 anchors.leftMargin: 20
1222                 anchors.verticalCenter: parent.verticalCenter
1223                 text: txt("pdfRaporOnizlemeBaslik")
1224                 color: "#dce8f5"
1225                 font.family: "Segoe UI"
1226                 font.pixelSize: 14
1227                 font.bold: true
1228             }
1229         }
1230
1231         Rectangle {
1232             width: parent.width
1233             height: pdfOnizlemePopup.height - 52 - 64
1234             color: "#f3f4f6"
1235
1236             Flickable {
1237                 anchors.fill: parent
1238                 anchors.margins: 20
1239                 contentWidth: width
1240                 contentHeight: onizlemeMetni.implicitHeight
1241                 clip: true
1242                 boundsBehavior: Flickable.StopAtBounds
1243
1244                 TextEdit {
1245                     id: onizlemeMetni
1246                     width: parent.width
1247                     readOnly: true
1248                     selectByMouse: true
1249                     textFormat: TextEdit.RichText
1250                     wrapMode: Text.WordWrap
1251                     text: pdfOnizlemePopup.onizlemeHtml
1252                     color: "#111827"
1253                     font.pixelSize: 13
1254                 }
1255             }
1256         }
1257
1258         Rectangle {
1259             width: parent.width
1260             height: 64
1261             color: "#0a0e17"
1262
1263             Row {
1264                 anchors.centerIn: parent
1265                 spacing: 12
1266
1267                 Button {

```

```

1268         width: 140
1269         height: 38
1270         text: txt("indirPdf")
1271         font.pixelSize: 13
1272         font.bold: true
1273
1274         onClicked: {
1275             var yol = reportManager.pdfOlustur(
1276                 olcumId,
1277                 musteriAdi,
1278                 receteAdi,
1279                 hesaplananTau0,
1280                 hesaplananMu,
1281                 parseFloat(r2Metni.text)
1282             )
1283             pdfOnizlemePopup.close()
1284             if (yol.length > 0) {
1285                 bildirimKutusu.goster(txt("pdfKaydedildi") + yol, false)
1286             } else {
1287                 bildirimKutusu.goster(txt("pdfOlusturulamadi"), true)
1288             }
1289         }
1290
1291         background: Rectangle {
1292             radius: 8
1293             color: parent.hovered ? "#4f8cf7" : "#3b82f6"
1294             Behavior on color { ColorAnimation { duration: 120 } }
1295         }
1296
1297         contentItem: Text {
1298             text: parent.text
1299             color: "#ffffff"
1300             font: parent.font
1301             horizontalAlignment: Text.AlignHCenter
1302             verticalAlignment: Text.AlignVCenter
1303         }
1304     }
1305
1306     Button {
1307         width: 100
1308         height: 38
1309         text: txt("kapat")
1310         font.pixelSize: 13
1311
1312         onClicked: pdfOnizlemePopup.close()
1313
1314         background: Rectangle {
1315             radius: 8
1316             color: parent.hovered ? "#1e2a3f" : "transparent"
1317             border.color: "#1e2a3f"
1318             border.width: 1
1319             Behavior on color { ColorAnimation { duration: 120 } }
1320         }
1321
1322         contentItem: Text {
1323             text: parent.text
1324             color: "#9ca3af"
1325             font: parent.font
1326             horizontalAlignment: Text.AlignHCenter
1327             verticalAlignment: Text.AlignVCenter
1328         }
1329     }
1330 }
1331
1332 }
1333
1334
1335 Popup {
1336     id: csvOnizlemePopup
1337     modal: true
1338     focus: true
1339     anchors.centerIn: parent
1340     width: 640
1341     height: Math.min(680, parent.height - 40)
1342     padding: 0
1343     closePolicy: Popup.CloseOnEscape
1344
1345     property var satirlar: []
1346
1347     background: Rectangle {

```

```

1348         color: "#0f1420"
1349         radius: 12
1350         border.color: "#1e2a3f"
1351         border.width: 1
1352     }
1353
1354     Overlay.modal: Rectangle {
1355         color: "#a6000000"
1356     }
1357
1358     contentItem: Column {
1359         width: csvOnizlemePopup.width
1360         spacing: 0
1361
1362         Rectangle {
1363             width: parent.width
1364             height: 52
1365             color: "#0a0e17"
1366
1367             Text {
1368                 anchors.left: parent.left
1369                 anchors.leftMargin: 20
1370                 anchors.verticalCenter: parent.verticalCenter
1371                 text: txt("excelCsvOnizlemeBaslik")
1372                 color: "#dce8f5"
1373                 font.family: "Segoe UI"
1374                 font.pixelSize: 14
1375                 font.bold: true
1376             }
1377         }
1378
1379         Item {
1380             width: parent.width
1381             height: csvBilgiSutunu.implicitHeight + 32
1382
1383             Column {
1384                 id: csvBilgiSutunu
1385                 anchors.left: parent.left
1386                 anchors.right: parent.right
1387                 anchors.top: parent.top
1388                 anchors.margins: 20
1389                 spacing: 4
1390
1391                 Text { text: txt("olcumIdEtiket") + olcumId; color: "#9ca3af"; font.family: "Segoe UI"; font.pixelSize: 12 }
1392                 Text { text: txt("tarihEtiket") + olcumTarihi; color: "#9ca3af"; font.family: "Segoe UI"; font.pixelSize: 12 }
1393                 Text { text: txt("musteriEtiket") + musteriAdi; color: "#9ca3af"; font.family: "Segoe UI"; font.pixelSize: 12 }
1394                 Text { text: txt("receteEtiket") + receteAdi; color: "#9ca3af"; font.family: "Segoe UI"; font.pixelSize: 12 }
1395                 Text { text: txt("agirlikEtiket") + agirlikDegeri.toFixed(1); color: "#9ca3af"; font.family: "Segoe UI";
font.pixelSize: 12 }
1396             }
1397         }
1398
1399         Rectangle {
1400             width: parent.width
1401             height: csvOnizlemePopup.height - 52 - 130 - 64
1402             color: "#0a0e17"
1403             border.color: "#1e2a3f"
1404             border.width: 1
1405
1406             Row {
1407                 id: csvTabloBasligi
1408                 anchors.top: parent.top
1409                 anchors.left: parent.left
1410                 anchors.right: parent.right
1411                 anchors.margins: 12
1412
1413                 Text { width: parent.width * 0.2; text: txt("strokeBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold: true }
1414                 Text { width: parent.width * 0.2; text: txt("basincBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold: true }
1415                 Text { width: parent.width * 0.2; text: txt("konumBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold: true }
1416                 Text { width: parent.width * 0.2; text: txt("debiBuyukBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold: true }
1417
1418                 Text { width: parent.width * 0.2; text: txt("gecerliBuyukBaslik"); color: "#6b7280"; font.pixelSize: 11; font.bold:
true }
1419             }
1420
1421             ListView {
1422                 anchors.top: csvTabloBasligi.bottom
1423                 anchors.left: parent.left
1424                 anchors.right: parent.right
1425                 anchors.bottom: parent.bottom

```

```

1425         anchors.margins: 12
1426         anchors.topMargin: 8
1427         clip: true
1428
1429         model: csvOnizlemePopup.satirlar
1430
1431         delegate: Rectangle {
1432             width: ListView.view.width
1433             height: 26
1434             color: index % 2 === 0 ? "transparent" : "#0f1420"
1435
1436             Row {
1437                 anchors.fill: parent
1438
1439                 Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter; text: modelData.stroke;
color: "#dce8f5"; font.pixelSize: 12 }
1440                 Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter; text:
modelData.basinc.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }
1441                 Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter; text:
modelData.konum.toFixed(1); color: "#dce8f5"; font.pixelSize: 12 }
1442                 Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter; text:
modelData.debi.toFixed(2); color: "#dce8f5"; font.pixelSize: 12 }
1443                 Text { width: parent.width * 0.2; anchors.verticalCenter: parent.verticalCenter; text: modelData.gecerli ?
txt("evet") : txt("hayir"); color: modelData.gecerli ? "#4ade80" : "#f87171"; font.pixelSize: 12 }
1444             }
1445         }
1446     }
1447 }
1448
1449 Rectangle {
1450     width: parent.width
1451     height: 64
1452     color: "#0a0e17"
1453
1454     Row {
1455         anchors.centerIn: parent
1456         spacing: 12
1457
1458         Button {
1459             width: 140
1460             height: 38
1461             text: txt("indirCsv")
1462             font.pixelSize: 13
1463             font.bold: true
1464
1465             onClicked: {
1466                 var yol = database.csvDisaAktar(olcumId)
1467                 csvOnizlemePopup.close()
1468                 if (yol.length > 0) {
1469                     bildirimKutusu.goster(txt("csvKaydedildi") + yol, false)
1470                 } else {
1471                     bildirimKutusu.goster(txt("csvOlusturulamadi"), true)
1472                 }
1473             }
1474
1475             background: Rectangle {
1476                 radius: 8
1477                 color: parent.hovered ? "#22c55e" : "#16a34a"
1478                 Behavior on color { ColorAnimation { duration: 120 } }
1479             }
1480
1481             contentItem: Text {
1482                 text: parent.text
1483                 color: "#ffffff"
1484                 font: parent.font
1485                 horizontalAlignment: Text.AlignHCenter
1486                 verticalAlignment: Text.AlignVCenter
1487             }
1488         }
1489
1490         Button {
1491             width: 100
1492             height: 38
1493             text: txt("kapat")
1494             font.pixelSize: 13
1495
1496             onClicked: csvOnizlemePopup.close()
1497
1498             background: Rectangle {
1499                 radius: 8

```

```

1500         color: parent.hovered ? "#1e2a3f" : "transparent"
1501         border.color: "#1e2a3f"
1502         border.width: 1
1503         Behavior on color { ColorAnimation { duration: 120 } }
1504     }
1505
1506     contentItem: Text {
1507         text: parent.text
1508         color: "#9ca3af"
1509         font: parent.font
1510         horizontalAlignment: Text.AlignHCenter
1511         verticalAlignment: Text.AlignVCenter
1512     }
1513 }
1514 }
1515 }
1516 }
1517 }
1518 }

```

src/Backend.cpp

```
1 #include "Backend.h"
2 #include <QDebug>
3
4 Backend::Backend(QObject *parent)
5     : QObject(parent)
6 {
7 }
8
9 QString Backend::durum() const
10 {
11     return "Baglanti bekleniyor...";
12 }
13
14 bool Backend::girisYap(const QString &kullaniciAdi, const QString &sifre)
15 {
16     if (kullaniciAdi == "admin" && sifre == "admin123") {
17         qDebug() << "Giris Basarili:" << kullaniciAdi;
18         return true;
19     }
20
21     qDebug() << "Giris Basarisiz:" << kullaniciAdi;
22     return false;
23 }
24
25 bool Backend::sifreDogrula(const QString &sifre)
26 {
27     return sifre == "admin123";
28 }
```

src/Backend.h

```
1  #pragma once
2  #include <QObject>
3
4  class Backend : public QObject
5  {
6      Q_OBJECT
7      Q_PROPERTY(QString durum READ durum CONSTANT)
8
9  public:
10     explicit Backend(QObject *parent = nullptr);
11     QString durum() const;
12     Q_INVOKABLE bool girisYap(const QString &kullaniciAdi, const QString &sifre);
13     Q_INVOKABLE bool sifreDogrula(const QString &sifre);
14 };
```

```

1  #include "Calculator.h"
2  #include <cmath>
3
4  Calculator::Calculator(QObject *parent)
5      : QObject(parent)
6  {
7  }
8
9  QString Calculator::durum() const
10 {
11     return m_durum;
12 }
13
14 int Calculator::strokeSayisi() const
15 {
16     return m_strokeSayisi;
17 }
18
19 bool Calculator::duraklatildi() const
20 {
21     return m_duraklatildi;
22 }
23
24 bool Calculator::sonStrokeGecerli() const
25 {
26     return m_sonStrokeGecerli;
27 }
28
29 void Calculator::konumGuncelle(double konum, double hiz)
30 {
31     if (m_duraklatildi) {
32         return;
33     }
34
35     m_sonHiz = hiz;
36
37     QString yeniDurum;
38
39     if (konum > 450) {
40         yeniDurum = "YUKARIDA";
41     } else if (konum > 20) {
42         yeniDurum = "INIYOR";
43     } else {
44         yeniDurum = "TAMAMLANDI";
45     }
46
47     if (yeniDurum == "YUKARIDA") {
48         m_yukaridaGorulduMu = true;
49     }
50
51     if (yeniDurum == "TAMAMLANDI" && m_durum != "TAMAMLANDI" && m_yukaridaGorulduMu) {
52         // Orijinal SLIPER'daki "invalid stroke" mantığı: hızı negatif veya
53         // sıfır olan stroke'lar geçersiz sayılır (boru gerçekte hareket
54         // etmemiş veya sensör verisi tutarsızdır).
55         m_sonStrokeGecerli = hiz > 0.0;
56         emit sonStrokeGecerliChanged();
57
58         m_strokeSayisi++;
59         m_yukaridaGorulduMu = false;
60         emit strokeSayisiChanged();
61     }
62
63     if (yeniDurum != m_durum) {
64         m_durum = yeniDurum;
65         emit durumChanged();
66     }
67 }
68
69 void Calculator::duraklat()
70 {
71     if (m_duraklatildi) {
72         return;
73     }
74
75     m_duraklatildi = true;
76     emit duraklatildiChanged();
77 }

```



```

78
79 void Calculator::devamEt()
80 {
81     if (!m_duraklatildi) {
82         return;
83     }
84
85     m_duraklatildi = false;
86     emit duraklatildiChanged();
87 }
88
89 void Calculator::sifirla()
90 {
91     m_strokeSayisi = 0;
92     m_durum = "BEKLENIYOR";
93     m_duraklatildi = false;
94     m_yukaridaGorulduMu = false;
95     m_sonStrokeGecerli = true;
96     m_sonHiz = 0.0;
97     emit strokeSayisiChanged();
98     emit durumChanged();
99     emit duraklatildiChanged();
100    emit sonStrokeGecerliChanged();
101 }
102
103 QVariantMap Calculator::boruHattiTahminHesapla(double tau0Mbar, double muMbarHm3,
104                                                double debiM3h, double boruCapiMm,
105                                                double boruUzunluguM,
106                                                double hataPayiYuzde) const
107 {
108     QVariantMap sonuc;
109     sonuc["gecerliGiris"] = false;
110     sonuc["basincBar"] = 0.0;
111     sonuc["altSinirBar"] = 0.0;
112     sonuc["ustSinirBar"] = 0.0;
113
114     if (boruCapiMm <= 0.0 || boruUzunluguM <= 0.0 || debiM3h < 0.0) {
115         return sonuc;
116     }
117
118     const double D_ref = REF_BORU_CAPI_MM / 1000.0; // m
119     const double L_ref = REF_BORU_UZUNLUGU_M; // m
120     const double A_ref = M_PI * (D_ref / 2.0) * (D_ref / 2.0); // m^2
121
122     // Referans borudaki (tau0, mu) katsayılarından "gerçek" Bingham
123     // reolojik parametrelerini (tau0': Pa, mu': Pa.s) geri hesapla.
124     // Bkz. Calculator.h başındaki formül açıklaması.
125     const double tau0Prime = (100.0 * tau0Mbar * D_ref) / (4.0 * L_ref);
126     const double muPrime = (100.0 * muMbarHm3 * A_ref * 3600.0 * D_ref * D_ref) / (32.0 * L_ref);
127
128     const double D_t = boruCapiMm / 1000.0; // m
129     const double L_t = boruUzunluguM; // m
130     const double A_t = M_PI * (D_t / 2.0) * (D_t / 2.0); // m^2
131     const double v_t = (debiM3h / 3600.0) / A_t; // m/s
132
133     const double basincPa = L_t * (4.0 * tau0Prime / D_t + 32.0 * muPrime * v_t / (D_t * D_t));
134     const double basincBar = basincPa / 1.0e5;
135
136     const double tolerans = qBound(0.0, hataPayiYuzde, 100.0) / 100.0;
137
138     sonuc["gecerliGiris"] = true;
139     sonuc["basincBar"] = basincBar;
140     sonuc["altSinirBar"] = basincBar * (1.0 - tolerans);
141     sonuc["ustSinirBar"] = basincBar * (1.0 + tolerans);
142     return sonuc;
143 }
144

```

```

1  #pragma once
2  #include <QObject>
3  #include <QVariantMap>
4
5  class Calculator : public QObject
6  {
7      Q_OBJECT
8      Q_PROPERTY(QString durum READ durum NOTIFY durumChanged)
9      Q_PROPERTY(int strokeSayisi READ strokeSayisi NOTIFY strokeSayisiChanged)
10     Q_PROPERTY(bool duraklatildi READ duraklatildi NOTIFY duraklatildiChanged)
11     Q_PROPERTY(bool sonStrokeGecerli READ sonStrokeGecerli NOTIFY sonStrokeGecerliChanged)
12
13 public:
14     explicit Calculator(QObject *parent = nullptr);
15     QString durum() const;
16     int strokeSayisi() const;
17     bool duraklatildi() const;
18     bool sonStrokeGecerli() const;
19
20     // konum: mesafe sensöründen gelen anlık konum (mm)
21     // hiz: sensorManager.hiz üzerinden hesaplanan anlık hız (m/s)
22     // Stroke tamamlandığında (YUKARIDA -> INIYOR -> TAMAMLANDI geçişinde),
23     // orijinal SLIPER cihazındaki "invalid stroke" mantığına paralel olarak
24     // hız negatif veya sıfırsa (yani boru gerçekte hareket etmediyse ya da
25     // yön/konum sensörü tutarsız veri verdiyse) stroke geçersiz sayılır.
26     Q_INVOKABLE void konumGuncelle(double konum, double hiz);
27     Q_INVOKABLE void duraklat();
28     Q_INVOKABLE void devamEt();
29     Q_INVOKABLE void sifirla();
30
31     // Bingham modelinden (tau0, mu) elde edilen değerleri, hedef pompa
32     // hattının çapı ve uzunluğuna göre ölçekleyerek beklenen basınç kaybını
33     // hesaplar. Referans değerler SLIPER'in kendi ölçüm borusuna aittir
34     // (boru çapı 126 mm, dolum yüksekliği/ölçüm uzunluğu 500 mm -
35     // bkz. Datasheet_SLIPER.pdf / slipermanV13.pdf "Technical specifications").
36     //
37     // NOT: Orijinal Schleibinger SLIPER yazılımının tam hesaplama modeli
38     // kamuya açık değildir. Burada kullanılan yaklaşım, Bingham akışkanlar
39     // için bilinen geliştirilmiş boru basınç kaybı formülüne dayanır:
40     //  $dP/dL = 4 \cdot \tau_0 / D + 32 \cdot \mu \cdot v / D^2$ 
41     // (v: ortalama akış hızı = Q/A). Referans borudaki ölçümden elde edilen
42     // tau0 ve mu, bu formülle referans geometriye göre "geri hesaplanır",
43     // sonra hedef boru geometrisine uygulanır. Bu, sahadaki gerçek pompa
44     // basıncıyla karşılaştırılarak doğrulanmalıdır - kesin mühendislik
45     // referansı olarak kullanılmadan önce gerçek verilerle kalibre edilmeli.
46     Q_INVOKABLE QVariantMap boruHattiTahminHesapla(double tau0Mbar, double muMbarHm3,
47                                                     double debiM3h, double boruCapıMm,
48                                                     double boruUzunluguM,
49                                                     double hataPayiYuzde) const;
50
51 signals:
52     void durumChanged();
53     void strokeSayisiChanged();
54     void duraklatildiChanged();
55     void sonStrokeGecerliChanged();
56
57 private:
58     QString m_durum = "BEKLENIYOR";
59     int m_strokeSayisi = 0;
60     bool m_duraklatildi = false;
61     bool m_yukaridaGorulduMu = false;
62     bool m_sonStrokeGecerli = true;
63     double m_sonHiz = 0.0;
64
65     // Referans geometri: SLIPER'in kendi ölçüm borusu (WifiManager::BORU_CAPI_M ile aynı)
66     static constexpr double REF_BORU_CAPI_MM = 126.0;
67     static constexpr double REF_BORU_UZUNLUGU_M = 0.5;
68 };
69

```

```

1  #include "Database.h"
2  #include <QSqlQuery>
3  #include <QSqlError>
4  #include <QDebug>
5  #include <QDateTime>
6  #include <QVector>
7  #include <QFile>
8  #include <QTextStream>
9  #include <QStandardPaths>
10 #include <QDir>
11
12 Database::Database(QObject *parent)
13     : QObject(parent)
14 {
15     const QString klasor = QStandardPaths::writableLocation(QStandardPaths::AppLocalDataLocation) + "/database";
16     QDir().mkpath(klasor);
17
18     m_veriTabaniDosyaYolu = klasor + "/sliper.db";
19
20     m_db = QSqlDatabase::addDatabase("SQLITE");
21     m_db.setDatabaseName(m_veriTabaniDosyaYolu);
22
23     if (!m_db.open()) {
24         qWarning() << "Veritabani acilamadi:" << m_db.lastError().text();
25     } else {
26         qDebug() << "Veritabani basariyla acildi.";
27         tablolariOlustur();
28     }
29 }
30
31 void Database::tablolariOlustur()
32 {
33     QSqlQuery sorgu;
34
35     sorgu.exec(
36         "CREATE TABLE IF NOT EXISTS Olcumler ("
37         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
38         "tarih TEXT, "
39         "musteri TEXT, "
40         "recete TEXT, "
41         "agirlik REAL"
42         ")");
43
44     sorgu.exec(
45         "CREATE TABLE IF NOT EXISTS Strokelar ("
46         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
47         "olcumId INTEGER, "
48         "basinc REAL, "
49         "konum REAL, "
50         "debi REAL, "
51         "gecerli INTEGER, "
52         "FOREIGN KEY(olcumId) REFERENCES Olcumler(id)"
53         ")");
54
55     );
56
57     QSqlQuery kolonKontrol("PRAGMA table_info(Strokelar)");
58     bool debiKolonuVar = false;
59     while (kolonKontrol.next()) {
60         if (kolonKontrol.value("name").toString() == "debi") {
61             debiKolonuVar = true;
62             break;
63         }
64     }
65
66     if (!debiKolonuVar) {
67         QSqlQuery migrasyon;
68         migrasyon.exec("ALTER TABLE Strokelar ADD COLUMN debi REAL DEFAULT 0");
69     }
70
71     sorgu.exec(
72         "CREATE TABLE IF NOT EXISTS Kalibrasyonlar ("
73         "sensor TEXT PRIMARY KEY, "
74         "deger1 REAL, "
75         "deger2 REAL, "
76         "tarih TEXT"
77         ")");

```

```

78     );
79
80     sorgu.exec(
81         "CREATE TABLE IF NOT EXISTS LoadCellNoktalari ("
82         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
83         "hedefKg REAL, "
84         "hamDeger REAL"
85         ")"
86     );
87
88     sorgu.exec(
89         "CREATE TABLE IF NOT EXISTS MesafeNoktalari ("
90         "id INTEGER PRIMARY KEY AUTOINCREMENT, "
91         "hedefMm REAL, "
92         "hamDeger REAL"
93         ")"
94     );
95
96     sorgu.exec(
97         "CREATE TABLE IF NOT EXISTS EgimKalibrasyon ("
98         "id INTEGER PRIMARY KEY CHECK (id = 1), "
99         "biasX REAL, biasY REAL, biasZ REAL, "
100        "gainX REAL, gainY REAL, gainZ REAL, "
101        "tarih TEXT"
102        ")"
103    );
104
105    sorgu.exec(
106        "CREATE TABLE IF NOT EXISTS KalibrasyonTarihleri ("
107        "sensor TEXT PRIMARY KEY, "
108        "tarih TEXT"
109        ")"
110    );
111
112    qDebug() << "Tablolar hazir.";
113 }
114
115 int Database::olcumBaslat(const QString &musteri, const QString &recete, double agirlik)
116 {
117     QSqlQuery sorgu;
118     sorgu.prepare(
119         "INSERT INTO Olcumler (tarih, muster_i, recete, agirlik) "
120         "VALUES (:tarih, :muster_i, :recete, :agirlik)"
121     );
122
123     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
124     sorgu.bindValue(":muster_i", muster_i);
125     sorgu.bindValue(":recete", recete);
126     sorgu.bindValue(":agirlik", agirlik);
127
128     if (!sorgu.exec()) {
129         qWarning() << "Olcum baslatilamadi:" << sorgu.lastError().text();
130         return -1;
131     }
132
133     int yeniId = sorgu.lastInsertId().toInt();
134     qDebug() << "Yeni olcum baslatildi, id:" << yeniId;
135     return yeniId;
136 }
137
138 void Database::strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli)
139 {
140     if (olcumId <= 0) {
141         qWarning() << "Gecersiz olcumId, stroke kaydedilemedi.";
142         return;
143     }
144
145     QSqlQuery sorgu;
146     sorgu.prepare(
147         "INSERT INTO Strokelar (olcumId, basinc, konum, debi, gecerli) "
148         "VALUES (:olcumId, :basinc, :konum, :debi, :gecerli)"
149     );
150
151     sorgu.bindValue(":olcumId", olcumId);
152     sorgu.bindValue(":basinc", basinc);
153     sorgu.bindValue(":konum", konum);
154     sorgu.bindValue(":debi", debi);
155     sorgu.bindValue(":gecerli", gecerli ? 1 : 0);
156
157     if (!sorgu.exec()) {

```

```

158         qWarning() << "Stroke kaydedilemedi:" << sorgu.lastError().text();
159     } else {
160         qDebug() << "Stroke kaydedildi, olcumId:" << olcumId;
161     }
162 }
163
164 bool Database::olcumSil(int olcumId)
165 {
166     if (olcumId <= 0) {
167         qWarning() << "Gecersiz olcumId, olcum silinemedi.";
168         return false;
169     }
170
171     if (!m_db.transaction()) {
172         qWarning() << "Olcum silme icin transaction baslatilamadi:" << m_db.lastError().text();
173         return false;
174     }
175
176     QSqlQuery strokeSil;
177     strokeSil.prepare("DELETE FROM Strokelar WHERE olcumId = :olcumId");
178     strokeSil.bindValue(":olcumId", olcumId);
179     if (!strokeSil.exec()) {
180         qWarning() << "Strokelar silinemedi:" << strokeSil.lastError().text();
181         m_db.rollback();
182         return false;
183     }
184
185     QSqlQuery olcumSil;
186     olcumSil.prepare("DELETE FROM Olcumler WHERE id = :id");
187     olcumSil.bindValue(":id", olcumId);
188     if (!olcumSil.exec()) {
189         qWarning() << "Olcum silinemedi:" << olcumSil.lastError().text();
190         m_db.rollback();
191         return false;
192     }
193
194     if (!m_db.commit()) {
195         qWarning() << "Olcum silme commit edilemedi:" << m_db.lastError().text();
196         return false;
197     }
198
199     qDebug() << "Olcum silindi, id:" << olcumId;
200     return true;
201 }
202
203 QVariantList Database::tumOlcumleriGetir()
204 {
205     QVariantList sonuclar;
206
207     QSqlQuery sorgu(
208         "SELECT o.id, o.tarih, o.musteri, o.recete, o.agirlik, "
209         "COUNT(s.id) AS strokeSayisi "
210         "FROM Olcumler o "
211         "LEFT JOIN Strokelar s ON s.olcumId = o.id "
212         "GROUP BY o.id "
213         "ORDER BY o.id DESC"
214     );
215
216     while (sorgu.next()) {
217         QVariantMap satir;
218         satir["id"] = sorgu.value("id");
219         satir["tarih"] = sorgu.value("tarih");
220         satir["musteri"] = sorgu.value("musteri");
221         satir["recete"] = sorgu.value("recete");
222         satir["agirlik"] = sorgu.value("agirlik");
223         satir["strokeSayisi"] = sorgu.value("strokeSayisi");
224         sonuclar.append(satir);
225     }
226
227     return sonuclar;
228 }
229
230 QVariantList Database::strokeVerileriGetir(int olcumId)
231 {
232     QVariantList sonuclar;
233
234     QSqlQuery sorgu;
235     sorgu.prepare(
236         "SELECT basinc, konum, debi, gecerli FROM Strokelar "
237         "WHERE olcumId = :olcumId ORDER BY id ASC"

```

```

238 );
239 sorgu.bindValue(":olcumId", olcumId);
240
241 if (!sorgu.exec()) {
242     qWarning() << "Stroke verileri alinamadi:" << sorgu.lastError().text();
243     return sonuclar;
244 }
245
246 int sir = 1;
247 while (sorgu.next()) {
248     QVariantMap satir;
249     satir["stroke"] = sir++;
250     satir["basinc"] = sorgu.value("basinc");
251     satir["konum"] = sorgu.value("konum");
252     satir["debi"] = sorgu.value("debi");
253     satir["gecerli"] = sorgu.value("gecerli").toBool();
254     sonuclar.append(satir);
255 }
256
257 return sonuclar;
258 }
259
260 QVariantMap Database::binghamHesapla(int olcumId)
261 {
262     QVariantMap sonuc;
263
264     QSqlQuery sorgu;
265     sorgu.prepare(
266         "SELECT basinc, debi FROM Strokeler "
267         "WHERE olcumId = :olcumId AND gecerli = 1"
268     );
269     sorgu.bindValue(":olcumId", olcumId);
270
271     QVector<double> pDegerleri;
272     QVector<double> qDegerleri;
273
274     if (sorgu.exec()) {
275         while (sorgu.next()) {
276             pDegerleri.append(sorgu.value("basinc").toDouble());
277             qDegerleri.append(sorgu.value("debi").toDouble());
278         }
279     }
280
281     const int n = pDegerleri.size();
282
283     if (n < 2) {
284         sonuc["tau0"] = 0.0;
285         sonuc["mu"] = 0.0;
286         sonuc["r2"] = 0.0;
287         sonuc["yeterliVeri"] = false;
288         return sonuc;
289     }
290
291     double qToplam = 0, pToplam = 0, qpToplam = 0, qKareToplam = 0;
292     for (int i = 0; i < n; ++i) {
293         qToplam += qDegerleri[i];
294         pToplam += pDegerleri[i];
295         qpToplam += qDegerleri[i] * pDegerleri[i];
296         qKareToplam += qDegerleri[i] * qDegerleri[i];
297     }
298
299     const double qOrt = qToplam / n;
300     const double pOrt = pToplam / n;
301
302     const double payda = qKareToplam - n * qOrt * qOrt;
303     const double mu = (payda != 0.0) ? (qpToplam - n * qOrt * pOrt) / payda : 0.0;
304     const double tau0 = pOrt - mu * qOrt;
305
306     double ssTot = 0, ssRes = 0;
307     for (int i = 0; i < n; ++i) {
308         const double tahmin = tau0 + mu * qDegerleri[i];
309         ssRes += (pDegerleri[i] - tahmin) * (pDegerleri[i] - tahmin);
310         ssTot += (pDegerleri[i] - pOrt) * (pDegerleri[i] - pOrt);
311     }
312
313     const double r2 = (ssTot != 0.0) ? (1.0 - ssRes / ssTot) : 0.0;
314
315     sonuc["tau0"] = tau0;
316     sonuc["mu"] = mu;
317     sonuc["r2"] = r2;

```

```

318     sonuc["yeterliVeri"] = true;
319     return sonuc;
320 }
321
322 bool Database::kalibrasyonKaydet(const QString &sensor, double deger1, double deger2)
323 {
324     QSqlQuery sorgu;
325     sorgu.prepare(
326         "INSERT INTO Kalibrasyonlar (sensor, deger1, deger2, tarih) "
327         "VALUES (:sensor, :deger1, :deger2, :tarih) "
328         "ON CONFLICT(sensor) DO UPDATE SET "
329         "deger1 = :deger1, deger2 = :deger2, tarih = :tarih"
330     );
331
332     sorgu.bindValue(":sensor", sensor);
333     sorgu.bindValue(":deger1", deger1);
334     sorgu.bindValue(":deger2", deger2);
335     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
336
337     if (!sorgu.exec()) {
338         qWarning() << "Kalibrasyon kaydedilemedi:" << sorgu.lastError().text();
339         return false;
340     }
341
342     qDebug() << "Kalibrasyon kaydedildi:" << sensor;
343     return true;
344 }
345
346 QVariantMap Database::kalibrasyonGetir(const QString &sensor)
347 {
348     QVariantMap sonuc;
349
350     QSqlQuery sorgu;
351     sorgu.prepare("SELECT deger1, deger2, tarih FROM Kalibrasyonlar WHERE sensor = :sensor");
352     sorgu.bindValue(":sensor", sensor);
353
354     if (sorgu.exec() && sorgu.next()) {
355         sonuc["deger1"] = sorgu.value("deger1");
356         sonuc["deger2"] = sorgu.value("deger2");
357         sonuc["tarih"] = sorgu.value("tarih");
358         sonuc["mevcut"] = true;
359     } else {
360         sonuc["mevcut"] = false;
361     }
362
363     return sonuc;
364 }
365
366 void Database::kalibrasyonTarihiKaydet(const QString &sensor)
367 {
368     QSqlQuery sorgu;
369     sorgu.prepare(
370         "INSERT INTO KalibrasyonTarihleri (sensor, tarih) VALUES (:sensor, :tarih) "
371         "ON CONFLICT(sensor) DO UPDATE SET tarih = :tarih"
372     );
373
374     sorgu.bindValue(":sensor", sensor);
375     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
376
377     if (!sorgu.exec()) {
378         qWarning() << "Kalibrasyon tarihi kaydedilemedi:" << sorgu.lastError().text();
379     }
380 }
381
382 QString Database::kalibrasyonTarihiGetir(const QString &sensor)
383 {
384     QSqlQuery sorgu;
385     sorgu.prepare("SELECT tarih FROM KalibrasyonTarihleri WHERE sensor = :sensor");
386     sorgu.bindValue(":sensor", sensor);
387
388     if (sorgu.exec() && sorgu.next()) {
389         return sorgu.value("tarih").toString();
390     }
391
392     return QString();
393 }
394
395 QVariantMap Database::olcumBilgisiGetir(int olcumId)
396 {
397     QVariantMap sonuc;

```

```

398
399 QSqlQuery sorgu;
400 sorgu.prepare("SELECT tarih, musteri, recete, agirlik FROM Olcumler WHERE id = :id");
401 sorgu.bindValue(":id", olcumId);
402
403 if (sorgu.exec() && sorgu.next()) {
404     sonuc["tarih"] = sorgu.value("tarih");
405     sonuc["musteri"] = sorgu.value("musteri");
406     sonuc["recete"] = sorgu.value("recete");
407     sonuc["agirlik"] = sorgu.value("agirlik");
408     sonuc["bulundu"] = true;
409 } else {
410     sonuc["bulundu"] = false;
411 }
412
413 return sonuc;
414 }
415
416 // Orjinal SLIPER'daki çok sayfalı Excel/XML dışa aktarıma (bkz. slipermanV13.pdf
417 // bölüm 5.6) yakın format: Office 2003 "SpreadsheetML" XML'i. Ek kütüphane
418 // gerektirmez, Excel bu formatı doğrudan açar. "Results" sayfasında temel
419 // veri + Bingham sonuçları, "Stroke_1".. "Stroke_N" sayfalarında ham stroke
420 // verisi bulunur.
421 QString Database::xmlDisaAktar(int olcumId)
422 {
423     QVariantMap bilgi = olcumBilgisiGetir(olcumId);
424     if (!bilgi["bulundu"].toBool()) {
425         qWarning() << "Olcum bulunamadi, xml olusturulamadi.";
426         return QString();
427     }
428
429     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) + "/SliperRaporlari";
430     if (!QDir().mkpath(klasor)) {
431         qWarning() << "Rapor klasoru olusturulamadi:" << klasor;
432         return QString();
433     }
434
435     QString dosyaAdi = klasor + QString("/SLIPER_Export_%1_%2.xml")
436         .arg(olcumId)
437         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
438
439     QFile dosya(dosyaAdi);
440     if (!dosya.open(QIODevice::WriteOnly | QIODevice::Text)) {
441         qWarning() << "XML dosyasi acilamadi:" << dosyaAdi;
442         return QString();
443     }
444
445     QVariantMap bingham = binghamHesapla(olcumId);
446     QVariantList strokelar = strokeVerileriGetir(olcumId);
447
448     auto kacisliMetin = [] (const QString &s) {
449         QString sonuc = s;
450         sonuc.replace("&", "&");
451         sonuc.replace("<", "<");
452         sonuc.replace(">", ">");
453         return sonuc;
454     };
455
456     QTextStream akis(&dosya);
457     akis.setEncoding(QStringConverter::Utf8);
458
459     akis << "<?xml version='1.0' encoding='UTF-8'?'>\n";
460     akis << "<?mso-application progid='Excel.Sheet'?'>\n";
461     akis << "<Workbook xmlns='urn:schemas-microsoft-com:office:spreadsheet' "
462         "xmlns:o='urn:schemas-microsoft-com:office:office' "
463         "xmlns:x='urn:schemas-microsoft-com:office:excel' "
464         "xmlns:ss='urn:schemas-microsoft-com:office:spreadsheet'>\n";
465
466     // --- Results sayfasi ---
467     akis << "<Worksheet ss:Name='Results'>\n<Table>\n";
468     akis << "<Row><Cell><Data ss:Type='String'>Olcum ID</Data><Cell><Data ss:Type='Number'>"
469         "<< olcumId << "</Data></Cell></Row>\n";
470     akis << "<Row><Cell><Data ss:Type='String'>Tarih</Data><Cell><Data ss:Type='String'>"
471         "<< kacisliMetin(bilgi["tarih"].toString()) << "</Data></Cell></Row>\n";
472     akis << "<Row><Cell><Data ss:Type='String'>Musteri</Data><Cell><Data ss:Type='String'>"
473         "<< kacisliMetin(bilgi["musteri"].toString()) << "</Data></Cell></Row>\n";
474     akis << "<Row><Cell><Data ss:Type='String'>Recete</Data><Cell><Data ss:Type='String'>"
475         "<< kacisliMetin(bilgi["recete"].toString()) << "</Data></Cell></Row>\n";
476     akis << "<Row><Cell><Data ss:Type='String'>Agirlik (kg)</Data><Cell><Data ss:Type='Number'>"
477         "<< bilgi["agirlik"].toDouble() << "</Data></Cell></Row>\n";

```



```

478 akis << "<Row><Cell><Data ss:Type=\"String\">Stroke Sayisi</Data></Cell><Cell><Data ss:Type=\"Number\">"
479 << strokelar.size() << "</Data></Cell></Row>\n";
480 akis << "<Row></Row>\n";
481 akis << "<Row><Cell><Data ss:Type=\"String\">Akma Gerilmesi tau0 (mbar)</Data></Cell><Cell><Data ss:Type=\"Number\">"
482 << bingham["tau0"].toDouble() << "</Data></Cell></Row>\n";
483 akis << "<Row><Cell><Data ss:Type=\"String\">Plastik Viskozite mu (mbar.h/m3)</Data></Cell><Cell><Data ss:Type=\"Number\">"
484 << bingham["mu"].toDouble() << "</Data></Cell></Row>\n";
485 akis << "<Row><Cell><Data ss:Type=\"String\">Uyum Kalitesi R2</Data></Cell><Cell><Data ss:Type=\"Number\">"
486 << bingham["r2"].toDouble() << "</Data></Cell></Row>\n";
487 akis << "</Table>\n</Worksheet>\n";
488
489 // --- Her stroke icin ayri sayfa ---
490 for (int i = 0; i < strokelar.size(); ++i) {
491     QVariantMap s = strokelar[i].toMap();
492     akis << "Worksheet ss:Name=\"Stroke_" << (i + 1) << "\">\n<Table>\n";
493     akis << "<Row><Cell><Data ss:Type=\"String\">Basinc (mbar)</Data></Cell>"
494     << "<Cell><Data ss:Type=\"String\">Konum (mm)</Data></Cell>"
495     << "<Cell><Data ss:Type=\"String\">Debi (m3/h)</Data></Cell>"
496     << "<Cell><Data ss:Type=\"String\">Gecerli</Data></Cell></Row>\n";
497     akis << "<Row><Cell><Data ss:Type=\"Number\">" << s["basinc"].toDouble() << "</Data></Cell>"
498     << "<Cell><Data ss:Type=\"Number\">" << s["konum"].toDouble() << "</Data></Cell>"
499     << "<Cell><Data ss:Type=\"Number\">" << s["debi"].toDouble() << "</Data></Cell>"
500     << "<Cell><Data ss:Type=\"String\">" << (s["gecerli"].toBool() ? "Evet" : "Hayir") << "</Data></Cell></Row>\n";
501     akis << "</Table>\n</Worksheet>\n";
502 }
503
504 akis << "</Workbook>\n";
505
506 dosya.close();
507 qDebug() << "XML disa aktarildi:" << dosyaAdi;
508 return dosyaAdi;
509 }
510
511 // Orijinal SLIPER'daki (Expert Settings > Database Export, sifre korumali)
512 // veritabani disa aktarim ozelligine karsilik gelir. Aktif SQLite baglantisi
513 // dosya uzerinde kilit tutabileceginden, kopyalamadan once WAL/journal'in
514 // diske yazilmasi icin basit bir bekleme (checkpoint) yapilir.
515 QString Database::veriTaniDisaAktar()
516 {
517     if (m_veriTaniDosyaYolu.isEmpty() || !QFile::exists(m_veriTaniDosyaYolu)) {
518         qDebug() << "Disa aktarilacak veritabani bulunamadi.";
519         return QString();
520     }
521
522     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) + "/SliperRaporlari";
523     if (!QDir().mkpath(klasor)) {
524         qDebug() << "Rapor klasoru olusturulamadi:" << klasor;
525         return QString();
526     }
527
528     // Bekleyen yazmalarin diske gitmesini garantiye almak icin.
529     QSqlQuery checkpoint(m_db);
530     checkpoint.exec("PRAGMA wal_checkpoint(FULL)");
531
532     QString hedefDosya = klasor + QString("/sliper_yedek_%1.db")
533     .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
534
535     if (!QFile::copy(m_veriTaniDosyaYolu, hedefDosya)) {
536         qDebug() << "Veritabani disa aktarilamadi.";
537         return QString();
538     }
539
540     qDebug() << "Veritabani disa aktarildi:" << hedefDosya;
541     return hedefDosya;
542 }
543
544 // Orijinal SLIPER'daki "Database Import" ozelligine karsilik gelir. Ice
545 // aktarmadan once mevcut veritabaninin bir yedegi otomatik alinir; sonra
546 // baglanti kapatilip yeni dosya yerine kopyalanir ve yeniden acilir.
547 bool Database::veriTaniIcaAktar(const QString &kaynakDosyaYolu)
548 {
549     if (!QFile::exists(kaynakDosyaYolu)) {
550         qDebug() << "Ice aktarilacak dosya bulunamadi:" << kaynakDosyaYolu;
551         return false;
552     }
553
554     // Guvenlik: mevcut veriyi kaybetmemek icin otomatik yedek al.
555     veriTaniDisaAktar();
556
557     m_db.close();

```

```

558
559 QFile hedef(m_veriTabaniDosyaYolu);
560 if (hedef.exists() && !hedef.remove()) {
561     qWarning() << "Eski veritabani silinemedi, ice aktarim iptal edildi.";
562     m_db.open();
563     return false;
564 }
565
566 if (!QFile::copy(kaynakDosyaYolu, m_veriTabaniDosyaYolu)) {
567     qWarning() << "Veritabani ice aktarilamadi.";
568     m_db.open();
569     return false;
570 }
571
572 if (!m_db.open()) {
573     qWarning() << "Ice aktarim sonrasi veritabani acilamadi:" << m_db.lastError().text();
574     return false;
575 }
576
577 tablolariOlustur();
578 qDebug() << "Veritabani ice aktarildi:" << kaynakDosyaYolu;
579 return true;
580 }
581
582 // Orjinal SLIPER'daki "Database Delete" ozelligine karsilik gelir. Sadece
583 // olcum/stroke verilerini siler; sensor kalibrasyonlari (load cell, mesafe,
584 // egim) korunur, cunku bunlar donanimsal kalibrasyon olup ayri bir islemdir.
585 bool Database::tumVeriyiSil()
586 {
587     if (!m_db.transaction()) {
588         qWarning() << "Tum veriyi silme icin transaction baslatilamadi:" << m_db.lastError().text();
589         return false;
590     }
591
592     QSqlQuery strokeSil;
593     if (!strokeSil.exec("DELETE FROM Strokelar")) {
594         qWarning() << "Strokelar silinemedi:" << strokeSil.lastError().text();
595         m_db.rollback();
596         return false;
597     }
598
599     QSqlQuery olcumSil;
600     if (!olcumSil.exec("DELETE FROM Olcumler")) {
601         qWarning() << "Olcumler silinemedi:" << olcumSil.lastError().text();
602         m_db.rollback();
603         return false;
604     }
605
606     if (!m_db.commit()) {
607         qWarning() << "Tum veriyi silme commit edilemedi:" << m_db.lastError().text();
608         return false;
609     }
610
611     qDebug() << "Tum olcum verileri silindi.";
612     return true;
613 }
614
615 QString Database::csvDisaAktar(int olcumId)
616 {
617     QVariantMap bilgi = olcumBilgisiGetir(olcumId);
618     if (!bilgi["bulundu"].toBool()) {
619         qWarning() << "Olcum bulunamadi, csv olusturulamadi.";
620         return QString();
621     }
622
623     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) + "/SliperRaporlari";
624     if (!QDir().mkpath(klasor)) {
625         qWarning() << "Rapor klasoru olusturulamadi:" << klasor;
626         return QString();
627     }
628
629     QString dosyaAdi = klasor + QString("/SLIPER_Veri_%1_%2.csv")
630         .arg(olcumId)
631         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
632
633     QFile dosya(dosyaAdi);
634     if (!dosya.open(QIODevice::WriteOnly | QIODevice::Text)) {
635         qWarning() << "CSV dosyasi acilamadi:" << dosyaAdi;
636         return QString();
637     }

```

```

638
639 QTextStream akis(&dosya);
640 akis.setEncoding(QStringConverter::Utf8);
641
642 akis << "Olcum ID;" << olcumId << "\n";
643 akis << "Tarih;" << bilgi["tarih"].toString() << "\n";
644 akis << "Musteri;" << bilgi["musteri"].toString() << "\n";
645 akis << "Recete;" << bilgi["recete"].toString() << "\n";
646 akis << "Agirlik (kg);" << bilgi["agirlik"].toDouble() << "\n";
647 akis << "\n";
648 akis << "Stroke;Basinc (mbar);Konum (mm);Debi (m3/h);Gecerli\n";
649
650 QVariantList strokelar = strokeVerileriGetir(olcumId);
651 for (const QVariant &v : strokelar) {
652     QVariantMap s = v.toMap();
653     akis << s["stroke"].toInt() << ";";
654     << s["basinc"].toDouble() << ";";
655     << s["konum"].toDouble() << ";";
656     << s["debi"].toDouble() << ";";
657     << (s["gecerli"].toBool() ? "Evet" : "Hayir") << "\n";
658 }
659
660 dosya.close();
661 qDebug() << "CSV disa aktarildi:" << dosyaAdi;
662 return dosyaAdi;
663 }
664
665 bool Database::loadCellNoktalarKaydet(const QVariantList &noktalar)
666 {
667     if (!m_db.transaction()) {
668         qWarning() << "Load cell noktaları için transaction başlatılamadı:" << m_db.lastError().text();
669         return false;
670     }
671
672     QSqlQuery temizle;
673     if (!temizle.exec("DELETE FROM LoadCellNoktaları")) {
674         qWarning() << "Load cell noktaları temizlenemedi:" << temizle.lastError().text();
675         m_db.rollback();
676         return false;
677     }
678
679     QSqlQuery sorgu;
680     sorgu.prepare(
681         "INSERT INTO LoadCellNoktaları (hedefKg, hamDeger) "
682         "VALUES (:hedefKg, :hamDeger)"
683     );
684
685     for (const QVariant &v : noktalar) {
686         QVariantMap nokta = v.toMap();
687         sorgu.bindValue(":hedefKg", nokta["hedefKg"].toDouble());
688         sorgu.bindValue(":hamDeger", nokta["hamDeger"].toDouble());
689         if (!sorgu.exec()) {
690             qWarning() << "Load cell noktasi kaydedilemedi:" << sorgu.lastError().text();
691             m_db.rollback();
692             return false;
693         }
694     }
695
696     if (!m_db.commit()) {
697         qWarning() << "Load cell noktaları commit edilemedi:" << m_db.lastError().text();
698         return false;
699     }
700
701     qDebug() << "Load cell" << noktalar.size() << "nokta kaydedildi.";
702     kalibrasyonTarihiKaydet("loadcell");
703     return true;
704 }
705
706 QVariantList Database::loadCellNoktalarıGetir()
707 {
708     QVariantList sonuclar;
709
710     QSqlQuery sorgu("SELECT hedefKg, hamDeger FROM LoadCellNoktaları ORDER BY hamDeger ASC");
711
712     while (sorgu.next()) {
713         QVariantMap satir;
714         satir["hedefKg"] = sorgu.value("hedefKg");
715         satir["hamDeger"] = sorgu.value("hamDeger");
716         sonuclar.append(satir);
717     }

```

```

718
719     return sonuclar;
720 }
721
722 bool Database::mesafeNoktalarKaydet(const QVariantList &noktalar)
723 {
724     if (!m_db.transaction()) {
725         qWarning() << "Mesafe noktalarini icin transaction baslatilamadi:" << m_db.lastError().text();
726         return false;
727     }
728
729     QSqlQuery temizle;
730     if (!temizle.exec("DELETE FROM MesafeNoktalar")) {
731         qWarning() << "Mesafe noktalarini temizlenemedi:" << temizle.lastError().text();
732         m_db.rollback();
733         return false;
734     }
735
736     QSqlQuery sorgu;
737     sorgu.prepare(
738         "INSERT INTO MesafeNoktalar (hedefMm, hamDeger) "
739         "VALUES (:hedefMm, :hamDeger)"
740     );
741
742     for (const QVariant &v : noktalar) {
743         QVariantMap nokta = v.toMap();
744         sorgu.bindValue(":hedefMm", nokta["hedefMm"].toDouble());
745         sorgu.bindValue(":hamDeger", nokta["hamDeger"].toDouble());
746         if (!sorgu.exec()) {
747             qWarning() << "Mesafe noktasi kaydedilemedi:" << sorgu.lastError().text();
748             m_db.rollback();
749             return false;
750         }
751     }
752
753     if (!m_db.commit()) {
754         qWarning() << "Mesafe noktalarini commit edilemedi:" << m_db.lastError().text();
755         return false;
756     }
757
758     qDebug() << "Mesafe" << noktalar.size() << "nokta kaydedildi.";
759     kalibrasyonTarihiKaydet("mesafe_olcum");
760     return true;
761 }
762
763 QVariantList Database::mesafeNoktalarGetir()
764 {
765     QVariantList sonuclar;
766
767     QSqlQuery sorgu("SELECT hedefMm, hamDeger FROM MesafeNoktalar ORDER BY hamDeger ASC");
768
769     while (sorgu.next()) {
770         QVariantMap satir;
771         satir["hedefMm"] = sorgu.value("hedefMm");
772         satir["hamDeger"] = sorgu.value("hamDeger");
773         sonuclar.append(satir);
774     }
775
776     return sonuclar;
777 }
778
779 bool Database::egimKalibrasyonuKaydet(double biasX, double biasY, double biasZ,
780                                     double gainX, double gainY, double gainZ)
781 {
782     QSqlQuery sorgu;
783     sorgu.prepare(
784         "INSERT OR REPLACE INTO EgimKalibrasyon (id, biasX, biasY, biasZ, gainX, gainY, gainZ, tarih) "
785         "VALUES (1, :biasX, :biasY, :biasZ, :gainX, :gainY, :gainZ, :tarih)"
786     );
787
788     sorgu.bindValue(":biasX", biasX);
789     sorgu.bindValue(":biasY", biasY);
790     sorgu.bindValue(":biasZ", biasZ);
791     sorgu.bindValue(":gainX", gainX);
792     sorgu.bindValue(":gainY", gainY);
793     sorgu.bindValue(":gainZ", gainZ);
794     sorgu.bindValue(":tarih", QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"));
795
796     if (!sorgu.exec()) {
797         qWarning() << "Egim kalibrasyonu kaydedilemedi:" << sorgu.lastError().text();

```

```
798         return false;
799     }
800
801     qDebug() << "Egim kalibrasyonu kaydedildi.";
802     return true;
803 }
804
805 QVariantMap Database::egimKalibrasyonuGetir()
806 {
807     QVariantMap sonuc;
808
809     QSqlQuery sorgu("SELECT biasX, biasY, biasZ, gainX, gainY, gainZ, tarih FROM EgimKalibrasyon WHERE id = 1");
810
811     if (sorgu.exec() && sorgu.next()) {
812         sonuc["biasX"] = sorgu.value("biasX");
813         sonuc["biasY"] = sorgu.value("biasY");
814         sonuc["biasZ"] = sorgu.value("biasZ");
815         sonuc["gainX"] = sorgu.value("gainX");
816         sonuc["gainY"] = sorgu.value("gainY");
817         sonuc["gainZ"] = sorgu.value("gainZ");
818         sonuc["tarih"] = sorgu.value("tarih");
819         sonuc["mevcut"] = true;
820     } else {
821         sonuc["mevcut"] = false;
822     }
823
824     return sonuc;
825 }
```

```

1  #pragma once
2  #include <QObject>
3  #include <QSqlDatabase>
4  #include <QVariantList>
5  #include <QVariantMap>
6
7  class Database : public QObject
8  {
9      Q_OBJECT
10
11 public:
12     explicit Database(QObject *parent = nullptr);
13
14     Q_INVOKABLE int olcumBaslat(const QString &musteri, const QString &recete, double agirlik);
15     Q_INVOKABLE void strokeKaydet(int olcumId, double basinc, double konum, double debi, bool gecerli);
16     Q_INVOKABLE bool olcumSil(int olcumId);
17     Q_INVOKABLE QVariantList tumOlcumleriGetir();
18     Q_INVOKABLE QVariantList strokeVerileriGetir(int olcumId);
19     Q_INVOKABLE QVariantMap binghamHesapla(int olcumId);
20     Q_INVOKABLE bool kalibrasyonKaydet(const QString &sensor, double deger1, double deger2);
21     Q_INVOKABLE QVariantMap kalibrasyonGetir(const QString &sensor);
22     Q_INVOKABLE QVariantMap olcumBilgisiGetir(int olcumId);
23     Q_INVOKABLE QString csvDisaAktar(int olcumId);
24     Q_INVOKABLE QString xmlDisaAktar(int olcumId);
25     Q_INVOKABLE QString veriTabaniDisaAktar();
26     Q_INVOKABLE bool veriTabaniIcaAktar(const QString &kaynakDosyaYolu);
27     Q_INVOKABLE bool tumVeriyiSil();
28     Q_INVOKABLE bool loadCellNoktalarKaydet(const QVariantList &noktalar);
29     Q_INVOKABLE QVariantList loadCellNoktalarGetir();
30     Q_INVOKABLE bool mesafeNoktalarKaydet(const QVariantList &noktalar);
31     Q_INVOKABLE QVariantList mesafeNoktalarGetir();
32     Q_INVOKABLE bool egimKalibrasyonuKaydet(double biasX, double biasY, double biasZ,
33                                             double gainX, double gainY, double gainZ);
34     Q_INVOKABLE QVariantMap egimKalibrasyonuGetir();
35     Q_INVOKABLE void kalibrasyonTarihiKaydet(const QString &sensor);
36     Q_INVOKABLE QString kalibrasyonTarihiGetir(const QString &sensor);
37
38 private:
39     void tablolariOlustur();
40     QSqlDatabase m_db;
41     QString m_veriTabaniDosyaYolu;
42 };

```

```

1  #include "ReportManager.h"
2  #include <QPrinter>
3  #include <QPainter>
4  #include <QTextDocument>
5  #include <QStandardPaths>
6  #include <QDir>
7  #include <QDateTime>
8  #include <QFile>
9  #include <QDebug>
10
11 ReportManager::ReportManager(QObject *parent)
12     : QObject(parent)
13 {
14 }
15
16 // Önizleme ve gerçek PDF çıktısı bu tek fonksiyondan beslenir; ikisi arasında
17 // tasarım/veri tutarsızlığı oluşmasın diye HTML üretimi burada merkezleştirildi.
18 QString ReportManager::raporHtmlOlustur(int olcumId, const QString &musteri, const QString &recete,
19     double tau0, double mu, double r2)
20 {
21     const bool pompalanabilir = tau0 < 5 && mu < 10;
22     const QString durumRengi = pompalanabilir ? "#16a34a" : "#dc2626";
23     const QString durumMetni = pompalanabilir ? "POMPALANABİLİR" : "POMPALAMA SORUNU";
24
25     return QString(
26         "<html><body style='font-family:Segoe UI, Arial, sans-serif; color:#111827;'>"
27         "<h1 style='color:#3b82f6; margin-bottom:4px;'>SLIPER Analiz Raporu</h1>"
28         "<p style='color:#6b7280; font-size:11px; margin-top:0;'>Liya Laboratuvar Test Cihazları</p>"
29         "<hr style='border:none; border-top:1px solid #d1d5db;'>"
30         "<table style='width:100%; border-collapse:collapse; margin-top:10px;'>"
31         "<tr><td style='padding:4px 0; width:140px;'><b>Ölçüm ID</b></td><td>%1</td></tr>"
32         "<tr><td style='padding:4px 0;'><b>Müşteri</b></td><td>%2</td></tr>"
33         "<tr><td style='padding:4px 0;'><b>Reçete</b></td><td>%3</td></tr>"
34         "<tr><td style='padding:4px 0;'><b>Tarih</b></td><td>%4</td></tr>"
35         "</table>"
36         "<h2 style='color:#111827; margin-top:24px; margin-bottom:8px; font-size:16px;'>Bingham Model Sonuçları</h2>"
37         "<table style='width:100%; border-collapse:collapse;' cellpadding='6'>"
38         "<tr style='background:#f3f4f6;'>"
39         "<th style='text-align:left; border:1px solid #d1d5db;'>Parametre</th>"
40         "<th style='text-align:left; border:1px solid #d1d5db;'>Değer</th></tr>"
41         "<tr><td style='border:1px solid #d1d5db;'>Akma Gerilmesi ( $\tau_0$ )</td><td style='border:1px solid #d1d5db;'>%5 mbar</td></tr>"
42         "<tr><td style='border:1px solid #d1d5db;'>Plastik Viskozite ( $\mu$ )</td><td style='border:1px solid #d1d5db;'>%6 mbar·h/m³</td></tr>"
43         "<tr><td style='border:1px solid #d1d5db;'>Uyum Kalitesi ( $R^2$ )</td><td style='border:1px solid #d1d5db;'>%7</td></tr>"
44         "</table>"
45         "<p style='margin-top:20px;'><b>Durum:</b> "
46         "<span style='color:%8; font-weight:bold;'>%9</span></p>"
47         "<hr style='border:none; border-top:1px solid #d1d5db; margin-top:24px;'>"
48         "<p style='color:#6b7280; font-size:10px;'>Liya Laboratuvar Test Cihazları - SLIPER Analiz Sistemi</p>"
49         "</body></html>"
50     ).arg(olcumId)
51     .arg(musteri.toHtmlEscaped())
52     .arg(recete.toHtmlEscaped())
53     .arg(QDateTime::currentDateTime().toString("dd.MM.yyyy HH:mm"))
54     .arg(QString::number(tau0, 'f', 2))
55     .arg(QString::number(mu, 'f', 2))
56     .arg(QString::number(r2, 'f', 2))
57     .arg(durumRengi)
58     .arg(durumMetni);
59 }
60
61 QString ReportManager::pdfOnizlemeHtml(int olcumId, const QString &musteri, const QString &recete,
62     double tau0, double mu, double r2)
63 {
64     return raporHtmlOlustur(olcumId, muster, recete, tau0, mu, r2);
65 }
66
67 QString ReportManager::pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
68     double tau0, double mu, double r2)
69 {
70     QString klasor = QStandardPaths::writableLocation(QStandardPaths::DocumentsLocation) + "/SliperRaporlari";
71     if (!QDir().mkpath(klasor)) {
72         qWarning() << "Rapor klasoru olusturulamadi:" << klasor;
73         return QString();
74     }
75
76     QString dosyaAdi = klasor + QString("/SLIPER_Rapor_%1_%2.pdf")
77         .arg(olcumId)

```

```
78         .arg(QDateTime::currentDateTime().toString("ddMMyyyy_HHmss"));
79
80     QTextDocument belge;
81     belge.setHtml(raporHtmlOlustur(olcumId, musteri, recete, tau0, mu, r2));
82
83     QPrinter yazici(QPrinter::HighResolution);
84     yazici.setOutputFormat(QPrinter::PdfFormat);
85     yazici.setOutputFileName(dosyaAdi);
86
87     belge.print(&yazici);
88
89     if (!QFile::exists(dosyaAdi)) {
90         qWarning() << "PDF dosyasi yazilamadi:" << dosyaAdi;
91         return QString();
92     }
93
94     qDebug() << "PDF rapor olusturuldu:" << dosyaAdi;
95     return dosyaAdi;
96 }
97
```


src/ReportManager.h

```
1  #pragma once
2  #include <QObject>
3
4  class ReportManager : public QObject
5  {
6      Q_OBJECT
7
8  public:
9      explicit ReportManager(QObject *parent = nullptr);
10     Q_INVOKABLE QString pdfOnizlemeHtml(int olcumId, const QString &musteri, const QString &recete,
11                                         double tau0, double mu, double r2);
12     Q_INVOKABLE QString pdfOlustur(int olcumId, const QString &musteri, const QString &recete,
13                                     double tau0, double mu, double r2);
14
15 private:
16     QString raporHtmlOlustur(int olcumId, const QString &musteri, const QString &recete,
17                             double tau0, double mu, double r2);
18 };
```

```

1  #include "SensorManager.h"
2
3  SensorManager::SensorManager(QObject *parent)
4      : QObject(parent)
5  {
6  }
7
8  double SensorManager::basinc() const { return m_basinc; }
9  double SensorManager::konum() const { return m_konum; }
10 double SensorManager::hiz() const { return m_hiz; }
11 double SensorManager::debi() const { return m_debi; }
12 double SensorManager::egimX() const { return m_egimX; }
13 double SensorManager::egimY() const { return m_egimY; }
14 double SensorManager::hamAgirlik() const { return m_hamAgirlik; }
15 double SensorManager::hamMesafe() const { return m_hamMesafe; }
16 double SensorManager::hamAccelX() const { return m_hamAccelX; }
17 double SensorManager::hamAccelY() const { return m_hamAccelY; }
18 double SensorManager::hamAccelZ() const { return m_hamAccelZ; }
19 bool SensorManager::veriGecerli() const { return m_veriGecerli; }
20 double SensorManager::bataryaVoltaj() const { return m_bataryaVoltaj; }
21
22 void SensorManager::veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX, double egimY)
23 {
24     const bool veriGecerliDegisti = !m_veriGecerli;
25     m_veriGecerli = true;
26
27     if (m_basinc != basinc) { m_basinc = basinc; emit basincChanged(); }
28     if (m_konum != konum) { m_konum = konum; emit konumChanged(); }
29     if (m_hiz != hiz) { m_hiz = hiz; emit hizChanged(); }
30     if (m_debi != debi) { m_debi = debi; emit debiChanged(); }
31     if (m_egimX != egimX) { m_egimX = egimX; emit egimXChanged(); }
32     if (m_egimY != egimY) { m_egimY = egimY; emit egimYChanged(); }
33     if (veriGecerliDegisti) { emit veriGecerliChanged(); }
34
35     emit veriGuncellendi();
36 }
37
38 void SensorManager::hamAgirlikGuncelle(double hamAgirlik)
39 {
40     if (m_hamAgirlik != hamAgirlik) {
41         m_hamAgirlik = hamAgirlik;
42         emit hamAgirlikChanged();
43     }
44 }
45
46 void SensorManager::hamMesafeGuncelle(double hamMesafe)
47 {
48     if (m_hamMesafe != hamMesafe) {
49         m_hamMesafe = hamMesafe;
50         emit hamMesafeChanged();
51     }
52 }
53
54 void SensorManager::hamAccelGuncelle(double x, double y, double z)
55 {
56     if (m_hamAccelX != x || m_hamAccelY != y || m_hamAccelZ != z) {
57         m_hamAccelX = x;
58         m_hamAccelY = y;
59         m_hamAccelZ = z;
60         emit hamAccelDegisti();
61     }
62 }
63
64 void SensorManager::bataryaVoltajGuncelle(double voltaj)
65 {
66     if (m_bataryaVoltaj != voltaj) {
67         m_bataryaVoltaj = voltaj;
68         emit bataryaVoltajChanged();
69     }
70 }
71
72 void SensorManager::veriyiGecersizYap()
73 {
74     if (!m_veriGecerli) return;
75     m_veriGecerli = false;
76     emit veriGecerliChanged();
77 }

```

```

1  #pragma once
2  #include <QObject>
3
4  class SensorManager : public QObject
5  {
6      Q_OBJECT
7      Q_PROPERTY(double basinc READ basinc NOTIFY basincChanged)
8      Q_PROPERTY(double konum READ konum NOTIFY konumChanged)
9      Q_PROPERTY(double hiz READ hiz NOTIFY hizChanged)
10     Q_PROPERTY(double debi READ debi NOTIFY debiChanged)
11     Q_PROPERTY(double egimX READ egimX NOTIFY egimXChanged)
12     Q_PROPERTY(double egimY READ egimY NOTIFY egimYChanged)
13     Q_PROPERTY(double hamAgirlik READ hamAgirlik NOTIFY hamAgirlikChanged)
14     Q_PROPERTY(double hamMesafe READ hamMesafe NOTIFY hamMesafeChanged)
15     Q_PROPERTY(double hamAccelX READ hamAccelX NOTIFY hamAccelDegisti)
16     Q_PROPERTY(double hamAccelY READ hamAccelY NOTIFY hamAccelDegisti)
17     Q_PROPERTY(double hamAccelZ READ hamAccelZ NOTIFY hamAccelDegisti)
18     Q_PROPERTY(bool veriGecerli READ veriGecerli NOTIFY veriGecerliChanged)
19     Q_PROPERTY(double bataryaVoltaj READ bataryaVoltaj NOTIFY bataryaVoltajChanged)
20
21 public:
22     explicit SensorManager(QObject *parent = nullptr);
23
24     double basinc() const;
25     double konum() const;
26     double hiz() const;
27     double debi() const;
28     double egimX() const;
29     double egimY() const;
30     double hamAgirlik() const;
31     double hamMesafe() const;
32     double hamAccelX() const;
33     double hamAccelY() const;
34     double hamAccelZ() const;
35     bool veriGecerli() const;
36     double bataryaVoltaj() const;
37
38     Q_INVOKABLE void veriGuncelle(double basinc, double konum, double hiz, double debi, double egimX, double egimY);
39     Q_INVOKABLE void hamAgirlikGuncelle(double hamAgirlik);
40     Q_INVOKABLE void hamMesafeGuncelle(double hamMesafe);
41     Q_INVOKABLE void hamAccelGuncelle(double x, double y, double z);
42     Q_INVOKABLE void bataryaVoltajGuncelle(double voltaj);
43     Q_INVOKABLE void veriyiGecersizYap();
44
45 signals:
46     void basincChanged();
47     void konumChanged();
48     void hizChanged();
49     void debiChanged();
50     void egimXChanged();
51     void egimYChanged();
52     void hamAgirlikChanged();
53     void hamMesafeChanged();
54     void hamAccelDegisti();
55     void veriGecerliChanged();
56     void veriGuncellendi();
57     void bataryaVoltajChanged();
58
59 private:
60     double m_basinc = 0.0;
61     double m_konum = 0.0;
62     double m_hiz = 0.0;
63     double m_debi = 0.0;
64     double m_egimX = 0.0;
65     double m_egimY = 0.0;
66     double m_hamAgirlik = 0.0;
67     double m_hamMesafe = 0.0;
68     double m_hamAccelX = 0.0;
69     double m_hamAccelY = 0.0;
70     double m_hamAccelZ = 0.0;
71     bool m_veriGecerli = false;
72     double m_bataryaVoltaj = 0.0;
73 };

```

```

1  #include "VoiceCommandManager.h"
2  #include "VoiceRecognitionWorker.h"
3
4  #include <QCoreApplication>
5  #include <QDir>
6  #include <QMetaObject>
7
8  VoiceCommandManager::VoiceCommandManager(QObject *parent)
9      : QObject(parent)
10 {
11     const QString modelYolu = QDir(QCoreApplication::applicationDirPath()).filePath("vosk-model-small-tr-0.3");
12
13     m_worker = new VoiceRecognitionWorker(modelYolu);
14     m_worker->moveToThread(&m_workerThread);
15
16     connect(&m_workerThread, &QThread::finished, m_worker, &QObject::deleteLater);
17
18     connect(m_worker, &VoiceRecognitionWorker::modelHazir, this, &VoiceCommandManager::workerModelHazir);
19     connect(m_worker, &VoiceRecognitionWorker::dinliyorDegisti, this, &VoiceCommandManager::workerDinliyorDegisti);
20     connect(m_worker, &VoiceRecognitionWorker::hataOlustu, this, &VoiceCommandManager::workerHataOlustu);
21     connect(m_worker, &VoiceRecognitionWorker::komutAlgilandi, this, &VoiceCommandManager::workerKomutAlgilandi);
22     connect(m_worker, &VoiceRecognitionWorker::anlikMetinDegisti, this, &VoiceCommandManager::workerAnlikMetinDegisti);
23
24     m_workerThread.start();
25     QMetaObject::invokeMethod(m_worker, "modeliYukle", Qt::QueuedConnection);
26 }
27
28 VoiceCommandManager::~VoiceCommandManager()
29 {
30     m_workerThread.quit();
31     m_workerThread.wait(3000);
32 }
33
34 bool VoiceCommandManager::etkin() const { return m_etkin; }
35
36 void VoiceCommandManager::setEtkin(bool etkin)
37 {
38     if (m_etkin == etkin)
39         return;
40     m_etkin = etkin;
41     emit etkinChanged();
42
43     if (m_etkin)
44         QMetaObject::invokeMethod(m_worker, "dinlemeyeBasla", Qt::QueuedConnection);
45     else
46         QMetaObject::invokeMethod(m_worker, "dinlemeyiDurdur", Qt::QueuedConnection);
47 }
48
49 bool VoiceCommandManager::dinliyor() const { return m_dinliyor; }
50 bool VoiceCommandManager::modelHazir() const { return m_modelHazir; }
51 QString VoiceCommandManager::sonKomutMetni() const { return m_sonKomutMetni; }
52 QString VoiceCommandManager::anlikMetin() const { return m_anlikMetin; }
53 QString VoiceCommandManager::sonHata() const { return m_sonHata; }
54
55 void VoiceCommandManager::workerModelHazir(bool hazir)
56 {
57     m_modelHazir = hazir;
58     emit modelHazirChanged();
59
60     if (hazir && m_etkin)
61         QMetaObject::invokeMethod(m_worker, "dinlemeyeBasla", Qt::QueuedConnection);
62 }
63
64 void VoiceCommandManager::workerDinliyorDegisti(bool dinliyor)
65 {
66     m_dinliyor = dinliyor;
67     emit dinliyorChanged();
68 }
69
70 void VoiceCommandManager::workerHataOlustu(const QString &mesaj)
71 {
72     m_sonHata = mesaj;
73     emit sonHataChanged();
74 }
75
76 void VoiceCommandManager::workerKomutAlgilandi(int komutTuru, const QString &metin)
77 {

```

```
78     m_sonKomutMetni = metin;
79     emit sonKomutMetniChanged();
80
81     switch (static_cast<VoiceRecognitionWorker::KomutTuru>(komutTuru)) {
82     case VoiceRecognitionWorker::KomutBaslat: emit baslatKomutu(); break;
83     case VoiceRecognitionWorker::KomutDurdur: emit durdurKomutu(); break;
84     case VoiceRecognitionWorker::KomutBitir:  emit bitirKomutu();  break;
85     case VoiceRecognitionWorker::KomutDevam:  emit devamKomutu();  break;
86     case VoiceRecognitionWorker::KomutEkle:   emit ekleKomutu();   break;
87     }
88 }
89
90 void VoiceCommandManager::workerAnlikMetinDegisti(const QString &metin)
91 {
92     m_anlikMetin = metin;
93     emit anlikMetinChanged();
94 }
95
```

```

1  #pragma once
2  #include <QObject>
3  #include <QString>
4  #include <QThread>
5
6  class VoiceRecognitionWorker;
7
8  // "Başlat", "Durdur", "Bitir" gibi sesli komutları dinlemek için kullanılır.
9  // Laboratuvar ortamında eller kirli/meşgulken dokunmadan ölçüm kontrolü
10 // yapılabilmesi amacıyla eklendi.
11 //
12 // Tanıma tamamen cihaz üzerinde (offline) Vosk kütüphanesi ile yapılır,
13 // internet bağlantısı gerekmez. Yanlışlıkla tetiklenmeyi önlemek için
14 // tanıyıcı sadece bu komut kelimelerini içeren kısıtlı bir sözlükle
15 // (grammar) çalışır; çevredeki gündelik konuşma bu dar kelime setine
16 // düşmediği için genelde göz ardı edilir (bkz. VoiceRecognitionWorker).
17 class VoiceCommandManager : public QObject
18 {
19     Q_OBJECT
20     Q_PROPERTY(bool etkin READ etkin WRITE setEtkin NOTIFY etkinChanged)
21     Q_PROPERTY(bool dinliyor READ dinliyor NOTIFY dinliyorChanged)
22     Q_PROPERTY(bool modelHazir READ modelHazir NOTIFY modelHazirChanged)
23     Q_PROPERTY(QString sonKomutMetni READ sonKomutMetni NOTIFY sonKomutMetniChanged)
24     Q_PROPERTY(QString anlikMetin READ anlikMetin NOTIFY anlikMetinChanged)
25     Q_PROPERTY(QString sonHata READ sonHata NOTIFY sonHataChanged)
26
27 public:
28     explicit VoiceCommandManager(QObject *parent = nullptr);
29     ~VoiceCommandManager() override;
30
31     bool etkin() const;
32     void setEtkin(bool etkin);
33     bool dinliyor() const;
34     bool modelHazir() const;
35     QString sonKomutMetni() const;
36     QString anlikMetin() const;
37     QString sonHata() const;
38
39 signals:
40     void etkinChanged();
41     void dinliyorChanged();
42     void modelHazirChanged();
43     void sonKomutMetniChanged();
44     void anlikMetinChanged();
45     void sonHataChanged();
46
47     // Sayfalar bu sinyallere bağlanıp ilgili aksiyonu tetikler.
48     void baslatKomutu();
49     void durdurKomutu();
50     void bitirKomutu();
51     void devamKomutu();
52     void ekleKomutu();
53
54 private slots:
55     void workerModelHazir(bool hazır);
56     void workerDinliyorDegisti(bool dinliyor);
57     void workerHataOlustu(const QString &mesaj);
58     void workerKomutAlgilandi(int komutTuru, const QString &metin);
59     void workerAnlikMetinDegisti(const QString &metin);
60
61 private:
62     QThread m_workerThread;
63     VoiceRecognitionWorker *m_worker = nullptr;
64
65     bool m_etkin = false;
66     bool m_dinliyor = false;
67     bool m_modelHazir = false;
68     QString m_sonKomutMetni;
69     QString m_anlikMetin;
70     QString m_sonHata;
71 };
72

```

```

1  #include "VoiceRecognitionWorker.h"
2  #include "vosk_api.h"
3
4  #include <QAudioSource>
5  #include <QAudioFormat>
6  #include <QMediaDevices>
7  #include <QAudioDevice>
8  #include <QIODevice>
9  #include <QJsonDocument>
10 #include <QJsonObject>
11 #include <QJsonArray>
12 #include <QDebug>
13 #include <QLocale>
14 #include <QVector>
15 #include <algorithm>
16 #include <cstdlib>
17
18 namespace {
19
20 // Vosk'un tanıyabileceği komut kelimeleri. Tanıyıcı, dinlemeye başlarken
21 // bu listeyle "grammar" (kısıtlı sözlük) moduna alınır: motor SADECE bu
22 // kelimeleri ya da "[unk]" (bilinmeyen) çıktısını üretebilir. Bu sayede
23 // etraftaki gündelik konuşma bu dar kelime setine düşmediği için genelde
24 // [unk] olarak elenir, ama gerçek komut söylenince net yakalanır - ayrıca
25 // arama uzayı kuculduğu için tanıma çok daha hızlı/dogru çalışır.
26 const QStringList &grammarKelimeleri()
27 {
28     static const QStringList kelimeler = {
29         "başlat", "başla",
30         "durdur",
31         "bitir", "bitti", "bitirdi", "bitiriyor", "bitirelim", "tamamla",
32         "devam",
33         "ekle"
34     };
35     return kelimeler;
36 }
37
38 QByteArray grammarJsonUret()
39 {
40     QJsonArray dizi;
41     for (const QString &kelime : grammarKelimeleri())
42         dizi.append(kelime);
43     dizi.append(QStringLiteral("[unk]"));
44     return QJsonDocument(dizi).toJson(QJsonDocument::Compact);
45 }
46
47 // Turkce karakterleri ASCII'ye indirger, kucuk harfe cevırir; boylece
48 // Vosk ciktisi ile aday kelime listeleri arasında esnek karsilastirma
49 // yapılabilir (aksan/karakter farkından etkilenmez).
50 QString sadelestir(const QString &metin)
51 {
52     QString kucuk = QLocale(QLocale::Turkish).toLower(metin);
53     QString sonuc;
54     sonuc.reserve(kucuk.size());
55     for (const QChar &ch : kucuk) {
56         switch (ch.unicode()) {
57             case u's': sonuc += 's'; break;
58             case u'ç': sonuc += 'c'; break;
59             case u'ğ': sonuc += 'g'; break;
60             case u'ü': sonuc += 'u'; break;
61             case u'ö': sonuc += 'o'; break;
62             case u'ı': sonuc += 'i'; break;
63             case u'İ': sonuc += 'i'; break;
64             default:
65                 if (ch.isLetterOrNumber() || ch.isSpace())
66                     sonuc += ch;
67                 break;
68         }
69     }
70     return sonuc;
71 }
72
73 int levenshtein(const QString &a, const QString &b)
74 {
75     const int n = a.size();
76     const int m = b.size();
77     QVector<int> onceki(m + 1), simdiki(m + 1);

```

```

78     for (int j = 0; j <= m; ++j) onceki[j] = j;
79
80     for (int i = 1; i <= n; ++i) {
81         simdiki[0] = i;
82         for (int j = 1; j <= m; ++j) {
83             int maliyet = (a[i - 1] == b[j - 1]) ? 0 : 1;
84             simdiki[j] = std::min({ onceki[j] + 1, simdiki[j - 1] + 1, onceki[j - 1] + maliyet });
85         }
86         onceki.swap(simdiki);
87     }
88     return onceki[m];
89 }
90
91 bool kelimeUyuyorMu(const QString &token, const QStringList &adaylar, int tolerans)
92 {
93     if (adaylar.contains(token))
94         return true;
95     for (const QString &aday : adaylar) {
96         if (std::abs(token.size() - aday.size()) <= tolerans && levenshtein(token, aday) <= tolerans)
97             return true;
98     }
99     return false;
100 }
101
102 } // namespace
103
104 VoiceRecognitionWorker::VoiceRecognitionWorker(const QString &modelYolu, QObject *parent)
105     : QObject(parent)
106     , m_modelYolu(modelYolu)
107 {
108     m_sonKomutZamani.start();
109 }
110
111 VoiceRecognitionWorker::~VoiceRecognitionWorker()
112 {
113     dinlemeyiDurdur();
114     if (m_recognizer) vok_recognizer_free(m_recognizer);
115     if (m_model) vok_model_free(m_model);
116 }
117
118 void VoiceRecognitionWorker::modeliYukle()
119 {
120     vok_set_log_level(-1);
121     m_model = vok_model_new(m_modelYolu.toUtf8().constData());
122     if (!m_model) {
123         emit hataOlustu(QStringLiteral("Ses tanima modeli yuklenemedi: ") + m_modelYolu);
124         emit modelHazir(false);
125         return;
126     }
127     emit modelHazir(true);
128 }
129
130 void VoiceRecognitionWorker::dinlemeyeBasla()
131 {
132     if (!m_model) {
133         emit hataOlustu(QStringLiteral("Ses tanima modeli hazir degil."));
134         return;
135     }
136     if (m_audioSource)
137         return; // zaten dinliyor
138
139     QAudioFormat format;
140     format.setSampleRate(16000);
141     format.setChannelCount(1);
142     format.setSampleFormat(QAudioFormat::Int16);
143
144     const QAudioDevice girisAygiti = QMediaDevices::defaultAudioInput();
145     if (girisAygiti.isNull()) {
146         emit hataOlustu(QStringLiteral("Mikrofon bulunamadi."));
147         return;
148     }
149
150     if (!m_recognizer) {
151         const QByteArray grammar = grammarJsonUret();
152         m_recognizer = vok_recognizer_new_grm(m_model, 16000.0f, grammar.constData());
153         if (!m_recognizer) {
154             emit hataOlustu(QStringLiteral("Ses tanima motoru baslatilamadi."));
155             return;
156         }
157         vok_recognizer_set_words(m_recognizer, 0);

```



```

158     } else {
159         vosk_recognizer_reset(m_recognizer);
160     }
161
162     m_audioSource = new QAudioSource(girisAygiti, format, this);
163     m_audioDevice = m_audioSource->start();
164     if (!m_audioDevice) {
165         emit hataOlustu(QStringLiteral("Mikrofon acilamadi."));
166         delete m_audioSource;
167         m_audioSource = nullptr;
168         return;
169     }
170
171     connect(m_audioDevice, &QIODevice::readyRead, this, &VoiceRecognitionWorker::sesVerisiHazir);
172     emit dinliyorDegisti(true);
173 }
174
175 void VoiceRecognitionWorker::dinlemeyiDurdur()
176 {
177     if (!m_audioSource)
178         return;
179
180     m_audioSource->stop();
181     m_audioSource->deleteLater();
182     m_audioSource = nullptr;
183     m_audioDevice = nullptr;
184
185     if (m_recognizer)
186         vosk_recognizer_reset(m_recognizer);
187
188     emit dinliyorDegisti(false);
189     emit anlikMetinDegisti(QString());
190 }
191
192 void VoiceRecognitionWorker::sesVerisiHazir()
193 {
194     if (!m_audioDevice || !m_recognizer)
195         return;
196
197     const QByteArray veri = m_audioDevice->readAll();
198     if (veri.isEmpty())
199         return;
200
201     const int bitti = vosk_recognizer_accept_waveform(m_recognizer, veri.constData(), veri.size());
202     if (bitti == 1) {
203         sonucuIsle(QString::fromUtf8(vosk_recognizer_result(m_recognizer)));
204     } else if (bitti == 0) {
205         const QString kismi = QString::fromUtf8(vosk_recognizer_partial_result(m_recognizer));
206         QJsonObject obj = QJsonDocument::fromJson(kismi.toUtf8()).object();
207         emit anlikMetinDegisti(obj.value(QStringLiteral("partial")).toString());
208     }
209 }
210
211 void VoiceRecognitionWorker::sonucuIsle(const QString &json)
212 {
213     QJsonObject obj = QJsonDocument::fromJson(json.toUtf8()).object();
214     const QString metin = obj.value(QStringLiteral("text")).toString().trimmed();
215     emit anlikMetinDegisti(QString());
216     if (metin.isEmpty())
217         return;
218
219     metniDegerlendir(metin);
220 }
221
222 void VoiceRecognitionWorker::metniDegerlendir(const QString &metin)
223 {
224     static const QStringList baslatAdaylari = { "baslat", "basla" };
225     static const QList durdurAdaylari = { "durdur" };
226     static const QStringList bitirAdaylari = { "bitir", "bitti", "bitirdi", "bitiriyor", "bitirelim", "tamamla" };
227     static const QStringList devamAdaylari = { "devam" };
228     static const QStringList ekleAdaylari = { "ekle" };
229
230     const QString sade = sadelestir(metin);
231     const QStringList tokenlar = sade.split(' ', Qt::SkipEmptyParts);
232
233     for (const QString &token : tokenlar) {
234         if (token == "unk")
235             continue; // grammar disina cikan / eslesmeyen ses
236
237         int komutTuru = -1;

```

```
238     if (kelimeUyuyorMu(token, bitirAdaylari, 1)) komutTuru = VoiceRecognitionWorker::KomutBitir;
239     else if (kelimeUyuyorMu(token, durdurAdaylari, 1)) komutTuru = VoiceRecognitionWorker::KomutDurdur;
240     else if (kelimeUyuyorMu(token, devamAdaylari, 1)) komutTuru = VoiceRecognitionWorker::KomutDevam;
241     else if (kelimeUyuyorMu(token, ekleAdaylari, 1)) komutTuru = VoiceRecognitionWorker::KomutEkle;
242     else if (kelimeUyuyorMu(token, baslatAdaylari, 1)) komutTuru = VoiceRecognitionWorker::KomutBaslat;
243
244     if (komutTuru >= 0) {
245         if (m_sonKomutZamani.elapsed() < KOMUT_BEKLEME_MS)
246             return; // cok yakin zamanda baska komut tetiklendi, tekrari engelle
247         m_sonKomutZamani.restart();
248         emit komutAlgilandi(komutTuru, metin);
249         return;
250     }
251 }
252 }
253
```

src/VoiceRecognitionWorker.h

```
1  #pragma once
2  #include <QObject>
3  #include <QString>
4  #include <QElapsedTimer>
5
6  class QAudioSource;
7  class QIODevice;
8  struct VoskModel;
9  struct VoskRecognizer;
10
11 // VoiceCommandManager'in ayri bir QThread icinde calistirdigi gercek isci.
12 // Mikrofon yakalama ve Vosk ile konusma tanima burada, arayuz thread'ini
13 // bloklamadan yapilir.
14 class VoiceRecognitionWorker : public QObject
15 {
16     Q_OBJECT
17
18 public:
19     enum KomutTuru {
20         KomutBaslat = 0,
21         KomutDurdur = 1,
22         KomutBitir = 2,
23         KomutDevam = 3,
24         KomutEkle = 4
25     };
26     Q_ENUM(KomutTuru)
27
28     explicit VoiceRecognitionWorker(const QString &modelYolu, QObject *parent = nullptr);
29     ~VoiceRecognitionWorker() override;
30
31 public slots:
32     void modeliYukle();
33     void dinlemeyeBasla();
34     void dinlemeyiDurdur();
35
36 signals:
37     void modelHazir(bool basarili);
38     void dinliyorDegisti(bool dinliyor);
39     void hataOlustu(const QString &mesaj);
40     void komutAlgilandi(int komutTuru, const QString &metin);
41     void anlikMetinDegisti(const QString &metin);
42
43 private slots:
44     void sesVerisiHazir();
45
46 private:
47     void sonucuIsle(const QString &json);
48     void metniDegerlendir(const QString &metin);
49
50     QString m_modelYolu;
51     VoskModel *m_model = nullptr;
52     VoskRecognizer *m_recognizer = nullptr;
53     QAudioSource *m_audioSource = nullptr;
54     QIODevice *m_audioDevice = nullptr;
55
56     QElapsedTimer m_sonKomutZamani;
57     static constexpr qint64 KOMUT_BEKLEME_MS = 1200;
58 };
59
```

```

1  #include "WifiManager.h"
2  #include <QJsonDocument>
3  #include <QJsonObject>
4  #include <QDebug>
5  #include <cmath>
6
7  WifiManager::WifiManager(SensorManager *sensorManager, Database *database, QObject *parent)
8      : QObject(parent)
9      , m_sensorManager(sensorManager)
10     , m_database(database)
11 {
12     m_soket = new QTcpSocket(this);
13
14     connect(m_soket, &QTcpSocket::connected, this, &WifiManager::soketBaglandi);
15     connect(m_soket, &QTcpSocket::disconnected, this, &WifiManager::soketAyrildi);
16     connect(m_soket, &QTcpSocket::errorOccurred, this, &WifiManager::soketHata);
17     connect(m_soket, &QTcpSocket::readyRead, this, &WifiManager::veriHazir);
18
19     // Hareketsizlik kontrolu: baglanti acikken 2 dakikada bir kontrol edilir,
20     // en son veri geldigi zamandan itibaren HAREKETSIZLIK_LIMIT_MS gecmisse
21     // baglanti otomatik kesilir (orijinal cihazin "2 saat sonra otomatik
22     // kapanma" davranisinin yazilimsal karsiligi).
23     m_hareketsizlikTimer.setInterval(2 * 60 * 1000);
24     connect(&m_hareketsizlikTimer, &QTimer::timeout, this, &WifiManager::hareketsizlikKontrolEt);
25 }
26
27 bool WifiManager::baglandi() const { return m_baglandi; }
28 bool WifiManager::baglaniyor() const { return m_baglaniyor; }
29 QString WifiManager::durumMesaji() const { return m_durumMesaji; }
30
31 void WifiManager::baglan()
32 {
33     if (m_baglandi || m_baglaniyor) return;
34
35     m_baglaniyor = true;
36     emit baglaniyorChanged();
37     durumGuncelle("SLIPER-ESP32'ye baglaniliyor...");
38
39     m_soket->connectToHost(QString(ESP32_IP), ESP32_PORT);
40 }
41
42 void WifiManager::baglantiyiKes()
43 {
44     m_soket->disconnectFromHost();
45 }
46
47 void WifiManager::kalibrasyonYenidenYukle()
48 {
49     kalibrasyonYukle();
50 }
51
52 void WifiManager::kalibrasyonYukle()
53 {
54     if (!m_database) return;
55
56     m_loadCellHamDegerleri.clear();
57     m_loadCellKiloDegerleri.clear();
58     const QVariantList noktalar = m_database->loadCellNoktalariniGetir();
59     for (const QVariant &v : noktalar) {
60         QVariantMap n = v.toMap();
61         m_loadCellHamDegerleri.append(n["hamDeger"].toDouble());
62         m_loadCellKiloDegerleri.append(n["hedefKg"].toDouble());
63     }
64     qDebug() << "Load cell" << m_loadCellHamDegerleri.size() << "nokta yuklendi.";
65
66     m_mesafeHamDegerleri.clear();
67     m_mesafeMmDegerleri.clear();
68     const QVariantList mesafeNoktalar = m_database->mesafeNoktalariniGetir();
69     for (const QVariant &v : mesafeNoktalar) {
70         QVariantMap n = v.toMap();
71         m_mesafeHamDegerleri.append(n["hamDeger"].toDouble());
72         m_mesafeMmDegerleri.append(n["hedefMm"].toDouble());
73     }
74     qDebug() << "Mesafe" << m_mesafeHamDegerleri.size() << "nokta yuklendi.";
75
76     QVariantMap egimKal = m_database->egimKalibrasyonuGetir();
77     if (egimKal.value("mevcut").toBool()) {

```

```

78     m_egimBiasX = egimKal.value("biasX").toDouble();
79     m_egimBiasY = egimKal.value("biasY").toDouble();
80     m_egimBiasZ = egimKal.value("biasZ").toDouble();
81     m_egimGainX = egimKal.value("gainX").toDouble();
82     m_egimGainY = egimKal.value("gainY").toDouble();
83     m_egimGainZ = egimKal.value("gainZ").toDouble();
84     qDebug() << "Egim ivme kalibrasyonu yuklendi.";
85 }
86 }
87
88 void WifiManager::soketBaglandi()
89 {
90     m_baglandi = true;
91     m_baglaniyor = false;
92     emit baglandiChanged();
93     emit baglaniyorChanged();
94     durumGuncelle("SLIPER-ESP32 Bağlı");
95
96     m_ilkPaket = true;
97     m_zamanlayici.restart();
98     m_hareketsizlikZamanlayici.restart();
99     m_hareketsizlikTimer.start();
100    kalibrasyonYukle();
101
102    if (m_sensorManager) {
103        m_sensorManager->veriyiGecersizYap();
104    }
105
106    qDebug() << "ESP32'ye baglanildi:" << ESP32_IP << ESP32_PORT;
107 }
108
109 void WifiManager::soketAyrildi()
110 {
111     m_baglandi = false;
112     m_baglaniyor = false;
113     m_hareketsizlikTimer.stop();
114     emit baglandiChanged();
115     emit baglaniyorChanged();
116     durumGuncelle("Bağlı Değil");
117
118     if (m_sensorManager) {
119         m_sensorManager->veriyiGecersizYap();
120     }
121
122     qDebug() << "ESP32 baglantisi kesildi.";
123 }
124
125 void WifiManager::hareketsizlikKontrolEt()
126 {
127     if (!m_baglandi) return;
128
129     if (m_hareketsizlikZamanlayici.elapsed() >= HAREKETSIZLIK_LIMIT_MS) {
130         qWarning() << "2 saatten uzun suredir veri akisi yok, baglanti otomatik kesiliyor.";
131         durumGuncelle("Hareketsizlik nedeniyle baglanti kesildi");
132         emit hareketsizlikNedeniyleBaglantiKesildi();
133         baglantiyiKes();
134     }
135 }
136
137 void WifiManager::soketHata(QAbstractSocket::SocketError hata)
138 {
139     m_baglandi = false;
140     m_baglaniyor = false;
141     emit baglandiChanged();
142     emit baglaniyorChanged();
143     durumGuncelle("Baglanti hatasi: " + m_soket->errorString());
144
145     qWarning() << "Wifi soket hatasi:" << hata << m_soket->errorString();
146 }
147
148 void WifiManager::veriHazir()
149 {
150     if (m_hareketsizlikZamanlayici.isValid()) {
151         m_hareketsizlikZamanlayici.restart();
152     }
153
154     m_tamponVeri.append(m_soket->readAll());
155
156     int satirSonu;
157     while ((satirSonu = m_tamponVeri.indexOf('\n')) != -1) {

```

```

158     QByteArray satir = m_tamponVeri.left(satirSonu).trimmed();
159     m_tamponVeri.remove(0, satirSonu + 1);
160
161     if (!satir.isEmpty()) {
162         jsonSatiriIsle(satir);
163     }
164 }
165 }
166
167 void WifiManager::egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY) const
168 {
169     egimX = std::atan2(accelY, accelZ) * 180.0 / M_PI;
170     egimY = std::atan2(-accelX, std::sqrt(accelY * accelY + accelZ * accelZ)) * 180.0 / M_PI;
171 }
172
173 void WifiManager::jsonSatiriIsle(const QByteArray &satir)
174 {
175     QJsonParseError hata;
176     QJsonDocument belge = QJsonDocument::fromJson(satir, &hata);
177
178     if (hata.error != QJsonParseError::NoError || !belge.isObject()) {
179         qWarning() << "JSON parse hatasi:" << hata.errorString() << satir;
180         return;
181     }
182
183     QJsonObject obj = belge.object();
184
185     const double hamAgirlik = obj.value("hamAgirlik").toDouble();
186     const double hamMesafe = obj.contains("hamMesafe")
187         ? obj.value("hamMesafe").toDouble()
188         : obj.value("konum").toDouble();
189     const double accelX = obj.value("accelX").toDouble();
190     const double accelY = obj.value("accelY").toDouble();
191     const double accelZ = obj.value("accelZ").toDouble();
192
193     // Kalibrasyon ekraninda gerekiyor: ham deger her zaman guncellensin
194     if (m_sensorManager) {
195         m_sensorManager->hamAgirlikGuncelle(hamAgirlik);
196         m_sensorManager->hamMesafeGuncelle(hamMesafe);
197         m_sensorManager->hamAccelGuncelle(accelX, accelY, accelZ);
198
199         // Batarya voltaji: firmware "hamBatarya" alanini göndermiyorsa
200         // (henuz donanim baglanmadiysa) 0 olarak kalir, QML tarafinda
201         // gosterge "bilinmiyor" durumunda birakilmalidir.
202         if (obj.contains("hamBatarya")) {
203             const double hamBatarya = obj.value("hamBatarya").toDouble();
204             const double voltaj = hamBatarya * ADS1115_LSB_VOLT * BATARYA_BOLUCU_ORANI;
205             m_sensorManager->bataryaVoltajGuncelle(voltaj);
206         }
207     }
208
209     const double konum = m_mesafeHamDegerleri.isEmpty()
210         ? hamMesafe
211         : mesafeInterpolasyon(hamMesafe);
212
213     // --- Basinc: kayitli kalibrasyon ile ---
214     const double agirlikKg = loadCellInterpolasyon(hamAgirlik);
215     const double basinc = agirlikKg * 98.1;
216
217     // --- Hiz ve debi: konum farkindan turetiliyor ---
218     double hiz = 0.0;
219     double debi = 0.0;
220
221     const qint64 simdikiZamanMs = m_zamanlayici.elapsed();
222
223     if (!m_ilkPaket) {
224         double dt = (simdikiZamanMs - m_oncekiZamanMs) / 1000.0;
225         if (dt <= 0.0) dt = 0.001;
226
227         const double boruAlanim2 = M_PI * (BORU_CAPI_M / 2.0) * (BORU_CAPI_M / 2.0);
228         hiz = std::fabs(konum - m_oncekiKonum) / 1000.0 / dt;
229         debi = hiz * boruAlanim2 * 3600.0;
230     } else {
231         m_ilkPaket = false;
232     }
233
234     m_oncekiKonum = konum;
235     m_oncekiZamanMs = simdikiZamanMs;
236
237     // --- Egim: once ham ivme bias/gain ile duzeltilir, sonra aciya cevrilir ---

```

```

238     const double duzeltilmisX = (accelX - m_egimBiasX) / m_egimGainX;
239     const double duzeltilmisY = (accelY - m_egimBiasY) / m_egimGainY;
240     const double duzeltilmisZ = (accelZ - m_egimBiasZ) / m_egimGainZ;
241
242     double egimX = 0.0, egimY = 0.0;
243     egimHesapla(duzeltilmisX, duzeltilmisY, duzeltilmisZ, egimX, egimY);
244
245     if (m_sensorManager) {
246         m_sensorManager->veriGuncelle(basinc, konum, hiz, debi, egimX, egimY);
247     }
248 }
249
250 void WifiManager::durumGuncelle(const QString &mesaj)
251 {
252     if (m_durumMesaji != mesaj) {
253         m_durumMesaji = mesaj;
254         emit durumMesajiChanged();
255     }
256 }
257
258 double WifiManager::loadCellInterpolasyon(double hamDeger) const
259 {
260     const int n = m_loadCellHamDegerleri.size();
261     if (n == 0) return 0.0;
262     if (n == 1) return m_loadCellKiloDegerleri[0];
263
264     int i;
265     if (hamDeger <= m_loadCellHamDegerleri[0]) {
266         i = 0;
267     } else if (hamDeger >= m_loadCellHamDegerleri[n - 1]) {
268         i = n - 2;
269     } else {
270         for (i = 0; i < n - 1; ++i) {
271             if (hamDeger <= m_loadCellHamDegerleri[i + 1]) break;
272         }
273     }
274
275     const double ham1 = m_loadCellHamDegerleri[i];
276     const double ham2 = m_loadCellHamDegerleri[i + 1];
277     const double kg1 = m_loadCellKiloDegerleri[i];
278     const double kg2 = m_loadCellKiloDegerleri[i + 1];
279
280     if (ham2 == ham1) return kg1;
281     return kg1 + (hamDeger - ham1) * (kg2 - kg1) / (ham2 - ham1);
282 }
283
284 double WifiManager::mesafeInterpolasyon(double hamDeger) const
285 {
286     const int n = m_mesafeHamDegerleri.size();
287     if (n == 0) return 0.0;
288     if (n == 1) return m_mesafeMmDegerleri[0];
289
290     int i;
291     if (hamDeger <= m_mesafeHamDegerleri[0]) {
292         i = 0;
293     } else if (hamDeger >= m_mesafeHamDegerleri[n - 1]) {
294         i = n - 2;
295     } else {
296         for (i = 0; i < n - 1; ++i) {
297             if (hamDeger <= m_mesafeHamDegerleri[i + 1]) break;
298         }
299     }
300
301     const double ham1 = m_mesafeHamDegerleri[i];
302     const double ham2 = m_mesafeHamDegerleri[i + 1];
303     const double mm1 = m_mesafeMmDegerleri[i];
304     const double mm2 = m_mesafeMmDegerleri[i + 1];
305
306     if (ham2 == ham1) return mm1;
307     return mm1 + (hamDeger - ham1) * (mm2 - mm1) / (ham2 - ham1);
308 }

```

```

1  #pragma once
2  #include <QObject>
3  #include <QTcpSocket>
4  #include <QElapsedTimer>
5  #include <QTimer>
6  #include <QVector>
7  #include "SensorManager.h"
8  #include "Database.h"
9
10 class WifiManager : public QObject
11 {
12     Q_OBJECT
13     Q_PROPERTY(bool baglandi READ baglandi NOTIFY baglandiChanged)
14     Q_PROPERTY(bool baglaniyor READ baglaniyor NOTIFY baglaniyorChanged)
15     Q_PROPERTY(QString durumMesaji READ durumMesaji NOTIFY durumMesajiChanged)
16
17 public:
18     explicit WifiManager(SensorManager *sensorManager, Database *database, QObject *parent = nullptr);
19
20     bool baglandi() const;
21     bool baglaniyor() const;
22     QString durumMesaji() const;
23
24     Q_INVOKABLE void baglan();
25     Q_INVOKABLE void baglantiyiKes();
26     Q_INVOKABLE void kalibrasyonYenidenYukle();
27
28 signals:
29     void baglandiChanged();
30     void baglaniyorChanged();
31     void durumMesajiChanged();
32     // Orjinal SLIPER'daki "2 saat hareketsizlikten sonra otomatik kapanma"
33     // davranışının yazılımsal karşılığı: gerçek donanımsal güç kesme yerine,
34     // uzun süre veri akışı olmayınca bağlantı otomatik kesilip kullanıcı
35     // bilgilendirilir.
36     void hareketsizlikNedeniyleBaglantiKesildi();
37
38 private slots:
39     void soketBaglandi();
40     void soketAyrildi();
41     void soketHata(QAbstractSocket::SocketError hata);
42     void veriHazir();
43     void hareketsizlikKontrolEt();
44
45 private:
46     void durumGuncelle(const QString &mesaj);
47     void jsonSatiriIsle(const QByteArray &satir);
48     void egimHesapla(double accelX, double accelY, double accelZ, double &egimX, double &egimY) const;
49     void kalibrasyonYukle();
50     double loadCellInterpolasyon(double hamDeger) const;
51     double mesafeInterpolasyon(double hamDeger) const;
52
53     QTcpSocket *m_soket = nullptr;
54     SensorManager *m_sensorManager = nullptr;
55     Database *m_database = nullptr;
56     QByteArray m_tamponVeri;
57
58     bool m_baglandi = false;
59     bool m_baglaniyor = false;
60     QString m_durumMesaji = "Bağlı Değil";
61
62     bool m_ilkPaket = true;
63     double m_oncekiKonum = 0.0;
64     quint64 m_oncekiZamanMs = 0;
65     QElapsedTimer m_zamanlayici;
66     QElapsedTimer m_hareketsizlikZamanlayici;
67
68     // --- Kayitli kalibrasyon degerleri (Database'den yuklenir) ---
69
70     QVector<double> m_loadCellHamDegerleri;
71     QVector<double> m_loadCellKiloDegerleri;
72
73     QVector<double> m_mesafeHamDegerleri;
74     QVector<double> m_mesafeMmDegerleri;
75
76     double m_egimBiasX = 0.0, m_egimBiasY = 0.0, m_egimBiasZ = 0.0;
77     double m_egimGainX = 1.0, m_egimGainY = 1.0, m_egimGainZ = 1.0;

```



```
78
79 static constexpr const char *ESP32_IP = "192.168.4.1";
80 static constexpr quint16 ESP32_PORT = 8888;
81
82 static constexpr double BORU_CAPI_M = 0.126;
83
84 // --- Batarya izleme ---
85 // NOT: ESP32 firmware'inde henuz gercek bir voltaj bolucu devre
86 // kalibrasyonu yapilmadi (bkz. sliper_esp32.ino icindeki TODO notu).
87 // Asagidaki katsayi, ADS1115'in GAIN_ONE modunda ( +-4.096V, 16 bit )
88 // okudugu ham degeri, varsayimsal 1:2 oranli bir bolucu ile pil
89 // voltajina cevirir. Gercek donanim baglaninca BATARYA_BOLUCU_ORANI
90 // olcum yapilarak duzeltilmelidir.
91 static constexpr double ADS1115_LSB_VOLT = 0.000125; // 4.096V / 32768
92 static constexpr double BATARYA_BOLUCU_ORANI = 2.0;
93 static constexpr double BATARYA_UYARI_VOLTAJ = 14.0;
94 static constexpr double BATARYA_IYI_VOLTAJ = 14.5;
95
96 QTimer m_hareketsizlikTimer;
97 static constexpr int HAREKETSIZLIK_LIMIT_MS = 2 * 60 * 60 * 1000; // 2 saat
98 };
```

```

1 // Copyright 2020-2021 Alpha Cephei Inc.
2 //
3 // Licensed under the Apache License, Version 2.0 (the "License");
4 // you may not use this file except in compliance with the License.
5 // You may obtain a copy of the License at
6 //
7 //     http://www.apache.org/licenses/LICENSE-2.0
8 //
9 // Unless required by applicable law or agreed to in writing, software
10 // distributed under the License is distributed on an "AS IS" BASIS,
11 // WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
12 // See the License for the specific language governing permissions and
13 // limitations under the license.
14
15 /* This header contains the C API for Vosk speech recognition system */
16
17 #ifndef VOSK_API_H
18 #define VOSK_API_H
19
20 #ifdef __cplusplus
21 extern "C" {
22 #endif
23
24 /** Model stores all the data required for recognition
25  * it contains static data and can be shared across processing
26  * threads. */
27 typedef struct VoskModel VoskModel;
28
29
30 /** Speaker model is the same as model but contains the data
31  * for speaker identification. */
32 typedef struct VoskSpkModel VoskSpkModel;
33
34
35 /** Recognizer object is the main object which processes data.
36  * Each recognizer usually runs in own thread and takes audio as input.
37  * Once audio is processed recognizer returns JSON object as a string
38  * which represent decoded information - words, confidences, times, n-best lists,
39  * speaker information and so on */
40 typedef struct VoskRecognizer VoskRecognizer;
41
42
43 /**
44  * Batch model object
45  */
46 typedef struct VoskBatchModel VoskBatchModel;
47
48 /**
49  * Batch recognizer object
50  */
51 typedef struct VoskBatchRecognizer VoskBatchRecognizer;
52
53
54 /** Loads model data from the file and returns the model object
55  *
56  * @param model_path: the path of the model on the filesystem
57  * @returns model object or NULL if problem occurred */
58 VoskModel *vosk_model_new(const char *model_path);
59
60
61 /** Releases the model memory
62  *
63  * The model object is reference-counted so if some recognizer
64  * depends on this model, model might still stay alive. When
65  * last recognizer is released, model will be released too. */
66 void vosk_model_free(VoskModel *model);
67
68
69 /** Check if a word can be recognized by the model
70  * @param word: the word
71  * @returns the word symbol if @param word exists inside the model
72  * or -1 otherwise.
73  * Reminding that word symbol 0 is for <epsilon> */
74 int vosk_model_find_word(VoskModel *model, const char *word);
75
76
77 /** Loads speaker model data from the file and returns the model object

```

```

78 *
79 * @param model_path: the path of the model on the filesystem
80 * @returns model object or NULL if problem occurred */
81 VoskSpkModel *vosk_spk_model_new(const char *model_path);
82
83
84 /** Releases the model memory
85 *
86 * The model object is reference-counted so if some recognizer
87 * depends on this model, model might still stay alive. When
88 * last recognizer is released, model will be released too. */
89 void vosk_spk_model_free(VoskSpkModel *model);
90
91 /** Creates the recognizer object
92 *
93 * The recognizers process the speech and return text using shared model data
94 * @param model VoskModel containing static data for recognizer. Model can be
95 * shared across recognizers, even running in different threads.
96 * @param sample_rate The sample rate of the audio you going to feed into the recognizer.
97 * Make sure this rate matches the audio content, it is a common
98 * issue causing accuracy problems.
99 * @returns recognizer object or NULL if problem occurred */
100 VoskRecognizer *vosk_recognizer_new(VoskModel *model, float sample_rate);
101
102
103 /** Creates the recognizer object with speaker recognition
104 *
105 * With the speaker recognition mode the recognizer not just recognize
106 * text but also return speaker vectors one can use for speaker identification
107 *
108 * @param model VoskModel containing static data for recognizer. Model can be
109 * shared across recognizers, even running in different threads.
110 * @param sample_rate The sample rate of the audio you going to feed into the recognizer.
111 * Make sure this rate matches the audio content, it is a common
112 * issue causing accuracy problems.
113 * @param spk_model speaker model for speaker identification
114 * @returns recognizer object or NULL if problem occurred */
115 VoskRecognizer *vosk_recognizer_new_spk(VoskModel *model, float sample_rate, VoskSpkModel *spk_model);
116
117
118 /** Creates the recognizer object with the phrase list
119 *
120 * Sometimes when you want to improve recognition accuracy and when you don't need
121 * to recognize large vocabulary you can specify a list of phrases to recognize. This
122 * will improve recognizer speed and accuracy but might return [unk] if user said
123 * something different.
124 *
125 * Only recognizers with lookahead models support this type of quick configuration.
126 * Precompiled HCLG graph models are not supported.
127 *
128 * @param model VoskModel containing static data for recognizer. Model can be
129 * shared across recognizers, even running in different threads.
130 * @param sample_rate The sample rate of the audio you going to feed into the recognizer.
131 * Make sure this rate matches the audio content, it is a common
132 * issue causing accuracy problems.
133 * @param grammar The string with the list of phrases to recognize as JSON array of strings,
134 * for example "[\"one two three four five\", \"[unk]\"]".
135 *
136 * @returns recognizer object or NULL if problem occurred */
137 VoskRecognizer *vosk_recognizer_new_grm(VoskModel *model, float sample_rate, const char *grammar);
138
139
140 /** Adds speaker model to already initialized recognizer
141 *
142 * Can add speaker recognition model to already created recognizer. Helps to initialize
143 * speaker recognition for grammar-based recognizer.
144 *
145 * @param spk_model Speaker recognition model */
146 void vosk_recognizer_set_spk_model(VoskRecognizer *recognizer, VoskSpkModel *spk_model);
147
148
149 /** Reconfigures recognizer to use grammar
150 *
151 * @param recognizer Already running VoskRecognizer
152 * @param grammar Set of phrases in JSON array of strings or "[ ]" to use default model graph.
153 * See also vosk_recognizer_new_grm
154 */
155 void vosk_recognizer_set_grm(VoskRecognizer *recognizer, char const *grammar);
156
157

```

```

158 /** Configures recognizer to output n-best results
159 *
160 * <pre>
161 * {
162 *     "alternatives": [
163 *         { "text": "one two three four five", "confidence": 0.97 },
164 *         { "text": "one two three for five", "confidence": 0.03 },
165 *     ]
166 * }
167 * </pre>
168 *
169 * @param max_alternatives - maximum alternatives to return from recognition results
170 */
171 void vosk_recognizer_set_max_alternatives(VoskRecognizer *recognizer, int max_alternatives);
172
173
174 /** Enables words with times in the output
175 *
176 * <pre>
177 * "result" : [{
178 *     "conf" : 1.000000,
179 *     "end" : 1.110000,
180 *     "start" : 0.870000,
181 *     "word" : "what"
182 * }, {
183 *     "conf" : 1.000000,
184 *     "end" : 1.530000,
185 *     "start" : 1.110000,
186 *     "word" : "zero"
187 * }, {
188 *     "conf" : 1.000000,
189 *     "end" : 1.950000,
190 *     "start" : 1.530000,
191 *     "word" : "zero"
192 * }, {
193 *     "conf" : 1.000000,
194 *     "end" : 2.340000,
195 *     "start" : 1.950000,
196 *     "word" : "zero"
197 * }, {
198 *     "conf" : 1.000000,
199 *     "end" : 2.610000,
200 *     "start" : 2.340000,
201 *     "word" : "one"
202 * }],
203 * </pre>
204 *
205 * @param words - boolean value
206 */
207 void vosk_recognizer_set_words(VoskRecognizer *recognizer, int words);
208
209 /** Like above return words and confidences in partial results
210 *
211 * @param partial_words - boolean value
212 */
213 void vosk_recognizer_set_partial_words(VoskRecognizer *recognizer, int partial_words);
214
215 /** Set NLSML output
216 * @param nlsmml - boolean value
217 */
218 void vosk_recognizer_set_nlsmml(VoskRecognizer *recognizer, int nlsmml);
219
220
221 /** Accept voice data
222 *
223 * accept and process new chunk of voice data
224 *
225 * @param data - audio data in PCM 16-bit mono format
226 * @param length - length of the audio data
227 * @returns 1 if silence is occurred and you can retrieve a new utterance with result method
228 *         0 if decoding continues
229 *        -1 if exception occurred */
230 int vosk_recognizer_accept_waveform(VoskRecognizer *recognizer, const char *data, int length);
231
232
233 /** Same as above but the version with the short data for language bindings where you have
234 * audio as array of shorts */
235 int vosk_recognizer_accept_waveform_s(VoskRecognizer *recognizer, const short *data, int length);
236
237

```

```

238 /** Same as above but the version with the float data for language bindings where you have
239 * audio as array of floats */
240 int vosk_recognizer_accept_waveform_f(VoskRecognizer *recognizer, const float *data, int length);
241
242
243 /** Returns speech recognition result
244 *
245 * @returns the result in JSON format which contains decoded line, decoded
246 * words, times in seconds and confidences. You can parse this result
247 * with any json parser
248 *
249 * <pre>
250 * {
251 *   "text" : "what zero zero zero one"
252 * }
253 * </pre>
254 *
255 * If alternatives enabled it returns result with alternatives, see also vosk_recognizer_set_max_alternatives().
256 *
257 * If word times enabled returns word time, see also vosk_recognizer_set_word_times().
258 */
259 const char *vosk_recognizer_result(VoskRecognizer *recognizer);
260
261
262 /** Returns partial speech recognition
263 *
264 * @returns partial speech recognition text which is not yet finalized.
265 * result may change as recognizer process more data.
266 *
267 * <pre>
268 * {
269 *   "partial" : "cyril one eight zero"
270 * }
271 * </pre>
272 */
273 const char *vosk_recognizer_partial_result(VoskRecognizer *recognizer);
274
275
276 /** Returns speech recognition result. Same as result, but doesn't wait for silence
277 * You usually call it in the end of the stream to get final bits of audio. It
278 * flushes the feature pipeline, so all remaining audio chunks got processed.
279 *
280 * @returns speech result in JSON format.
281 */
282 const char *vosk_recognizer_final_result(VoskRecognizer *recognizer);
283
284
285 /** Resets the recognizer
286 *
287 * Resets current results so the recognition can continue from scratch */
288 void vosk_recognizer_reset(VoskRecognizer *recognizer);
289
290
291 /** Releases recognizer object
292 *
293 * Underlying model is also unreferenced and if needed released */
294 void vosk_recognizer_free(VoskRecognizer *recognizer);
295
296 /** Set log level for Kaldi messages
297 *
298 * @param log_level the level
299 * 0 - default value to print info and error messages but no debug
300 * less than 0 - don't print info messages
301 * greather than 0 - more verbose mode
302 */
303 void vosk_set_log_level(int log_level);
304
305 /**
306 * Init, automatically select a CUDA device and allow multithreading.
307 * Must be called once from the main thread.
308 * Has no effect if HAVE_CUDA flag is not set.
309 */
310 void vosk_gpu_init();
311
312 /**
313 * Init CUDA device in a multi-threaded environment.
314 * Must be called for each thread.
315 * Has no effect if HAVE_CUDA flag is not set.
316 */
317 void vosk_gpu_thread_init();

```

```

318
319 /** Creates the batch recognizer object
320 *
321 * @returns model object or NULL if problem occurred */
322 VoskBatchModel *vosk_batch_model_new(const char *model_path);
323
324 /** Releases batch model object */
325 void vosk_batch_model_free(VoskBatchModel *model);
326
327 /** Wait for the processing */
328 void vosk_batch_model_wait(VoskBatchModel *model);
329
330 /** Creates batch recognizer object
331 * @returns recognizer object or NULL if problem occurred */
332 VoskBatchRecognizer *vosk_batch_recognizer_new(VoskBatchModel *model, float sample_rate);
333
334 /** Releases batch recognizer object */
335 void vosk_batch_recognizer_free(VoskBatchRecognizer *recognizer);
336
337 /** Accept batch voice data */
338 void vosk_batch_recognizer_accept_waveform(VoskBatchRecognizer *recognizer, const char *data, int length);
339
340 /** Set NLSML output
341 * @param nlsml - boolean value
342 */
343 void vosk_batch_recognizer_set_nlsml(VoskBatchRecognizer *recognizer, int nlsml);
344
345 /** Closes the stream */
346 void vosk_batch_recognizer_finish_stream(VoskBatchRecognizer *recognizer);
347
348 /** Return results */
349 const char *vosk_batch_recognizer_front_result(VoskBatchRecognizer *recognizer);
350
351 /** Release and free first retrieved result */
352 void vosk_batch_recognizer_pop(VoskBatchRecognizer *recognizer);
353
354 /** Get amount of pending chunks for more intelligent waiting */
355 int vosk_batch_recognizer_get_pending_chunks(VoskBatchRecognizer *recognizer);
356
357 #ifdef __cplusplus
358 }
359 #endif
360
361 #endif /* VOSK_API_H */
362

```